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**LSM6DS3: always-on 3D accelerometer and 3D gyroscope**

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**Introduction**

This document is intended to provide usage information and application hints related to ST's LSM6DS3 iNEMO inertial module.

The LSM6DS3 is a 3D digital accelerometer and 3D digital gyroscope system-in-package with a digital I<sup>2</sup>C/SPI serial interface standard output, performing at 0.9 mA in combo Normal mode and 1.25 mA (up to 1.6 kHz) in combo High-Performance mode. Thanks to the ultra-low noise performance of both the gyroscope and the accelerometer, the device combines always-on low-power features with superior sensing precision for an optimal motion experience for the consumer. Furthermore, the accelerometer features smart sleep-to-wake-up (Activity) and return-to-sleep (Inactivity) functions that allow advanced power saving.

The device has a dynamic user-selectable full-scale acceleration range of  $\pm 2/\pm 4/\pm 8/\pm 16\text{ g}$  and an angular rate range of  $\pm 125/\pm 245/\pm 500/\pm 1000/\pm 2000\text{ dps}$ .

The LSM6DS3 can be configured to generate interrupt signals by using hardware recognition of free-fall events, 6D orientation, tap and double-tap sensing, activity or inactivity, wake-up events.

The availability of different connection modes to external sensors allows implementing additional functionalities such as sensor hub, auxiliary SPI, etc.

The LSM6DS3 is compatible with the requirements of Android KitKat, Android L and the leading OSs, offering real, virtual and batch-mode sensors. It has been designed to implement in hardware significant motion, tilt, pedometer functions, time stamp and to support the data acquisition of an external magnetometer with ironing correction (hard, soft).

The LSM6DS3 has an integrated smart first-in first-out (FIFO) buffer of up to 8 kbyte size, allowing dynamic batching of significant data (i.e. external sensors, step counter, time stamp and temperature).

The LSM6DS3 is available in a small plastic land grid array package (LGA-14L) and it is guaranteed to operate over an extended temperature range from  $-40\text{ }^{\circ}\text{C}$  to  $+85\text{ }^{\circ}\text{C}$ .

The ultra-small size and weight of the SMD package make it an ideal choice for handheld portable applications such as smartphones, IoT connected devices, and wearables or any other application where reduced package size and weight are required.

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# 1 Registers

Table 1. Registers

| Register name          | Address | Bit7               | Bit6                 | Bit5           | Bit4           | Bit3           | Bit2           | Bit1          | Bit0          |
|------------------------|---------|--------------------|----------------------|----------------|----------------|----------------|----------------|---------------|---------------|
| FUNC_CFG_ACCESS        | 01h     | FUNC_CFG_EN        | 0                    | 0              | 0              | 0              | 0              | 0             | 0             |
| SENSOR_SYNC_TIME_FRAME | 04h     | TPH_7              | TPH_6                | TPH_5          | TPH_4          | TPH_3          | TPH_2          | TPH_1         | TPH_0         |
| FIFO_CTRL1             | 06h     | FTH_7              | FTH_6                | FTH_5          | FTH_4          | FTH_3          | FTH_2          | FTH_1         | FTH_0         |
| FIFO_CTRL2             | 07h     | TIMER_PEDO_FIFO_EN | TIMER_PEDO_FIFO_DRDY | 0              | 0              | FTH_11         | FTH_10         | FTH_9         | FTH_8         |
| FIFO_CTRL3             | 08h     | 0                  | 0                    | DEC_FIFO_GYRO2 | DEC_FIFO_GYRO1 | DEC_FIFO_GYRO0 | DEC_FIFO_XL2   | DEC_FIFO_XL1  | DEC_FIFO_XL0  |
| FIFO_CTRL4             | 09h     | 0                  | ONLY_HIGH_DATA       | DEC_DS4_FIFO2  | DEC_DS4_FIFO1  | DEC_DS4_FIFO0  | DEC_DS3_FIFO2  | DEC_DS3_FIFO1 | DEC_DS3_FIFO0 |
| FIFO_CTRL5             | 0Ah     | 0                  | ODR_FIFO_3           | ODR_FIFO_2     | ODR_FIFO_1     | ODR_FIFO_0     | FIFO_MODE_2    | FIFO_MODE_1   | FIFO_MODE_0   |
| ORIENT_CFG_G           | 0Bh     | 0                  | 0                    | SignX_G        | SignY_G        | SignZ_G        | Orient_2       | Orient_1      | Orient_0      |
| INT1_CTRL              | 0Dh     | INT1_STEP_DETECTOR | INT1_SIG_MOT         | INT1_FULL_FLAG | INT1_FIFO_OVR  | INT1_FTH       | INT1_BOOT      | INT1_DRDY_G   | INT1_DRDY_XL  |
| INT2_CTRL              | 0Eh     | IINT2_STEP_DELTA   | INT2_STEP_COUNT_OV   | INT2_FULL_FLAG | INT2_FIFO_OVR  | INT2_FTH       | INT2_DRDY_TEMP | INT2_DRDY_G   | INT2_DRDY_XL  |
| WHO_AM_I               | 0Fh     | 0                  | 1                    | 1              | 0              | 1              | 0              | 0             | 1             |
| CTRL1_XL               | 10h     | ODR_XL3            | ODR_XL2              | ODR_XL1        | ODR_XL0        | FS_XL1         | FS_XL0         | BW_XL1        | BW_XL0        |
| CTRL2_G                | 11h     | ODR_G3             | ODR_G2               | ODR_G1         | ODR_G0         | FS_G1          | FS_G0          | FS_125        | 0             |
| CTRL3_C                | 12h     | BOOT               | BDU                  | H_LACTIVE      | PP_OD          | SIM            | IF_INC         | BLE           | SW_RESET      |
| CTRL4_C                | 13h     | XL_BW_SCAL_ODR     | SLEEP_G              | INT2_on_INT1   | FIFO_TEMP_EN   | DRDY_MASK      | I2C_disable    | MODE3_EN      | STOP_ON_FTH   |
| CTRL5_C                | 14h     | ROUNDING2          | ROUNDING1            | ROUNDING0      | 0              | ST1_G          | ST0_G          | ST1_XL        | ST0_XL        |

**Table 1. Registers (continued)**

| Register name | Address | Bit7         | Bit6                | Bit5       | Bit4           | Bit3       | Bit2              | Bit1          | Bit0           |
|---------------|---------|--------------|---------------------|------------|----------------|------------|-------------------|---------------|----------------|
| CTRL6_C       | 15h     | TRIG_EN      | LVL <sub>en</sub>   | LVL2_EN    | XL_HM_MODE     | 0          | 0                 | 0             | 0              |
| CTRL7_G       | 16h     | G_HM_MODE    | HP_G_EN             | HPCF_G1    | HPCF_G0        | HP_G_RST   | ROUNDING_STATUS   | 0             | 0              |
| CTRL8_XL      | 17h     | LPF2_XL_EN   | HPCF_XL1            | HPCF_XL0   | 0              | 0          | HP_SLOPE_XL_EN    | 0             | LOW_PASS_ON_6D |
| CTRL9_XL      | 18h     | 0            | 0                   | Zen_XL     | Yen_XL         | Xen_XL     | SOFT_EN           | 0             | 0              |
| CTRL10_C      | 19h     | 0            | 0                   | Zen_G      | Yen_G          | Xen_G      | FUNC_EN           | PEDO_RST_STEP | SIGN_MOTION_EN |
| MASTER_CONFIG | 1Ah     | DRDY_ON_INT1 | DATA_VALID_SEL_FIFO | 0          | START_CONFIG   | PULL_UP_EN | PASS_THROUGH_MODE | IRON_EN       | MASTER_ON      |
| WAKE_UP_SRC   | 1Bh     | 0            | 0                   | FF_IA      | SLEEP_STATE_IA | WU_IA      | X_WU              | Y_WU          | Z_WU           |
| TAP_SRC       | 1Ch     | 0            | TAP_IA              | SINGLE_TAP | DOUBLE_TAP     | TAP_SIGN   | X_TAP             | Y_TAP         | Z_TAP          |
| D6D_SRC       | 1Dh     | 0            | D6D_IA              | ZH         | ZL             | YH         | YL                | XH            | XL             |
| STATUS_REG    | 1Eh     | -            | -                   | -          | -              | EV_BOOT    | TDA               | GDA           | XLDA           |
| OUT_TEMP_L    | 20h     | Temp7        | Temp6               | Temp5      | Temp4          | Temp3      | Temp2             | Temp1         | Temp0          |
| OUT_TEMP_H    | 21h     | Temp15       | Temp14              | Temp13     | Temp12         | Temp11     | Temp10            | Temp9         | Temp8          |
| OUTX_L_G      | 22h     | D7           | D6                  | D5         | D4             | D3         | D2                | D1            | D0             |
| OUTX_H_G      | 23h     | D15          | D14                 | D13        | D12            | D11        | D10               | D9            | D8             |
| OUTY_L_G      | 24h     | D7           | D6                  | D5         | D4             | D3         | D2                | D1            | D0             |
| OUTY_H_G      | 25h     | D15          | D14                 | D13        | D12            | D11        | D10               | D9            | D8             |
| OUTZ_L_G      | 26h     | D7           | D6                  | D5         | D4             | D3         | D2                | D1            | D0             |
| OUTZ_H_G      | 27h     | D15          | D14                 | D13        | D12            | D11        | D10               | D9            | D8             |
| OUTX_L_XL     | 28h     | D7           | D6                  | D5         | D4             | D3         | D2                | D1            | D0             |

**Table 1. Registers (continued)**

| Register name   | Address | Bit7           | Bit6           | Bit5           | Bit4           | Bit3           | Bit2           | Bit1           | Bit0           |
|-----------------|---------|----------------|----------------|----------------|----------------|----------------|----------------|----------------|----------------|
| OUTX_H_XL       | 29h     | D15            | D14            | D13            | D12            | D11            | D10            | D9             | D8             |
| OUTY_L_XL       | 2Ah     | D7             | D6             | D5             | D4             | D3             | D2             | D1             | D0             |
| OUTY_H_XL       | 2Bh     | D15            | D14            | D13            | D12            | D11            | D10            | D9             | D8             |
| OUTZ_L_XL       | 2Ch     | D7             | D6             | D5             | D4             | D3             | D2             | D1             | D0             |
| OUTZ_H_XL       | 2Dh     | D15            | D14            | D13            | D12            | D11            | D10            | D9             | D8             |
| SENSORHUB1_REG  | 2Eh     | SHub1_7        | SHub1_6        | SHub1_5        | SHub1_4        | SHub1_3        | SHub1_2        | SHub1_1        | SHub1_0        |
| SENSORHUB2_REG  | 2Fh     | SHub2_7        | SHub2_6        | SHub2_5        | SHub2_4        | SHub2_3        | SHub2_2        | SHub2_1        | SHub2_0        |
| SENSORHUB3_REG  | 30h     | SHub3_7        | SHub3_6        | SHub3_5        | SHub3_4        | SHub3_3        | SHub3_2        | SHub3_1        | SHub3_0        |
| SENSORHUB4_REG  | 31h     | SHub4_7        | SHub4_6        | SHub4_5        | SHub4_4        | SHub4_3        | SHub4_2        | SHub4_1        | SHub4_0        |
| SENSORHUB5_REG  | 32h     | SHub5_7        | SHub5_6        | SHub5_5        | SHub5_4        | SHub5_3        | SHub5_2        | SHub5_1        | SHub5_0        |
| SENSORHUB6_REG  | 33h     | SHub6_7        | SHub6_6        | SHub6_5        | SHub6_4        | SHub6_3        | SHub6_2        | SHub6_1        | SHub6_0        |
| SENSORHUB7_REG  | 34h     | SHub7_7        | SHub7_6        | SHub7_5        | SHub7_4        | SHub7_3        | SHub7_2        | SHub7_1        | SHub7_0        |
| SENSORHUB8_REG  | 35h     | SHub8_7        | SHub8_6        | SHub8_5        | SHub8_4        | SHub8_3        | SHub8_2        | SHub8_1        | SHub8_0        |
| SENSORHUB9_REG  | 36h     | SHub9_7        | SHub9_6        | SHub9_5        | SHub9_4        | SHub9_3        | SHub9_2        | SHub9_1        | SHub9_0        |
| SENSORHUB10_REG | 37h     | SHub10_7       | SHub10_6       | SHub10_5       | SHub10_4       | SHub10_3       | SHub10_2       | SHub10_1       | SHub10_0       |
| SENSORHUB11_REG | 38h     | SHub11_7       | SHub11_6       | SHub11_5       | SHub11_4       | SHub11_3       | SHub11_2       | SHub11_1       | SHub11_0       |
| SENSORHUB12_REG | 39h     | SHub12_7       | SHub12_6       | SHub12_5       | SHub12_4       | SHub12_3       | SHub12_2       | SHub12_1       | SHub12_0       |
| FIFO_STATUS1    | 3Ah     | DIFF_FIFO_7    | DIFF_FIFO_6    | DIFF_FIFO_5    | DIFF_FIFO_4    | DIFF_FIFO_3    | DIFF_FIFO_2    | DIFF_FIFO_1    | DIFF_FIFO_0    |
| FIFO_STATUS2    | 3Bh     | FTH            | FIFO_OVER_RUN  | FIFO_FULL      | FIFO_EMPTY     | DIFF_FIFO_11   | DIFF_FIFO_10   | DIFF_FIFO_9    | DIFF_FIFO_8    |
| FIFO_STATUS3    | 3Ch     | FIFO_PATTERN_7 | FIFO_PATTERN_6 | FIFO_PATTERN_5 | FIFO_PATTERN_4 | FIFO_PATTERN_3 | FIFO_PATTERN_2 | FIFO_PATTERN_1 | FIFO_PATTERN_0 |
| FIFO_STATUS4    | 3Dh     | 0              | 0              | 0              | 0              | 0              | 0              | FIFO_PATTERN_9 | FIFO_PATTERN_8 |

**Table 1. Registers (continued)**

| Register name    | Address | Bit7                  | Bit6               | Bit5               | Bit4               | Bit3               | Bit2               | Bit1               | Bit0               |
|------------------|---------|-----------------------|--------------------|--------------------|--------------------|--------------------|--------------------|--------------------|--------------------|
| FIFO_DATA_OUT_L  | 3Eh     | DATA_OUT_FIFO_L_7     | DATA_OUT_FIFO_L_6  | DATA_OUT_FIFO_L_5  | DATA_OUT_FIFO_L_4  | DATA_OUT_FIFO_L_3  | DATA_OUT_FIFO_L_2  | DATA_OUT_FIFO_L_1  | DATA_OUT_FIFO_L_0  |
| FIFO_DATA_OUT_H  | 3Fh     | DATA_OUT_FIFO_H_7     | DATA_OUT_FIFO_H_6  | DATA_OUT_FIFO_H_5  | DATA_OUT_FIFO_H_4  | DATA_OUT_FIFO_H_3  | DATA_OUT_FIFO_H_2  | DATA_OUT_FIFO_H_1  | DATA_OUT_FIFO_H_0  |
| TIMESTAMP0_REG   | 40h     | TIMESTAMP_0_7         | TIMESTAMP_0_6      | TIMESTAMP_0_5      | TIMESTAMP_0_4      | TIMESTAMP_0_3      | TIMESTAMP_0_2      | TIMESTAMP_0_1      | TIMESTAMP_0_0      |
| TIMESTAMP1_REG   | 41h     | TIMESTAMP_1_7         | TIMESTAMP_1_6      | TIMESTAMP_1_5      | TIMESTAMP_1_4      | TIMESTAMP_1_3      | TIMESTAMP_1_2      | TIMESTAMP_1_1      | TIMESTAMP_1_0      |
| TIMESTAMP2_REG   | 42h     | TIMESTAMP_2_7         | TIMESTAMP_2_6      | TIMESTAMP_2_5      | TIMESTAMP_2_4      | TIMESTAMP_2_3      | TIMESTAMP_2_2      | TIMESTAMP_2_1      | TIMESTAMP_2_0      |
| STEP_TIMESTAMP_L | 49h     | STEP_TIMESTAMP_L_7    | STEP_TIMESTAMP_L_6 | STEP_TIMESTAMP_L_5 | STEP_TIMESTAMP_L_4 | STEP_TIMESTAMP_L_3 | STEP_TIMESTAMP_L_2 | STEP_TIMESTAMP_L_1 | STEP_TIMESTAMP_L_0 |
| STEP_TIMESTAMP_H | 4Ah     | STEP_TIMESTAMP_L_7    | STEP_TIMESTAMP_L_6 | STEP_TIMESTAMP_L_5 | STEP_TIMESTAMP_L_4 | STEP_TIMESTAMP_L_3 | STEP_TIMESTAMP_L_2 | STEP_TIMESTAMP_L_1 | STEP_TIMESTAMP_L_0 |
| STEP_COUNTER_L   | 4Bh     | STEP_COUNTER_L_7      | STEP_COUNTER_L_6   | STEP_COUNTER_L_5   | STEP_COUNTER_L_4   | STEP_COUNTER_L_3   | STEP_COUNTER_L_2   | STEP_COUNTER_L_1   | STEP_COUNTER_L_0   |
| STEP_COUNTER_H   | 4Ch     | STEP_COUNTER_H_7      | STEP_COUNTER_H_6   | STEP_COUNTER_H_5   | STEP_COUNTER_H_4   | STEP_COUNTER_H_3   | STEP_COUNTER_H_2   | STEP_COUNTER_H_1   | STEP_COUNTER_H_0   |
| SENSORHUB13_REG  | 4Dh     | SHub13_7              | SHub13_6           | SHub13_5           | SHub13_4           | SHub13_3           | SHub13_2           | SHub13_1           | SHub13_0           |
| SENSORHUB14_REG  | 4Eh     | SHub14_7              | SHub14_6           | SHub14_5           | SHub14_4           | SHub14_3           | SHub14_2           | SHub14_1           | SHub14_0           |
| SENSORHUB15_REG  | 4Fh     | SHub15_7              | SHub15_6           | SHub15_5           | SHub15_4           | SHub15_3           | SHub15_2           | SHub15_1           | SHub15_0           |
| SENSORHUB16_REG  | 50h     | SHub16_7              | SHub16_6           | SHub16_5           | SHub16_4           | SHub16_3           | SHub16_2           | SHub16_1           | SHub16_0           |
| SENSORHUB17_REG  | 51h     | SHub17_7              | SHub17_6           | SHub17_5           | SHub17_4           | SHub17_3           | SHub17_2           | SHub17_1           | SHub17_0           |
| SENSORHUB18_REG  | 52h     | SHub18_7              | SHub18_6           | SHub18_5           | SHub18_4           | SHub18_3           | SHub18_2           | SHub18_1           | SHub18_0           |
| FUNC_SRC         | 53h     | STEP_COUNTER_DELTA_IA | SIGN_MOTION_IA     | TILT_IA            | STEP_DETECTED      | STEP_OVERFLOW      | -                  | SI_END_OP          | SENSORHUB_END_OP   |
| TAP_CFG          | 58h     | TIMER_EN              | PEDO_EN            | TILT_EN            | SLOPE_FDS          | TAP_X_EN           | TAP_Y_EN           | TAP_Z_EN           | LIR                |

**Table 1. Registers (continued)**

| Register name   | Address | Bit7              | Bit6            | Bit5      | Bit4     | Bit3            | Bit2       | Bit1       | Bit0       |
|-----------------|---------|-------------------|-----------------|-----------|----------|-----------------|------------|------------|------------|
| TAP_THS_6D      | 59h     | D4D_EN            | SIXD_THS1       | SIXD_THS0 | TAP_THS4 | TAP_THS3        | TAP_THS2   | TAP_THS1   | TAP_THS0   |
| INT_DUR2        | 5Ah     | DUR3              | DUR2            | DUR1      | DUR0     | QUIET1          | QUIET0     | SHOCK1     | SHOCK0     |
| WAKE_UP_THS     | 5Bh     | SINGLE_DOUBLE_TAP | INACTIVITY      | WK_THS5   | WK_THS4  | WK_THS3         | WK_THS2    | WK_THS1    | WK_THS0    |
| WAKE_UP_DUR     | 5Ch     | FF_DUR5           | WAKE_DUR1       | WAKE_DUR0 | TIMER_HR | SLEEP_DUR3      | SLEEP_DUR2 | SLEEP_DUR1 | SLEEP_DUR0 |
| FREE_FALL       | 5Dh     | FF_DUR4           | FF_DUR3         | FF_DUR2   | FF_DUR1  | FF_DUR0         | FF_THS2    | FF_THS1    | FF_THS0    |
| MD1_CFG         | 5Eh     | INT1_INACT_STATE  | INT1_SINGLE_TAP | INT1_WU   | INT1_FF  | INT1_DOUBLE_TAP | INT1_6D    | INT1_TILT  | INT1_TIMER |
| MD2_CFG         | 5Fh     | INT2_INACT_STATE  | INT2_SINGLE_TAP | INT2_WU   | INT2_FF  | INT2_DOUBLE_TAP | INT2_6D    | INT2_TILT  | INT2_IRON  |
| OUT_MAG_RAW_X_L | 66h     | D7                | D6              | D5        | D4       | D3              | D2         | D1         | D0         |
| OUT_MAG_RAW_X_H | 67h     | D15               | D14             | D13       | D12      | D11             | D10        | D9         | D8         |
| OUT_MAG_RAW_Y_L | 68h     | D7                | D6              | D5        | D4       | D3              | D2         | D1         | D0         |
| OUT_MAG_RAW_Y_H | 69h     | D15               | D14             | D13       | D12      | D11             | D10        | D9         | D8         |
| OUT_MAG_RAW_Z_L | 6Ah     | D7                | D6              | D5        | D4       | D3              | D2         | D1         | D0         |
| OUT_MAG_RAW_Z_H | 6Bh     | D15               | D14             | D13       | D12      | D11             | D10        | D9         | D8         |

## 1.1 Embedded functions registers

The list of the registers for embedded functions available in the device is given in [Table 2](#).

Embedded functions registers are accessible when the FUNC\_CFG\_EN bit is set to '1' in the FUNC\_CFG\_ACCESS register.

**Table 2. Embedded functions registers**

| Register name | Address | Bit7             | Bit6             | Bit5             | Bit4             | Bit3            | Bit2              | Bit1              | Bit0              |
|---------------|---------|------------------|------------------|------------------|------------------|-----------------|-------------------|-------------------|-------------------|
| SLV0_ADD      | 02h     | Slave0<br>_add6  | Slave0<br>_add5  | Slave0<br>_add4  | Slave0<br>_add3  | Slave0<br>_add2 | Slave0<br>_add1   | Slave0<br>_add0   | rw_0              |
| SLV0_SUBADD   | 03h     | Slave0<br>_reg7  | Slave0<br>_reg6  | Slave0<br>_reg5  | Slave0<br>_reg4  | Slave0<br>_reg3 | Slave0<br>_reg2   | Slave0<br>_reg1   | Slave0<br>_reg0   |
| SLAVE0_CONFIG | 04h     | Slave0<br>_rate1 | Slave0<br>_rate0 | Aux_sens<br>_on1 | Aux_sens<br>_on0 | Src_mode        | Slave0<br>_numop2 | Slave0<br>_numop1 | Slave0<br>_numop0 |
| SLV1_ADD      | 05h     | Slave1<br>_add6  | Slave1<br>_add5  | Slave1<br>_add4  | Slave1<br>_add3  | Slave1<br>_add2 | Slave1<br>_add1   | Slave1<br>_add0   | r_1               |
| SLV1_SUBADD   | 06h     | Slave1<br>_reg7  | Slave1<br>_reg6  | Slave1<br>_reg5  | Slave1<br>_reg4  | Slave1<br>_reg3 | Slave1<br>_reg2   | Slave1<br>_reg1   | Slave1<br>_reg0   |
| SLAVE1_CONFIG | 07h     | Slave1<br>_rate1 | Slave1<br>_rate0 | 0                | 0                | 0               | Slave1<br>_numop2 | Slave1<br>_numop1 | Slave1<br>_numop0 |
| SLV2_ADD      | 08h     | Slave2<br>_add6  | Slave2<br>_add5  | Slave2<br>_add4  | Slave2<br>_add3  | Slave2<br>_add2 | Slave2<br>_add1   | Slave2<br>_add0   | r_2               |
| SLV2_SUBADD   | 09h     | Slave2<br>_reg7  | Slave2<br>_reg6  | Slave2<br>_reg5  | Slave2<br>_reg4  | Slave2<br>_reg3 | Slave2<br>_reg2   | Slave2<br>_reg1   | Slave2<br>_reg0   |
| SLAVE2_CONFIG | 0Ah     | Slave2<br>_rate1 | Slave2<br>_rate0 | 0                | 0                | 0               | Slave2<br>_numop2 | Slave2<br>_numop1 | Slave2<br>_numop0 |
| SLV3_ADD      | 0Bh     | Slave3<br>_add6  | Slave3<br>_add5  | Slave3<br>_add4  | Slave3<br>_add3  | Slave3<br>_add2 | Slave3<br>_add1   | Slave3<br>_add0   | r_3               |
| SLV3_SUBADD   | 0Ch     | Slave3<br>_reg7  | Slave3<br>_reg6  | Slave3<br>_reg5  | Slave3<br>_reg4  | Slave3<br>_reg3 | Slave3<br>_reg2   | Slave3<br>_reg1   | Slave3<br>_reg0   |

**Table 2. Embedded functions registers (continued)**

| Register name               | Address | Bit7         | Bit6         | Bit5         | Bit4         | Bit3         | Bit2          | Bit1          | Bit0          |
|-----------------------------|---------|--------------|--------------|--------------|--------------|--------------|---------------|---------------|---------------|
| SLAVE3_CONFIG               | 0Dh     | Slave3_rate1 | Slave3_rate0 | 0            | 0            | 0            | Slave3_numop2 | Slave3_numop1 | Slave3_numop0 |
| DATAWRITE_SRC_MODE_SUB_SLV0 | 0Eh     | Slave_dataw7 | Slave_dataw6 | Slave_dataw5 | Slave_dataw4 | Slave_dataw3 | Slave_dataw2  | Slave_dataw1  | Slave_dataw0  |
| SM_THS                      | 13h     | SM_THS_7     | SM_THS_6     | SM_THS_5     | SM_THS_4     | SM_THS_3     | SM_THS_2      | SM_THS_1      | SM_THS_0      |
| STEP_COUNT_DELTA            | 15h     | SC_DELTA_7   | SC_DELTA_6   | SC_DELTA_5   | SC_DELTA_4   | SC_DELTA_3   | SC_DELTA_2    | SC_DELTA_1    | SC_DELTA_0    |
| MAG_SI_XX                   | 24h     | MAG_SI_XX_7  | MAG_SI_XX_6  | MAG_SI_XX_5  | MAG_SI_XX_4  | MAG_SI_XX_3  | MAG_SI_XX_2   | MAG_SI_XX_1   | MAG_SI_XX_0   |
| MAG_SI_XY                   | 25h     | MAG_SI_XY_7  | MAG_SI_XY_6  | MAG_SI_XY_5  | MAG_SI_XY_4  | MAG_SI_XY_3  | MAG_SI_XY_2   | MAG_SI_XY_1   | MAG_SI_XY_0   |
| MAG_SI_XZ                   | 26h     | MAG_SI_XZ_7  | MAG_SI_XZ_6  | MAG_SI_XZ_5  | MAG_SI_XZ_4  | MAG_SI_XZ_3  | MAG_SI_XZ_2   | MAG_SI_XZ_1   | MAG_SI_XZ_0   |
| MAG_SI_YX                   | 27h     | MAG_SI_YX_7  | MAG_SI_YX_6  | MAG_SI_YX_5  | MAG_SI_YX_4  | MAG_SI_YX_3  | MAG_SI_YX_2   | MAG_SI_YX_1   | MAG_SI_YX_0   |
| MAG_SI_YZ                   | 28h     | MAG_SI_YZ_7  | MAG_SI_YZ_6  | MAG_SI_YZ_5  | MAG_SI_YZ_4  | MAG_SI_YZ_3  | MAG_SI_YZ_2   | MAG_SI_YZ_1   | MAG_SI_YZ_0   |
| MAG_SI_YX                   | 29h     | MAG_SI_YX_7  | MAG_SI_YX_6  | MAG_SI_YX_5  | MAG_SI_YX_4  | MAG_SI_YX_3  | MAG_SI_YX_2   | MAG_SI_YX_1   | MAG_SI_YX_0   |
| MAG_SI_ZX                   | 2Ah     | MAG_SI_ZX_7  | MAG_SI_ZX_6  | MAG_SI_ZX_5  | MAG_SI_ZX_4  | MAG_SI_ZX_3  | MAG_SI_ZX_2   | MAG_SI_ZX_1   | MAG_SI_ZX_0   |
| MAG_SI_ZY                   | 2Bh     | MAG_SI_ZY_7  | MAG_SI_ZY_6  | MAG_SI_ZY_5  | MAG_SI_ZY_4  | MAG_SI_ZY_3  | MAG_SI_ZY_2   | MAG_SI_ZY_1   | MAG_SI_ZY_0   |
| MAG_SI_ZZ                   | 2Ch     | MAG_SI_ZZ_7  | MAG_SI_ZZ_6  | MAG_SI_ZZ_5  | MAG_SI_ZZ_4  | MAG_SI_ZZ_3  | MAG_SI_ZZ_2   | MAG_SI_ZZ_1   | MAG_SI_ZZ_0   |
| MAG_OFFX_L                  | 2Dh     | MAG_OFFX_L_7 | MAG_OFFX_L_6 | MAG_OFFX_L_5 | MAG_OFFX_L_4 | MAG_OFFX_L_3 | MAG_OFFX_L_2  | MAG_OFFX_L_1  | MAG_OFFX_L_0  |
| MAG_OFFX_H                  | 2Eh     | MAG_OFFX_H_7 | MAG_OFFX_H_6 | MAG_OFFX_H_5 | MAG_OFFX_H_4 | MAG_OFFX_H_3 | MAG_OFFX_H_2  | MAG_OFFX_H_1  | MAG_OFFX_H_0  |

**Table 2. Embedded functions registers (continued)**

| Register name | Address | Bit7         | Bit6         | Bit5         | Bit4         | Bit3         | Bit2         | Bit1         | Bit0         |
|---------------|---------|--------------|--------------|--------------|--------------|--------------|--------------|--------------|--------------|
| MAG_OFFY_L    | 2Fh     | MAG_OFFY_L_7 | MAG_OFFY_L_6 | MAG_OFFY_L_5 | MAG_OFFY_L_4 | MAG_OFFY_L_3 | MAG_OFFY_L_2 | MAG_OFFY_L_1 | MAG_OFFY_L_0 |
| MAG_OFFY_H    | 30h     | MAG_OFFY_H_7 | MAG_OFFY_H_6 | MAG_OFFY_H_5 | MAG_OFFY_H_4 | MAG_OFFY_H_3 | MAG_OFFY_H_2 | MAG_OFFY_H_1 | MAG_OFFY_H_0 |
| MAG_OFFZ_L    | 31h     | MAG_OFFZ_L_7 | MAG_OFFZ_L_6 | MAG_OFFZ_L_5 | MAG_OFFZ_L_4 | MAG_OFFZ_L_3 | MAG_OFFZ_L_2 | MAG_OFFZ_L_1 | MAG_OFFZ_L_0 |
| MAG_OFFZ_H    | 32h     | MAG_OFFZ_H_7 | MAG_OFFZ_H_6 | MAG_OFFZ_H_5 | MAG_OFFZ_H_4 | MAG_OFFZ_H_3 | MAG_OFFZ_H_2 | MAG_OFFZ_H_1 | MAG_OFFZ_H_0 |



## 2 Operating modes

The LSM6DS3 provides three possible operating configurations:

- only accelerometer active and gyroscope in Power-Down;
- only gyroscope active and accelerometer in Power-Down;
- both accelerometer and gyroscope active with independent ODR.

After the power supply is applied, the LSM6DS3 performs a 20 ms boot procedure to load the trimming parameters. After the boot is completed, both the accelerometer and the gyroscope are automatically configured in Power-Down mode.

The accelerometer and the gyroscope can be independently configured in four different power modes: Power-Down, Low-Power, Normal and High-Performance mode. They are allowed to have different data rates without any limit. The gyroscope sensor can also be set in Sleep mode to reduce its power consumption.

When both the accelerometer and gyroscope are on, the accelerometer is synchronized with the gyroscope and the data rates of the two sensors are integer multiples of each other. If the accelerometer and the gyroscope have been configured with the same output data rate, the gyroscope data-ready signal (DRDY\_G) is always subsequent to the accelerometer data-ready signal (DRDY\_XL); in this case, for synchronous reading of the two sensors, it is convenient to use the gyroscope data-ready signal.

Referring to the LSM6DS3 datasheet, the output data rate (ODR\_XL) bits of CTRL1\_XL register and the High-Performance disable (XL\_HM\_MODE) bit of CTRL6\_C register are used to select the power mode and the output data rate of the accelerometer sensor ([Table 3](#)).

**Table 3. Accelerometer ODR and power mode selection**

| ODR_XL [3:0] | ODR [Hz] when<br>XL_HM_MODE = 1 | ODR [Hz] when<br>XL_HM_MODE = 0 |
|--------------|---------------------------------|---------------------------------|
| 0000         | Power Down                      | Power Down                      |
| 0001         | 13 Hz (Low Power)               | 13 Hz (High Performance)        |
| 0010         | 26 Hz (Low Power)               | 26 Hz (High Performance)        |
| 0011         | 52 Hz (Low Power)               | 52 Hz (High Performance)        |
| 0100         | 104 Hz (Normal mode)            | 104 Hz (High Performance)       |
| 0101         | 208 Hz (Normal mode)            | 208 Hz (High Performance)       |
| 0110         | 416 Hz (High Performance)       | 416 Hz (High Performance)       |
| 0111         | 833 Hz (High Performance)       | 833 Hz (High Performance)       |
| 1000         | 1.66 kHz (High Performance)     | 1.66 kHz (High Performance)     |
| 1001         | 3.33 kHz (High Performance)     | 3.33 kHz (High Performance)     |
| 1010         | 6.66 kHz (High Performance)     | 6.66 kHz (High Performance)     |

The output data rate (ODR\_G) bits of CTRL2\_G register and the High-Performance disable (G\_HM\_MODE) bit of CTRL7\_G register are used to select the power mode and output data rate of the gyroscope sensor ([Table 4](#)).

**Table 4. Gyroscope ODR and power mode selection**

| ODR_G [3:0] | ODR [Hz] when<br>G_HM_MODE = 1 | ODR [Hz] when<br>G_HM_MODE = 0 |
|-------------|--------------------------------|--------------------------------|
| 0000        | Power Down                     | Power Down                     |
| 0001        | 13 Hz (Low Power)              | 13 Hz (High Performance)       |
| 0010        | 26 Hz (Low Power)              | 26 Hz (High Performance)       |
| 0011        | 52 Hz (Low Power)              | 52 Hz (High Performance)       |
| 0100        | 104 Hz (Normal mode)           | 104 Hz (High Performance)      |
| 0101        | 208 Hz (Normal mode)           | 208 Hz (High Performance)      |
| 0110        | 416 Hz (High Performance)      | 416 Hz (High Performance)      |
| 0111        | 833 Hz (High Performance)      | 833 Hz (High Performance)      |
| 1000        | 1.66 kHz (High Performance)    | 1.66 kHz (High Performance)    |

*Table 5* shows the typical values of LSM6DS3 power consumption for the different operating modes.

**Table 5. Power consumption**

| ODR [Hz]              | Accelerometer only<br>(at Vdd = 1.8 V) | Gyroscope only<br>(at Vdd = 1.8 V) | Combo [Acc + Gyro]<br>(at Vdd = 1.8 V) |
|-----------------------|----------------------------------------|------------------------------------|----------------------------------------|
| Power Down            | -                                      | -                                  | 6 $\mu$ A                              |
| 13 Hz (Low Power)     | 24 $\mu$ A                             | 470 $\mu$ A                        | 425 $\mu$ A                            |
| 26 Hz (Low Power)     | 31 $\mu$ A                             | 500 $\mu$ A                        | 450 $\mu$ A                            |
| 52 Hz (Low Power)     | 45 $\mu$ A                             | 540 $\mu$ A                        | 500 $\mu$ A                            |
| 104 Hz (Normal mode)  | 70 $\mu$ A                             | 625 $\mu$ A                        | 600 $\mu$ A                            |
| 208 Hz (Normal mode)  | 120 $\mu$ A                            | 880 $\mu$ A                        | 900 $\mu$ A                            |
| 13 Hz (High Perf.)    | 240 $\mu$ A                            | 1.15 mA                            | 1.25 mA                                |
| 26 Hz (High Perf.)    | 240 $\mu$ A                            | 1.15 mA                            | 1.25 mA                                |
| 52 Hz (High Perf.)    | 240 $\mu$ A                            | 1.15 mA                            | 1.25 mA                                |
| 104 Hz (High Perf.)   | 240 $\mu$ A                            | 1.15 mA                            | 1.25 mA                                |
| 208 Hz (High Perf.)   | 240 $\mu$ A                            | 1.15 mA                            | 1.25 mA                                |
| 416 Hz (High Perf.)   | 240 $\mu$ A                            | 1.15 mA                            | 1.25 mA                                |
| 833 Hz (High Perf.)   | 240 $\mu$ A                            | 1.15 mA                            | 1.25 mA                                |
| 1.66 kHz (High Perf.) | 240 $\mu$ A                            | 1.15 mA                            | 1.25 mA                                |
| 3.33 kHz (High Perf.) | 325 $\mu$ A                            | N.A.                               | N.A.                                   |
| 6.66 kHz (High Perf.) | 325 $\mu$ A                            | N.A.                               | N.A.                                   |

## 2.1 Power-Down mode

When the accelerometer/gyroscope is in Power-Down mode, almost all internal blocks of the device are switched off to minimize power consumption. Digital interfaces (I<sup>2</sup>C and SPI) are still active to allow communication with the device. The content of the configuration registers is preserved and the output data registers are not updated, keeping the last data sampled in memory before going into Power-Down mode.

## 2.2 High-Performance mode

In High-Performance mode, all accelerometer/gyroscope circuitry is always on and data are generated at the data rate selected through the ODR\_XL/ODR\_G bits.

Data interrupt generation is active.

## 2.3 Normal mode

While High-Performance mode guarantees the best performance in terms of noise, Normal mode further reduces the current consumption. The accelerometer/gyroscope data reading chain is automatically turned on and off to save power. In the gyroscope device, only the driving circuitry is always on.

Data interrupt generation is active.

## 2.4 Low-Power mode

Low-Power mode differs from Normal mode in the available output data rates. In Low-Power mode low-speed ODRs are enabled; three low-speed ODRs can be chosen through the ODR\_XL/ODR\_G bits: 13 Hz, 26 Hz and 52 Hz.

Data interrupt generation is active.

## 2.5 Gyroscope Sleep mode

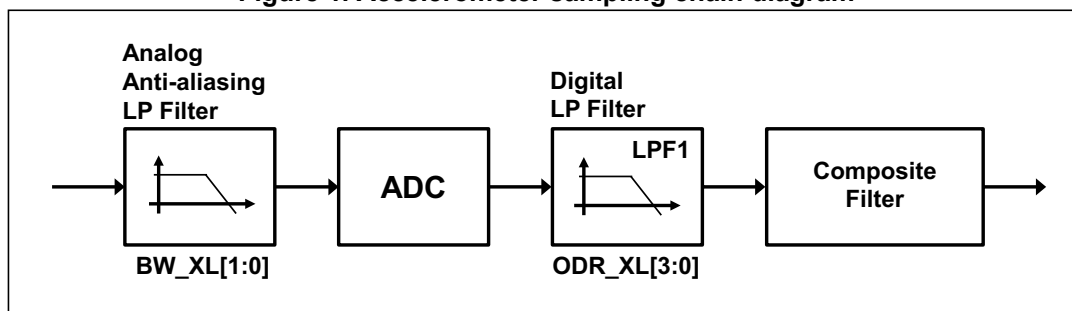
While the gyroscope is in Sleep mode the circuitry that drives the oscillation of the gyroscope mass is kept active. Compared to gyroscope Power-Down, turn-on time from Sleep mode to Low-Power/Normal/High-Performance mode is drastically reduced.

If the gyroscope is not configured in Power-Down mode, it enters in Sleep mode when the Sleep mode enable (SLEEP\_G) bit of CTRL4\_C register is set to 1, regardless of the selected gyroscope ODR.

## 2.6 Accelerometer bandwidth

The accelerometer sampling chain (*Figure 1*) is represented by a cascade of four blocks: an analog low-pass filter, an ADC converter, a digital low-pass filter and the composite group of digital filters described in *Figure 2*.

**Figure 1. Accelerometer sampling chain diagram**



The analog signal coming from the mechanical parts is filtered by a low-pass anti-aliasing filter before being converted by the ADC. The anti-aliasing filter is enabled in High-Performance mode only.

If the XL\_BW\_SCAL\_ODR bit in CTRL4\_C register is set to 1, the bandwidth of this analog filter is determined by setting the BW\_XL bits of CTRL1\_XL register; relative filter cutoff frequency values are given in *Table 6*. If the XL\_BW\_SCAL\_ODR bit is set to 0, the bandwidth of the analog filter is determined by the ODR\_XL selection (*Table 7*).

**Table 6. Accelerometer anti-aliasing filter bandwidth selection (XL\_BW\_SCAL\_ODR=1)**

| BW_XL[1:0] | Bandwidth [Hz] |
|------------|----------------|
| 00         | 400            |
| 01         | 200            |
| 10         | 100            |
| 11         | 50             |

**Table 7. Accelerometer anti-aliasing bandwidth options (High-Performance mode)**

| Accelerometer ODR [Hz]      | Analog filter cutoff [Hz]<br>XL_BW_SCAL_ODR = 0 | Analog filter cutoff [Hz]<br>XL_BW_SCAL_ODR = 1 |
|-----------------------------|-------------------------------------------------|-------------------------------------------------|
| 13 Hz (High Performance)    | 50                                              | BW_XL[1:0]                                      |
| 26 Hz (High Performance)    | 50                                              | BW_XL[1:0]                                      |
| 52 Hz (High Performance)    | 50                                              | BW_XL[1:0]                                      |
| 104 Hz (High Performance)   | 50                                              | BW_XL[1:0]                                      |
| 208 Hz (High Performance)   | 100                                             | BW_XL[1:0]                                      |
| 416 Hz (High Performance)   | 200                                             | BW_XL[1:0]                                      |
| 833 Hz (High Performance)   | 400                                             | BW_XL[1:0]                                      |
| 1.66 kHz (High Performance) | 400                                             | BW_XL[1:0]                                      |
| 3.33 kHz (High Performance) | FILTER NOT USED                                 | BW_XL[1:0]                                      |
| 6.66 kHz (High Performance) | FILTER NOT USED                                 | BW_XL[1:0]                                      |

The digital signal is then filtered by a low-pass digital filter (LPF1) whose cutoff frequency depends on the selected accelerometer ODR, as shown in [Table 8](#).

**Table 8. Accelerometer LPF1 cutoff frequency**

| Accelerometer ODR [Hz]      | LPF1 digital Filter cutoff frequency [Hz] |
|-----------------------------|-------------------------------------------|
| 13 Hz (Low Power)           | 742                                       |
| 26 Hz (Low Power)           | 742                                       |
| 52 Hz (Low Power)           | 742                                       |
| 104 Hz (Normal mode)        | 742                                       |
| 208 Hz (Normal mode)        | 742                                       |
| 13 Hz (High Performance)    | 23                                        |
| 26 Hz (High Performance)    | 46                                        |
| 52 Hz (High Performance)    | 92                                        |
| 104 Hz (High Performance)   | 184                                       |
| 208 Hz (High Performance)   | 369                                       |
| 416 Hz (High Performance)   | 742                                       |
| 833 Hz (High Performance)   | 1517                                      |
| 1.66 kHz (High Performance) | 3320                                      |
| 3.33 kHz (High Performance) | 1517                                      |
| 6.66 kHz (High Performance) | 3320                                      |

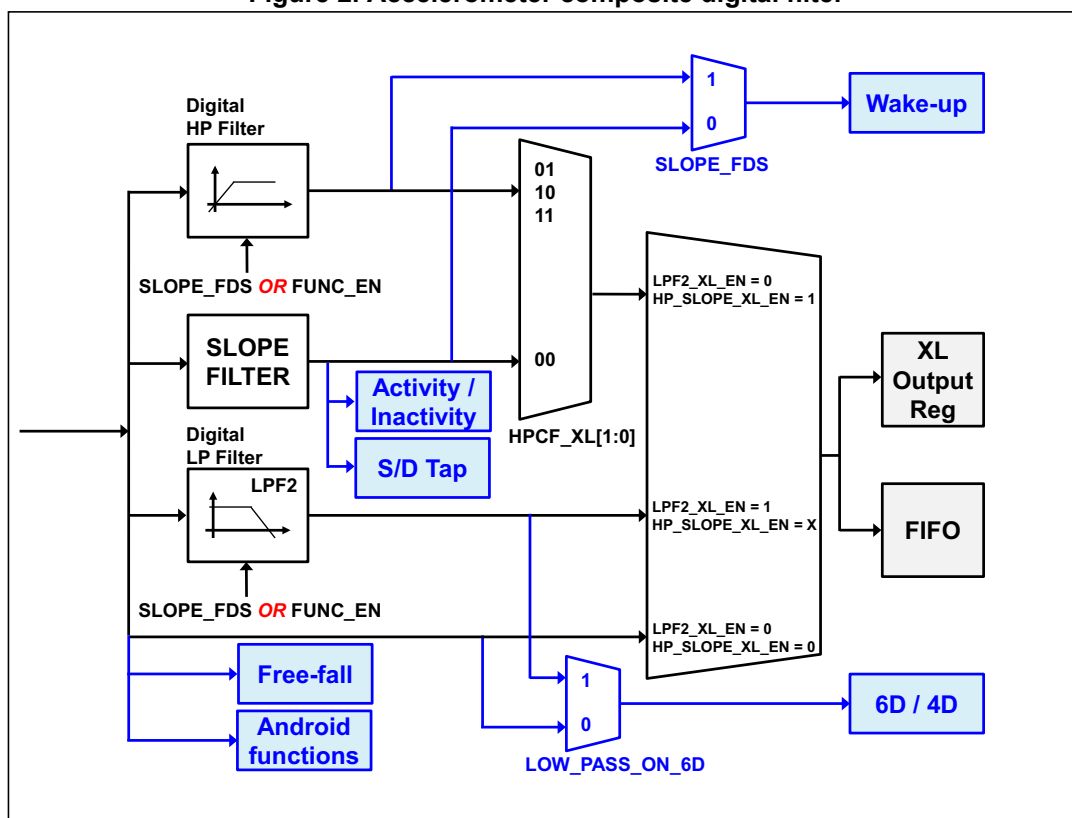
Finally, the digital signal is processed by the composite group of filters composed of a low-pass digital filter (LPF2), a high-pass digital filter and a slope filter. As shown in [Figure 2](#), it is possible to independently apply these filters to the accelerometer output data (and to the FIFO data) and/or to the interrupt generators.

The enable signal of these high-pass and low-pass digital filters is the logic “OR” of the SLOPE\_FDS bit of the TAP\_CFG register and the FUNC\_EN bit of the CTRL10\_C register.

The SLOPE\_FDS bit is also used to select the filter (high-pass or slope) used for the wake-up interrupt functionality. For this reason, if the wake-up functionality is implemented using the slope filter and also the LPF2 filter is required, the latter has to be enabled by setting the FUNC\_EN bit to 1.

In all other cases, to enable the high-pass and low-pass digital filters it's recommended to set to 1 the SLOPE\_FDS bit and set to 0 the FUNC\_EN bit (if the embedded functions, such as the Android functions and the sensor hub, are not used).

Figure 2. Accelerometer composite digital filter



The bits LPF2\_XL\_EN, HP\_SLOPE\_XL\_EN and HPCF\_XL [1:0] of CTRL8\_XL are used to select the filter applied to the accelerometer output data and to the FIFO data:

- if both the LPF2\_XL\_EN bit and the HP\_SLOPE\_XL\_EN bit are set to 0, no filter is applied;
- if the LPF2\_XL\_EN bit is set to 1, the LP digital filter is applied, regardless of the HP\_SLOPE\_XL\_EN bit configuration;
- if the LPF2\_XL\_EN bit is set to 0 and the HP\_SLOPE\_XL\_EN bit is set to 1, the applied filter depends on the configuration of the HPCF\_XL [1:0] bits, as shown in [Table 9](#).

Table 9. Accelerometer slope and high-pass filter selection and cutoff frequency

| HPCF_XL[1:0] | Applied filter | HP digital filter cutoff frequency [Hz] |
|--------------|----------------|-----------------------------------------|
| 00           | Slope          | $ODR_{XL} / 4$                          |
| 01           | High-Pass      | $ODR_{XL} / 100$                        |
| 10           | High-Pass      | $ODR_{XL} / 9$                          |
| 11           | High-Pass      | $ODR_{XL} / 400$                        |

The HPCF\_XL [1:0] bits of CTRL8\_XL are also used to select the cutoff frequency of the LPF2 filter, as shown in [Table 10](#). This low-pass filter can also be used in the 6D/4D functionality by setting the LOW\_PASS\_ON\_6D bit of CTRL8\_XL register to 1.

Table 10. Accelerometer LPF2 cutoff frequency

| HPCF_XL[1:0] | LPF2 digital filter cutoff frequency [Hz] |
|--------------|-------------------------------------------|
| 00           | ODR_XL / 50                               |
| 01           | ODR_XL / 100                              |
| 10           | ODR_XL / 9                                |
| 11           | ODR_XL / 400                              |

### 2.6.1 Accelerometer slope filter

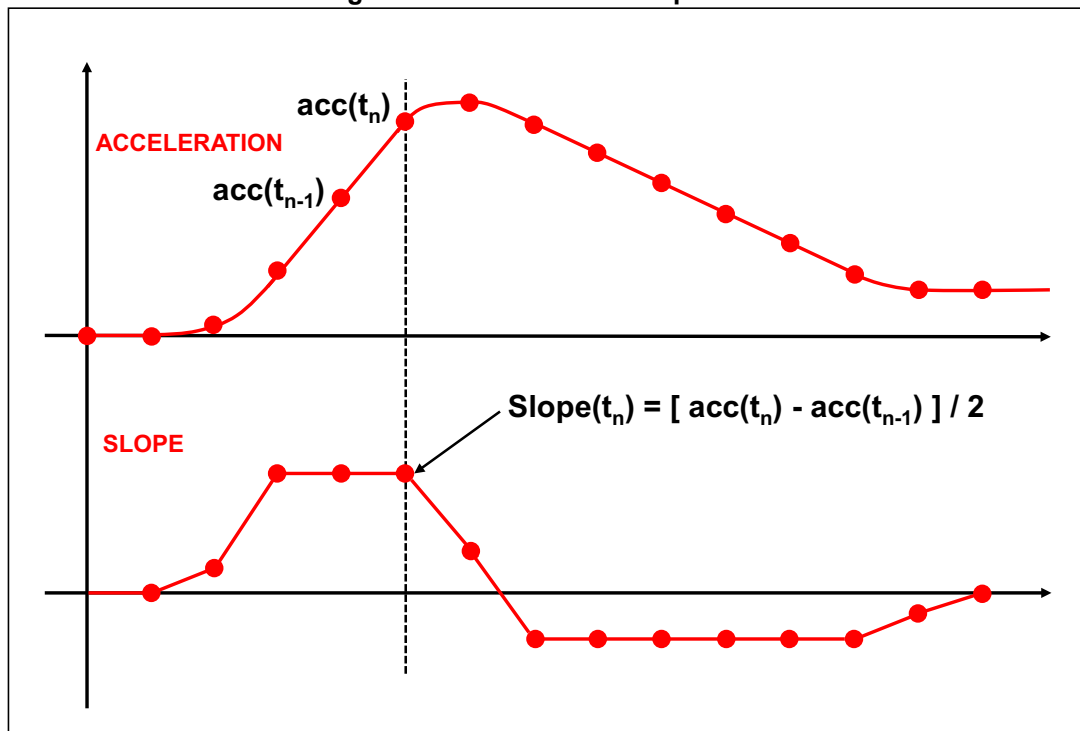
As shown in [Figure 2](#), the LSM6DS3 device embeds a digital slope filter which is used for activity/inactivity and single/double-tap features; it can also be used for wake-up detection when the SLOPE\_FDS bit of the TAP\_CFG register is set to 0.

The slope filter output data is computed using the following formula:

$$\text{slope}(t_n) = [ \text{acc}(t_n) - \text{acc}(t_{n-1}) ] / 2$$

An example of a slope data signal is illustrated in [Figure 3](#).

Figure 3. Accelerometer slope filter



## 2.6.2 Accelerometer turn-on/off time

The accelerometer reading chain contains low-pass filtering to improve signal-to-noise performance and to reduce aliasing effects. For this reason it is needed to take into account the settling time of the filter when the accelerometer / gyroscope power mode is switched or when the accelerometer / gyroscope ODR is changed.

The delay in order to switch accelerometer modes is shown in [Table 11](#).

No anti-aliasing filter is used in Low-Power mode and in Normal mode: in these cases no samples have to be discarded.

The anti-aliasing filter is bypassed also when the XL\_BW\_SCAL\_ODR bit of the CTRL4\_C register is set to 0 and the accelerometer ODR is set to 3.33 kHz or 6.66 kHz: in these cases the number of samples to be discarded is equal to 2 and 4, respectively.

[Table 12](#) clarifies how many accelerometer samples have to be discarded in High-Performance mode depending on the internal filter bandwidth and output data rate selection. Bandwidth value selection is described in [Table 7](#).

Setting the DRDY\_MASK bit of the CTRL4\_C register to 1, the accelerometer and gyroscope data-ready signals are masked until the settling of the sensor filters is completed: this feature allows automatically ignoring the samples to be discarded.

**Table 11. Accelerometer turn-on/off time**

| Starting mode                         | Target mode        | Max turn-on/off time         |
|---------------------------------------|--------------------|------------------------------|
| Power Down                            | Low Power / Normal | First sample correct         |
| Power Down                            | High Performance   | see <a href="#">Table 12</a> |
| Low Power / Normal                    | High Performance   | see <a href="#">Table 12</a> |
| Low Power / Normal / High Performance | Power Down         | 1 $\mu$ s                    |

**Table 12. Accelerometer number of samples to be discarded (High-Perf. mode)**

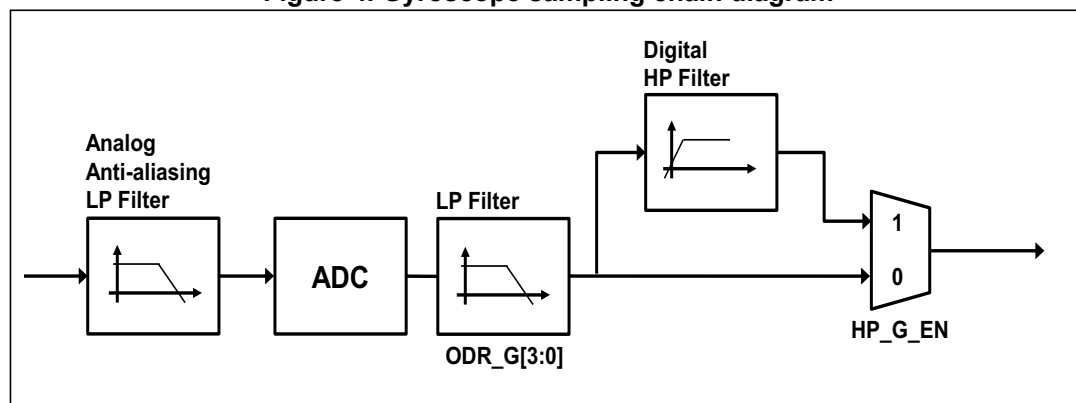
| Accelerometer ODR [Hz] | BW = 400 Hz | BW = 200 Hz | BW = 100 Hz | BW = 50 Hz | No filter |
|------------------------|-------------|-------------|-------------|------------|-----------|
| 13 Hz (High Perf.)     | 1           | 1           | 1           | 1          | 1         |
| 26 Hz (High Perf.)     | 1           | 1           | 1           | 1          | N.A.      |
| 52 Hz (High Perf.)     | 1           | 1           | 1           | 1          | N.A.      |
| 104 Hz (High Perf.)    | 1           | 1           | 1           | 2          | N.A.      |
| 208 Hz (High Perf.)    | 1           | 1           | 2           | 4          | N.A.      |
| 416 Hz (High Perf.)    | 1           | 2           | 3           | 7          | N.A.      |
| 833 Hz (High Perf.)    | 2           | 4           | 7           | 14         | N.A.      |
| 1.66 kHz (High Perf.)  | 4           | 8           | 14          | 28         | N.A.      |
| 3.33 kHz (High Perf.)  | 8           | 16          | 28          | 56         | 2         |
| 6.66 kHz (High Perf.)  | 16          | 32          | 56          | 112        | 4         |



## 2.7 Gyroscope bandwidth

The gyroscope sampling chain is represented by a cascade of four blocks: analog low-pass anti-aliasing filter, ADC converter, digital low-pass filter and a selectable high-pass filter ([Figure 4](#)).

**Figure 4. Gyroscope sampling chain diagram**



The analog signal coming from the mechanical parts is filtered by a low-pass anti-aliasing filter (having a constant bandwidth) before being converted by the ADC. The digital signal is then filtered by a low-pass digital filter whose cutoff frequency depends on the selected gyroscope ODR: the cutoff values in Low-Power and Normal mode are shown in [Table 13](#); the cutoff values related to High-Performance mode are indicated in [Table 14](#).

**Table 13. Gyroscope digital low-pass filter cutoff in Low-Power / Normal mode**

| Gyroscope ODR [Hz]   | Cutoff [Hz] |
|----------------------|-------------|
| 13 Hz (Low Power)    | 3.9         |
| 26 Hz (Low Power)    | 7.9         |
| 52 Hz (Low Power)    | 15.8        |
| 104 Hz (Normal mode) | 31.4        |
| 208 Hz (Normal mode) | 60.2        |

**Table 14. Gyroscope digital low-pass filter cutoff in High-Performance mode**

| Gyroscope ODR [Hz]    | Cutoff [Hz] |
|-----------------------|-------------|
| 13 Hz (High Perf.)    | 4.2         |
| 26 Hz (High Perf.)    | 8.3         |
| 52 Hz (High Perf.)    | 16.6        |
| 104 Hz (High Perf.)   | 33.4        |
| 208 Hz (High Perf.)   | 66.7        |
| 416 Hz (High Perf.)   | 135.9       |
| 833 Hz (High Perf.)   | 295.4       |
| 1.66 kHz (High Perf.) | 1057.0      |

The LSM6DS3 gyroscope provides embedded high-pass filtering capability to easily delete the DC component of the measured angular rate. As shown in [Figure 4](#), through the HP\_G\_EN bit of the CTRL7\_G register, it is possible to apply the filter on the gyroscope output data and on FIFO stored data.

The bandwidth of the high-pass filter depends on the settings of the HPCF\_G[1:0] bits of the CTRL7\_G register. The high-pass filter cutoff frequencies are shown in [Table 15](#).

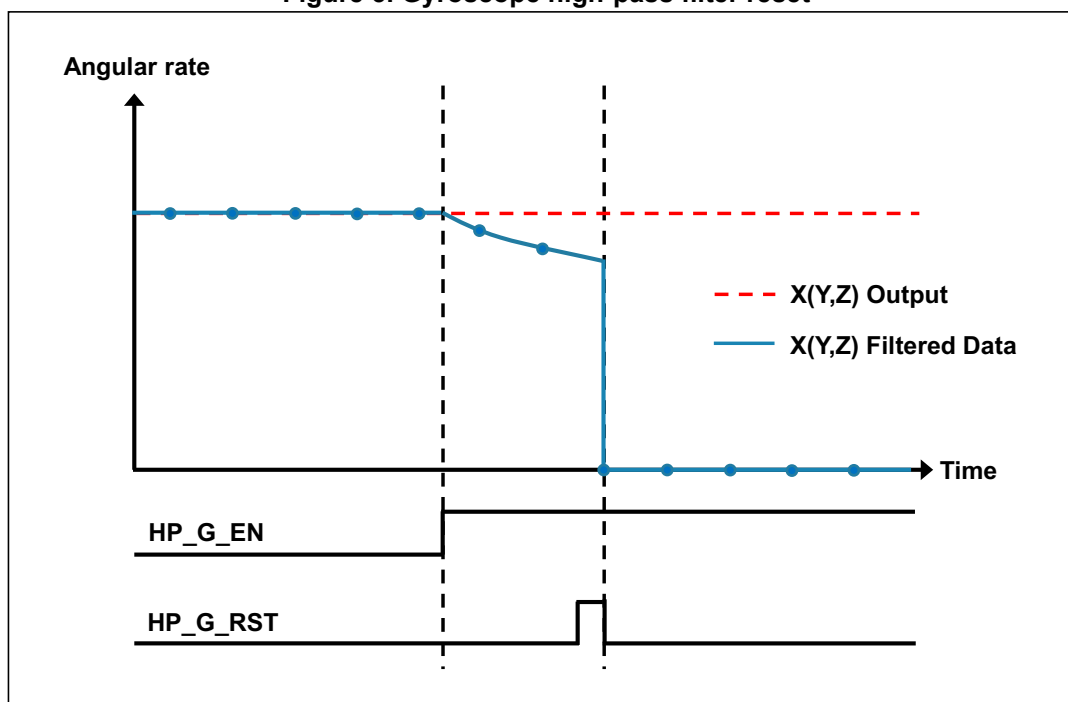
**Table 15. Gyroscope high-pass filter cutoff frequency [Hz]**

| HPCF_G1 | HPCF_G0 | High-pass filter cutoff frequency [Hz] |
|---------|---------|----------------------------------------|
| 0       | 0       | 0.0081                                 |
| 0       | 1       | 0.0324                                 |
| 1       | 0       | 2.07                                   |
| 1       | 1       | 16.32                                  |

The High-Pass filter can be reset by setting the HP\_G\_RST bit of the CTRL7\_G register to 1. The reset operation instantly deletes the DC component of the angular rate from the next generated X, Y, Z output value ([Figure 5](#)).

After the filter resets, the HP\_G\_RST bit is automatically set back to 0.

**Figure 5. Gyroscope high-pass filter reset**



### 2.7.1 Gyroscope turn-on/off time

The delay in order to switch gyroscope modes is shown in [Table 16](#).

[Table 17](#) clarifies how many gyroscope samples have to be discarded when switching from gyroscope Sleep mode to Low-Power / Normal / High-Performance mode or when the accelerometer / gyroscope ODR is changed, depending on the output data rate selection.

Setting the DRDY\_MASK bit of the CTRL4\_C register to 1, the accelerometer and gyroscope data-ready signals are masked until the settling of the sensor filters is completed: this feature allows automatically ignoring the samples to be discarded.

**Table 16. Gyroscope turn-on/off time**

| Starting mode                         | Target mode                           | Max turn-on/off time         |
|---------------------------------------|---------------------------------------|------------------------------|
| Power Down                            | Low Power / Normal / High Performance | 80 ms                        |
| Power Down                            | Gyro Sleep                            | 80 ms                        |
| Gyro Sleep                            | Low Power / Normal / High Performance | see <a href="#">Table 17</a> |
| Low Power / Normal / High Performance | Power Down                            | 1 $\mu$ s                    |

**Table 17. Gyroscope number of samples to be discarded**

| Gyroscope ODR [Hz] | Number of samples |
|--------------------|-------------------|
| 13 Hz              | 2                 |
| 26 Hz              | 2                 |
| 52 Hz              | 2                 |
| 104 Hz             | 2                 |
| 208 Hz             | 2                 |
| 416 Hz             | 2                 |
| 833 Hz             | 3                 |
| 1.66 kHz           | 4                 |

## 3 Reading output data

### 3.1 Startup sequence

Once the device is powered up, it automatically downloads the calibration coefficients from the embedded flash to the internal registers. When the boot procedure is completed, i.e. after approximately 20 milliseconds, the accelerometer and gyroscope automatically enter Power-Down mode.

To turn on the accelerometer and gather acceleration data, it is necessary to select one of the operating modes through the CTRL1\_XL register and to enable at least one of the axes through CTRL9\_XL.

The following general-purpose sequence can be used to configure the accelerometer:

- 1 Write CTRL9\_XL = 38h // Acc X, Y, Z axes enabled
- 2 Write CTRL1\_XL = 60h // Acc = 416Hz (High-Performance mode)
- 3 Write INT1\_CTRL = 01h // Acc Data Ready interrupt on INT1

To turn on the gyroscope and gather angular rate data, it is necessary to select one of the operating modes through the CTRL2\_G register and to enable at least one of the axes through CTRL10\_C.

The following general-purpose sequence can be used to configure the gyroscope:

- 1 Write CTRL10\_C = 38h // Gyro X, Y, Z axes enabled
- 2 Write CTRL2\_G = 60h // Gyro = 416Hz (High-Performance mode)
- 3 Write INT1\_CTRL = 02h // Gyro Data Ready interrupt on INT1

### 3.2 Using the status register

The device is provided with a STATUS\_REG register which should be polled to check when a new set of data is available. The XLDA bit is set to 1 when a new set of data is available at accelerometer output; the GDA bit is set to 1 when a new set of data is available at gyroscope output.

For the accelerometer (the gyroscope is similar), the reads should be performed as follows:

- 1 Read STATUS
- 2 If XLDA = 0, then go to 1
- 3 Read OUTX\_L\_XL
- 4 Read OUTX\_H\_XL
- 5 Read OUTY\_L\_XL
- 6 Read OUTY\_H\_XL
- 7 Read OUTZ\_L\_XL
- 8 Read OUTZ\_H\_XL
- 9 Data processing
- 10 Go to 1

### 3.3 Using the data-ready signal

The device can be configured to have one HW signal to determine when a new set of measurement data is available for reading.

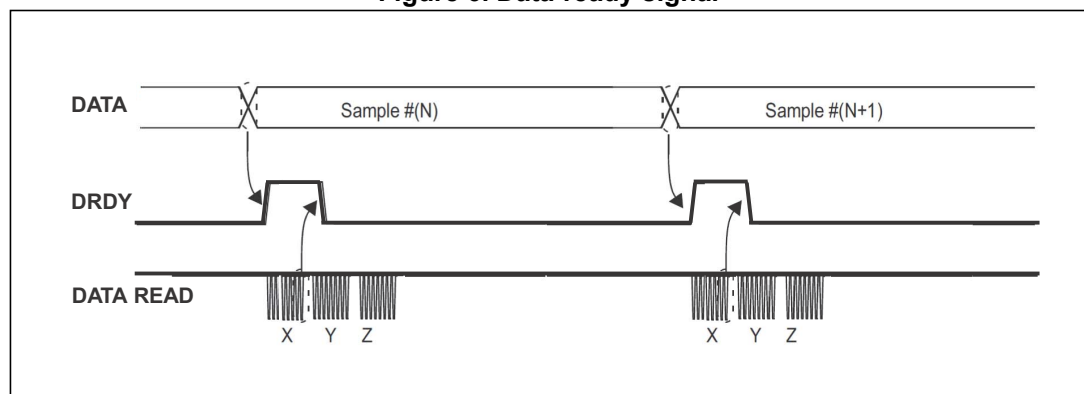
For the accelerometer sensor, the data-ready signal is represented by the XLDA bit of the STATUS\_REG register. The signal can be driven to the INT1 pin by setting to 1 the INT1\_DRDY\_XL bit of the INT1\_CTRL register and to the INT2 pin by setting to 1 the INT2\_DRDY\_XL bit of the INT2\_CTRL register.

For the gyroscope sensor, the data-ready signal is represented by the GDA bit of the STATUS\_REG register. The signal can be driven to the INT1 pin by setting to 1 the INT1\_DRDY\_G bit of the INT1\_CTRL register and to the INT2 pin by setting to 1 the INT2\_DRDY\_G bit of the INT2\_CTRL register.

The data-ready signal rises to 1 when a new set of data has been generated and it is available for reading. The interrupt is reset when the higher part of one of the enabled channels has been read (29h, 2Bh, 2Dh for the accelerometer; 23h, 25h, 27h for the gyroscope).

Setting the DRDY\_MASK bit of the CTRL4\_C register to 1, the accelerometer and gyroscope data-ready signals are masked until the settling of the sensor filters is completed.

**Figure 6. Data-ready signal**



### 3.4 Using the block data update (BDU) feature

If reading the accelerometer/gyroscope data is particularly slow and cannot be synchronized (or it is not required) with either the XLDA/GDA bits in the STATUS\_REG register or with the DRDY signal driven to the INT1/INT2 pins, it is strongly recommended to set the BDU (block data update) bit to 1 in the CTRL3\_C register.

This feature avoids reading values (most significant and least significant parts of output data) related to different samples. In particular, when the BDU is activated, the data registers related to each channel always contain the most recent output data produced by the device, but, in case the read of a given pair (i.e. OUTX\_H\_XL(G) and OUTX\_L\_XL(G), OUTY\_H\_XL(G) and OUTY\_L\_XL(G), OUTZ\_H\_XL(G) and OUTZ\_L\_XL(G)) is initiated, the refresh for that pair is blocked until both MSB and LSB parts of the data are read.

**Note:** *BDU only guarantees that the LSB part and MSB part have been sampled at the same moment. For example, if the reading speed is too slow, X and Y can be read at T1 and Z sampled at T2.*

### 3.5 Understanding output data

The measured acceleration data are sent to the OUTX\_H\_XL, OUTX\_L\_XL, OUTY\_H\_XL, OUTY\_L\_XL, OUTZ\_H\_XL, and OUTZ\_L\_XL registers. These registers contain, respectively, the most significant part and the least significant part of the acceleration signals acting on the X, Y, and Z axes.

The measured angular rate data are sent to the OUTX\_H\_G, OUTX\_L\_G, OUTY\_H\_G, OUTY\_L\_G, OUTZ\_H\_G, and OUTZ\_L\_G registers. These registers contain, respectively, the most significant part and the least significant part of the angular rate signals acting on the X, Y, and Z axes.

The complete output data for the X, Y, Z channels is given by the concatenation OUTX\_H\_XL(G) & OUTX\_L\_XL(G), OUTY\_H\_XL(G) & OUTY\_L\_XL(G), OUTZ\_H\_XL(G) & OUTZ\_L\_XL(G) and it is expressed as a two's complement number.

Both acceleration data and angular rate data are represented as 16-bit numbers.

#### 3.5.1 Big-little endian selection

The LSM6DS3 allows swapping the content of the lower and the upper part of the output data registers (i.e. OUTX\_H\_XL(G) with OUTX\_L\_XL(G), and OUT\_TEMP\_H with OUT\_TEMP\_L) in order to be compliant with both little-endian and big-endian data representations.

“Little Endian” means that the low-order byte of the number is stored in memory at the lowest address, and the high-order byte at the highest address. This mode corresponds to the BLE bit of the CTRL3\_C register set to 0 (default configuration).

On the contrary, “Big Endian” means that the high-order byte of the number is stored in memory at the lowest address, and the low-order byte at the highest address. This mode corresponds to the BLE bit of the CTRL3\_C register set to 1.

#### 3.5.2 Example of output data

*Table 18* provides a few basic examples of the accelerometer data that is read in the data registers when the device is subject to a given acceleration.

*Table 19* provides a few basic examples of the gyroscope data that is read in the data registers when the device is subject to a given angular rate.

The values listed in the following tables are given under the hypothesis of perfect device calibration (i.e. no offset, no gain error,...) and practically show the effect of the BLE bit.

Table 18. Output data registers content vs. acceleration (FS\_XL = 2 g)

| Acceleration values | BLE = 0            |                    | BLE = 1            |                    |
|---------------------|--------------------|--------------------|--------------------|--------------------|
|                     | Register address   |                    |                    |                    |
|                     | OUTX_H_XL<br>(29h) | OUTX_L_XL<br>(28h) | OUTX_H_XL<br>(29h) | OUTX_L_XL<br>(28h) |
| 0 <i>g</i>          | 00h                | 00h                | 00h                | 00h                |
| 350 <i>mg</i>       | 16h                | 69h                | 69h                | 16h                |
| 1 <i>g</i>          | 40h                | 09h                | 09h                | 40h                |
| -350 <i>mg</i>      | E9h                | 97h                | 97h                | E9h                |
| -1 <i>g</i>         | BFh                | F7h                | F7h                | BFh                |

Table 19. Output data registers content vs. angular rate (FS\_G = 245 dps)

| Angular rate values | BLE = 0           |                   | BLE = 1           |                   |
|---------------------|-------------------|-------------------|-------------------|-------------------|
|                     | Register address  |                   |                   |                   |
|                     | OUTX_H_G<br>(23h) | OUTX_L_G<br>(22h) | OUTX_H_G<br>(23h) | OUTX_L_G<br>(22h) |
| 0 dps               | 00h               | 00h               | 00h               | 00h               |
| 100 dps             | 2Ch               | A4h               | A4h               | 2Ch               |
| 200 dps             | 59h               | 49h               | 49h               | 59h               |
| -100 dps            | D3h               | 5Ch               | 5Ch               | D3h               |
| -200 dps            | A6h               | B7h               | B7h               | A6h               |

## 3.6 Rounding functions

The rounding function can be used to auto address the LSM6DS3 registers for a circular burst-mode read. Basically, with a multiple read operation the address of the register that is being read goes automatically from the first register to the last register of the pattern and then goes back to the first one.

### 3.6.1 Rounding of FIFO output registers

The rounding function is automatically enabled when performing a multiple read operation of the FIFO output registers FIFO\_DATA\_OUT\_L (3Eh) and FIFO\_DATA\_OUT\_H (3Fh).

### 3.6.2 Rounding of source registers

It's possible to apply the rounding function also to the source registers of the LSM6DS3 device, in order to verify with one multiple read whether new data was generated or a new interrupt event was detected.

The rounding function on the source registers can be enabled by setting to 1 the ROUNDING\_STATUS bit of the CTRL7\_G register: when this function is enabled, with a multiple read operation the address of the register that is being read goes automatically from STATUS\_REG (1Eh) to FUNC\_SRC (53h) and goes back to WAKE\_UP\_SRC (1Bh).

### 3.6.3 Rounding of sensor output registers

The rounding function can also be enabled for the following groups of output registers:

- Gyroscope output registers, from OUTX\_L\_G (22h) to OUTZ\_H\_G (27h);
- Accelerometer output registers, from OUTX\_L\_XL (28h) to OUTZ\_H\_XL (2Dh);
- First group of sensor hub output registers, from SENSORHUB1\_REG (2Eh) to SENSORHUB6\_REG (33h);
- Second group of sensor hub output registers, from SENSORHUB7\_REG (34h) to SENSORHUB12\_REG (39h).

The output registers rounding pattern can be configured using the bits ROUNDING[2:0] of the CTRL5\_C register, as indicated in [Table 20](#).

**Table 20. Output registers rounding pattern**

| ROUNDING[2:0] | Rounding pattern                                                                                         |
|---------------|----------------------------------------------------------------------------------------------------------|
| 000           | No rounding                                                                                              |
| 001           | Accelerometer only                                                                                       |
| 010           | Gyroscope only                                                                                           |
| 011           | Gyroscope + Accelerometer                                                                                |
| 100           | 1 <sup>st</sup> group of Sensor Hub only                                                                 |
| 101           | Accelerometer + 1 <sup>st</sup> group of Sensor Hub                                                      |
| 110           | Gyroscope + Accelerometer +<br>1 <sup>st</sup> group of Sensor Hub + 2 <sup>nd</sup> group of Sensor Hub |
| 111           | Gyroscope + Accelerometer + 1 <sup>st</sup> group of Sensor Hub                                          |



### 3.7 Gyroscope edge-sensitive/level-sensitive/impulse-sensitive data enable (DEN)

The LSM6DS3 allows external trigger level recognition by configuring the TRIG\_EN, LVLen and LVL2\_EN bits of the CTRL6\_C register ([Table 21](#)). The default value for these three bits is 0 (external trigger is disabled). Three different trigger modes can be used: edge-, level-, or impulse-sensitive trigger; the Data Enable (DEN) input signal is driven on the INT2 pin, which is configured as an input pin when one of the gyroscope trigger modes is enabled.

**Table 21. DEN configurations**

| TRIG_EN | LVLen | LVL2_EN | Function          |
|---------|-------|---------|-------------------|
| 1       | 0     | 0       | Edge-sensitive    |
| 0       | 1     | 0       | Level-sensitive   |
| 0       | 1     | 1       | Impulse-sensitive |

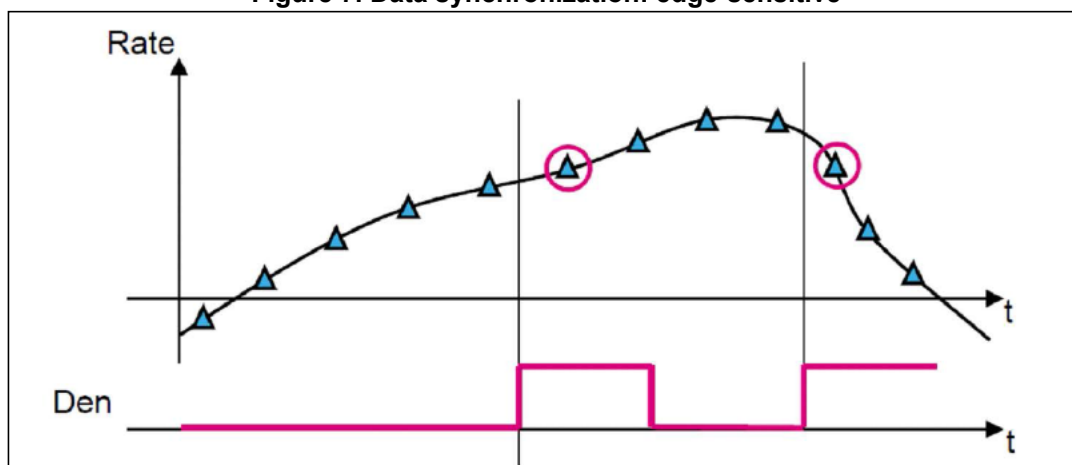
Edge-sensitive and impulse-sensitive triggers need both accelerometer and gyroscope sensors to be active (not in Power-Down mode) and configured with the same output data rate value. The level-sensitive trigger can also work with only the gyroscope in active mode, regardless of the selected gyroscope ODR.

#### 3.7.1 Edge-sensitive trigger

The edge-sensitive trigger is enabled when the TRIG\_EN bit of CTRL6\_C register is set to 1 and the LVLen and LVL2\_EN bits are set to 0. Furthermore, both accelerometer and gyroscope sensors have to be in active mode and configured with the same ODR value.

Once enabled, the gyroscope output registers are updated with the next generated X, Y, Z gyroscope data at the rising edge of the DEN (INT2 pin) input signal. If no rising edge occurs, the gyroscope output registers are not updated.

**Figure 7. Data synchronization: edge-sensitive**

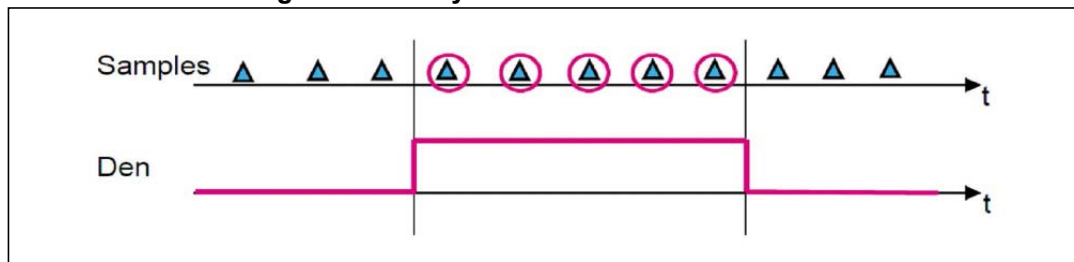


### 3.7.2 Level-sensitive trigger stamping

The level-sensitive trigger is enabled when the LVLen bit of the CTRL6\_C register is set to 1 and the TRIG\_EN and LVL2\_EN bits are set to 0.

Once enabled, the LSB of the generated gyroscope X, Y, Z output data is replaced with the current DEN (INT2 pin) level.

Figure 8. Data synchronization: level-sensitive



### 3.7.3 Impulse-sensitive trigger stamping

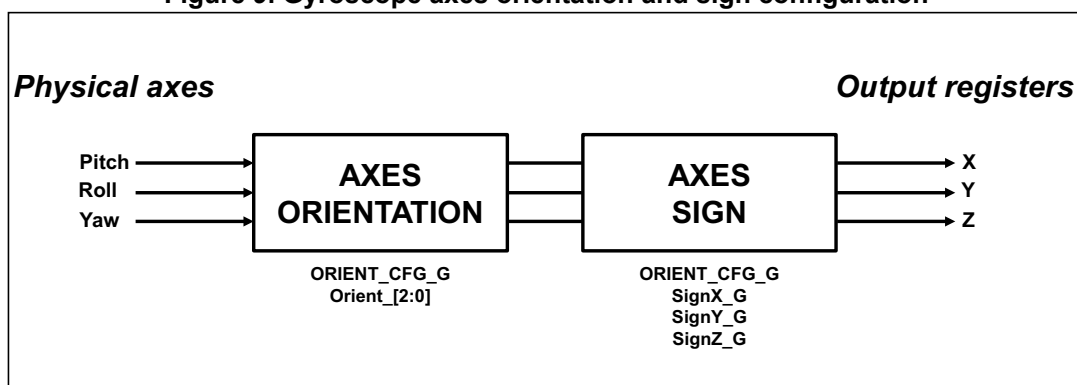
The impulse-sensitive trigger is enabled when the LVLen and LVL2\_EN bits of the CTRL6\_C register are set to 1 and the TRIG\_EN bit is set to 0. Furthermore, both accelerometer and gyroscope sensors have to be in active mode and configured with the same ODR value.

The impulse-sensitive trigger is similar to the level-sensitive trigger and has to be used if the duration of the DEN pulse is shorter than the selected gyroscope ODR. Once enabled, the LSB bit of the gyroscope X, Y, Z output data generated after the pulse is set to 1; if no pulse occurs, the LSB bit of the next generated gyroscope X, Y, Z output data is set to 0.

## 3.8 Gyroscope axes orientation

Axes orientation and sign of the gyroscope sensor can be changed by software using the ORIENT\_CFG\_G register, as illustrated in [Figure 9](#).

Figure 9. Gyroscope axes orientation and sign configuration



The Orient\_[2:0] bits of the ORIENT\_CFG\_G register allows driving the pitch, roll and yaw physical axes to the X, Y and Z output registers as indicated in [Table 23](#).

Table 22. ORIENT\_CFG\_G register

| b7 | b6 | b5      | b4      | b3      | b2       | b1       | b0       |
|----|----|---------|---------|---------|----------|----------|----------|
| 0  | 0  | SignX_G | SignY_G | SignZ_G | Orient_2 | Orient_1 | Orient_0 |

Table 23. Settings for gyroscope axes orientation

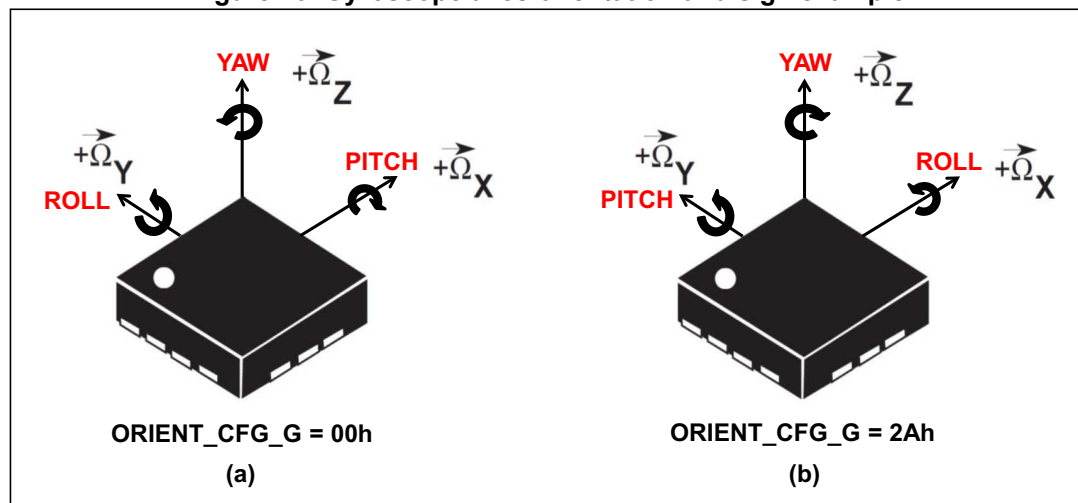
| Orient_[2:0] | Pitch | Roll | Yaw |
|--------------|-------|------|-----|
| 000          | X     | Y    | Z   |
| 001          | X     | Z    | Y   |
| 010          | Y     | X    | Z   |
| 011          | Y     | Z    | X   |
| 100          | Z     | X    | Y   |
| 101          | Z     | Y    | X   |

The SignX\_G, SignY\_G and SignZ\_G bits of the ORIENT\_CFG\_G register allow respectively changing (setting the bit to 1) the sign of the X, Y and Z output registers data.

Case (a) of example in [Figure 10](#) corresponds to the default case for axes orientation and sign, with all Orient\_[2:0], SignX\_G, SignY\_G and SignZ\_G bits of the ORIENT\_CFG\_G register set to 0.

In case (b) the Orient\_[2:0] bits have been set to 010b, driving Pitch and Roll data on different axes than the default case; furthermore the sign on X and Z data has been changed by setting the SignX\_G and SignZ\_G bits to 1.

Figure 10. Gyroscope axes orientation and sign example



## 4 Interrupt generation

In the LSM6DS3 device the interrupt generation is based on accelerometer data only, so, for interrupt generation purposes, the accelerometer sensor has to be set in an active operating mode (not in Power-Down); the gyroscope sensor can be configured in Power-Down mode since it's not involved in interrupt generation.

The interrupt generator can be configured to detect:

- Free-fall;
- Wake-up;
- 6D/4D orientation detection;
- Single-tap and double-tap sensing;
- Activity/Inactivity recognition.

In addition, the LSM6DS3 can efficiently run the sensor-related features specified in Android, saving power and enabling faster reaction time. In particular, it has been designed to implement in hardware:

- Significant motion;
- Tilt;
- Pedometer functions;
- Time stamp.

All these interrupt signals, together with FIFO interrupt signals, can be independently driven to the INT1 and INT2 interrupt pins or checked by reading the dedicated source register bits.

The H\_LACTIVE bit of the CTRL3\_C register must be used to select the polarity of the interrupt pins. If this bit is set to 0 (default value), the interrupt pins are active high and they change from low to high level when the related interrupt condition is verified. Otherwise, if the H\_LACTIVE bit is set to 1 (active low), the interrupt pins are normally at high level and they change from high to low when interrupt condition is reached.

The PP\_OD bit of CTRL3\_C allows changing the behavior of the interrupt pins from push-pull to open drain. If the PP\_OD bit is set to 0, the interrupt pins are in push-pull configuration (low-impedance output for both high and low level). When the PP\_OD bit is set to 1, only the interrupt active state is a low-impedance output.

The LIR bit of TAP\_CFG allows applying the latched mode to the interrupt signals. When the LIR bit is set to 1, once the interrupt pin is asserted, it must be reset by reading the related interrupt source register. If the LIR bit is set to 0, the interrupt signal is automatically reset when the interrupt condition is no longer verified or after a certain amount of time.

### 4.1 Interrupt pin configuration

The device is provided with two pins that can be activated to generate either Data Ready or interrupt signals. The functionality of these pins is selected through the MD1\_CFG and INT1\_CTRL registers for the INT1 pin, and through the MD2\_CFG and INT2\_CTRL registers for the INT2 pin.

Hereafter the description of these interrupt control registers; the default value of their bits is equal to 0, which corresponds to 'disable'. In order to enable the routing of a specific interrupt signal on the pin, the related bit has to be set to 1.

Table 24. INT1\_CTRL register

| b7                 | b6            | b5             | b4            | b3       | b2        | b1          | b0           |
|--------------------|---------------|----------------|---------------|----------|-----------|-------------|--------------|
| INT1_STEP_DETECTOR | INT1_SIGN_MOT | INT1_FULL_FLAG | INT1_FIFO_OVR | INT1_FTH | INT1_BOOT | INT1_DRDY_G | INT1_DRDY_XL |

- INT1\_STEP\_DETECTOR: Pedometer step recognition interrupt on INT1.
- INT1\_SIGN\_MOT: Significant motion interrupt on INT1.
- INT1\_FULL\_FLAG: FIFO full flag interrupt on INT1.
- INT1\_FIFO\_OVR: FIFO overrun flag interrupt on INT1.
- INT1\_FTH: FIFO threshold interrupt on INT1.
- INT1\_BOOT: Boot interrupt on INT1.
- INT1\_DRDY\_G: Gyroscope Data-Ready on INT1.
- INT1\_DRDY\_XL: Accelerometer Data-Ready on INT1.

Table 25. MD1\_CFG register

| b7               | b6              | b5      | b4      | b3              | b2      | b1        | b0         |
|------------------|-----------------|---------|---------|-----------------|---------|-----------|------------|
| INT1_INACT_STATE | INT1_SINGLE_TAP | INT1_WU | INT1_FF | INT1_DOUBLE_TAP | INT1_6D | INT1_TILT | INT1_TIMER |

- INT1\_INACT\_STATE: Inactivity interrupt on INT1.
- INT1\_SINGLE\_TAP: Single-tap interrupt on INT1.
- INT1\_WU: Wake-up interrupt on INT1.
- INT1\_FF: Free-fall interrupt on INT1.
- INT1\_DOUBLE\_TAP: Double-tap interrupt on INT1.
- INT1\_6D: 6D detection interrupt on INT1.
- INT1\_TILT: Tilt interrupt on INT1.
- INT1\_TIMER: Timer interrupt on INT1.

Table 26. INT2\_CTRL register

| b7              | b6                 | b5             | b4            | b3       | b2             | b1          | b0           |
|-----------------|--------------------|----------------|---------------|----------|----------------|-------------|--------------|
| INT2_STEP_DELTA | INT2_STEP_COUNT_OV | INT2_FULL_FLAG | INT2_FIFO_OVR | INT2_FTH | INT2_DRDY_TEMP | INT2_DRDY_G | INT2_DRDY_XL |

- INT2\_STEP\_DELTA: Pedometer step recognition on delta time interrupt on INT2.
- INT2\_STEP\_COUNT\_OV: Step counter overflow interrupt on INT2.
- INT2\_FULL\_FLAG: FIFO full flag interrupt on INT2.
- INT2\_FIFO\_OVR: FIFO overrun flag interrupt on INT2.
- INT2\_FTH: FIFO threshold interrupt on INT2.
- INT2\_DRDY\_TEMP: Temperature Data-Ready on INT2.
- INT2\_DRDY\_G: Gyroscope Data-Ready on INT2.
- INT2\_DRDY\_XL: Accelerometer Data-Ready on INT2.

Table 27. MD2\_CFG register

| b7               | b6              | b5      | b4      | b3              | b2      | b1        | b0        |
|------------------|-----------------|---------|---------|-----------------|---------|-----------|-----------|
| INT2_INACT_STATE | INT2_SINGLE_TAP | INT2_WU | INT2_FF | INT2_DOUBLE_TAP | INT2_6D | INT2_TILT | INT2_IRON |

- INT2\_INACT\_STATE: Inactivity interrupt on INT2.
- INT2\_SINGLE\_TAP: Single-tap interrupt on INT2.
- INT2\_WU: Wake-up interrupt on INT2.
- INT2\_FF: Free-fall interrupt on INT2.
- INT2\_DOUBLE\_TAP: Double-tap interrupt on INT2.
- INT2\_6D: 6D detection interrupt on INT2.
- INT2\_TILT: Tilt interrupt on INT2.
- INT2\_IRON: Soft-iron / hard-iron interrupt on INT2.

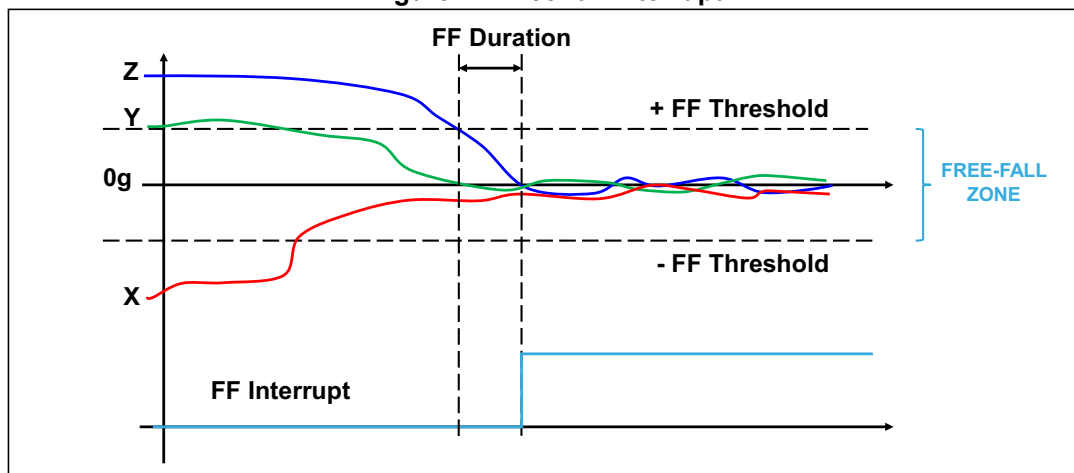
If multiple interrupt signals are routed on the same pin (INTx), the logic level of this pin is the “OR” combination of the selected interrupt signals. In order to know which event has generated the interrupt condition, the related source registers have to be read: WAKE\_UP\_SRC, D6D\_SRC, TAP\_SRC and FUNC\_SRC.

The INT2\_on\_INT1 pin of CTRL4\_C register allows driving all the enabled interrupt signals in logic “OR” on the INT1 pin (by setting this bit to 1). When this bit is set to 0, the interrupt signals are divided between the INT1 and INT2 pins.

## 4.2 Free-fall interrupt

Free-fall detection refers to a specific register configuration that allows recognizing when the device is in free-fall: the acceleration measured along all the axes goes to zero. In a real case a “free-fall zone” is defined around the zero-g level where all the accelerations are small enough to generate the interrupt. Configurable threshold and duration parameters are associated to free-fall event detection: the threshold parameter defines the free-fall zone amplitude; the duration parameter defines the minimum duration of the free-fall interrupt event to be recognized (*Figure 11*).

Figure 11. Free-fall interrupt



The free-fall interrupt signal can be driven to the two interrupt pins by setting to 1 the INT1\_FF bit of the MD1\_CFG register or the INT2\_FF bit of the MD2\_CFG register; it can also be checked by reading the FF\_IA bit of the WAKE\_UP\_SRC register.

If latch mode is disabled (LIR bit of TAP\_CFG is set to 0), the interrupt signal is automatically reset when the free-fall condition is no longer verified. If latch mode is enabled and the free-fall interrupt signal is driven to the interrupt pins, once a free-fall event has occurred and the interrupt pin is asserted, it must be reset by reading the WAKE\_UP\_SRC register. If the latch mode is enabled but the interrupt signal is not driven to the interrupt pins, the latch feature does not take effect.

The register used to configure the threshold parameter is named FREE\_FALL; the unsigned threshold value is related to the value of the FF\_THS[2:0] field value as indicated in [Table 28](#). The values given in this table are valid for each accelerometer full-scale value.

**Table 28. Free-fall threshold LSB value**

| FREE_FALL - FF_THS[2:0] | Threshold LSB value [mg] |
|-------------------------|--------------------------|
| 000                     | 156                      |
| 001                     | 219                      |
| 010                     | 250                      |
| 011                     | 312                      |
| 100                     | 344                      |
| 101                     | 406                      |
| 110                     | 469                      |
| 111                     | 500                      |

Duration time is measured in N/ODR\_XL, where N is the content of the FF\_DUR[5:0] field of the FREE\_FALL / WAKE\_UP\_DUR registers and ODR\_XL is the accelerometer data rate.

A basic SW routine for the free-fall event recognition is given below.

```

1   Write 60h into CTRL1_XL           // Turn on the accelerometer
                                       // ODR_XL = 416 Hz, FS_XL = 2g
2   Write 00h into WAKE_UP_DUR        // Set event duration (FF_DUR5 bit)
3   Write 33h into FREE_FALL          // Set FF threshold (FF_THS[2:0] = 011b)
                                       // Set six samples event duration (FF_DUR[5:0] = 000110b)
4   Write 10h into MD1_CFG            // FF interrupt driven to INT1 pin
5   Write 01h into TAP_CFG            // Latch interrupt

```

The sample code exploits a threshold set to 312 mg for free-fall recognition and the event is notified by hardware through the INT1 pin. The FF\_DUR[5:0] field of FREE\_FALL / WAKE\_UP\_DUR registers is configured like this to ignore events that are shorter than  $6/ODR\_XL = 6/412 \text{ Hz} \approx 15 \text{ msec}$  in order to avoid false detections.

### 4.3 Wake-up interrupt

In the LSM6DS3 device the wake-up feature can be implemented using either the slope filter (see [Section 2.6.1](#) for more details) or the high-pass digital filter, as illustrated in [Figure 2](#). The filter to be applied can be selected using the SLOPE\_FDS bit of the TAP\_CFG register: if this bit is set to 0 (default value), the slope filter is used; if it's set to 1, the high-pass digital filter is used.

The wake-up interrupt signal is generated if a certain number of consecutive filtered data exceed the configured threshold ([Figure 12](#)).

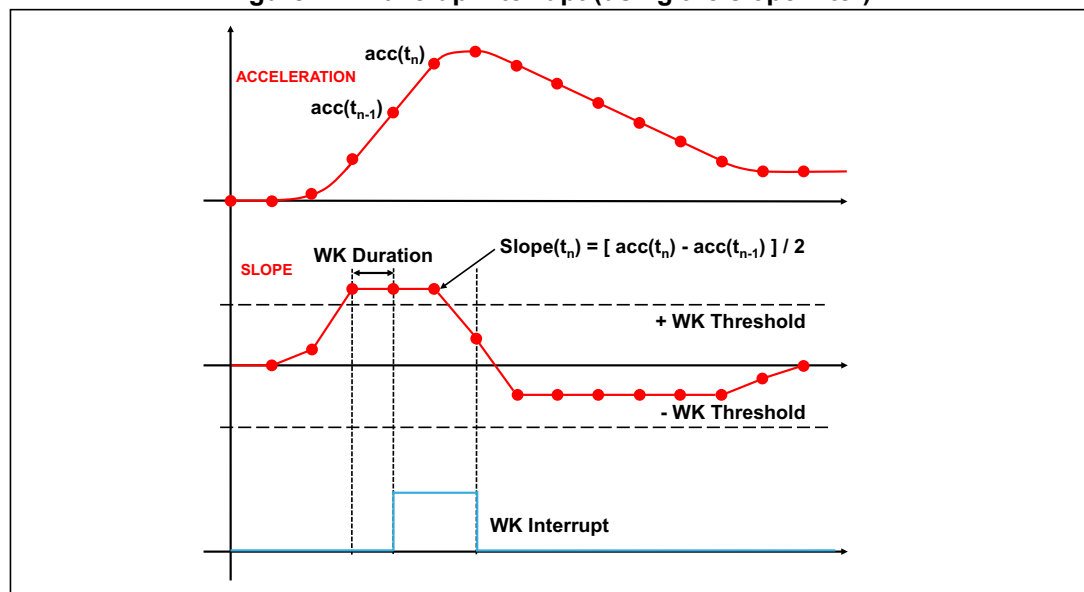
The unsigned threshold value is defined using the WK\_THS[5:0] bits of the WAKE\_UP\_THS register; the value of 1 LSB of these 6 bits depends on the selected accelerometer full scale: 1 LSB =  $(FS\_XL)/(2^6)$ . The threshold is applied to both positive and negative data: for a wake-up interrupt generation, the module of the filtered data must be bigger than the threshold.

The duration parameter defines the minimum duration of the wake-up event to be recognized; its value is set using the WAKE\_DUR[1:0] bits of the WAKE\_UP\_DUR register: 1 LSB corresponds to  $1 \cdot ODR\_XL$  time, where  $ODR\_XL$  is the accelerometer output data rate. It is important to appropriately define the duration parameter to avoid unwanted wake-up interrupts due to spurious spikes of the input signal.

This interrupt signal can be driven to the two interrupt pins setting to 1 the INT1\_WU bit of the MD1\_CFG register or the INT2\_WU bit of the MD2\_CFG register; it can also be checked by reading the WU\_IA bit of the WAKE\_UP\_SRC register. The X\_WU, Y\_WU, Z\_WU bits of the WAKE\_UP\_SRC register indicate which axis has triggered the wake-up event.

If latch mode is disabled (LIR bit of TAP\_CFG is set to 0), the interrupt signal is automatically reset when the filtered data falls below the threshold. If latch mode is enabled and the wake-up interrupt signal is driven to the interrupt pins, once a wake-up event has occurred and the interrupt pin is asserted, it must be reset by reading the WAKE\_UP\_SRC register. If the latch mode is enabled but the interrupt signal is not driven to the interrupt pins, the latch feature does not take effect.

**Figure 12. Wake-up interrupt (using the slope filter)**





The example code which implements the SW routine for the wake-up event recognition using the slope filter is given below.

```

1   Write 60h into CTRL1_XL           // Turn on the accelerometer
                                     // ODR_XL = 416 Hz, FS_XL = 2g
2   Write 00h into TAP_CFG            // Apply slope filter; latch mode disabled
3   Write 00h into WAKE_UP_DUR        // No duration
4   Write 02h into WAKE_UP_THS        // Set wake-up threshold
5   Write 20h into MD1_CFG            // Wake-up interrupt driven to INT1 pin

```

Since the duration time is set to zero, the wake-up interrupt signal is generated for each X,Y,Z slope data exceeding the configured threshold. The WK\_THS field of the WAKE\_UP\_THS register is set to 000010b, therefore the wake-up threshold is 62.5 mg ( $= 2 * FS\_XL / 2^6$ ).

## 4.4 6D/4D orientation detection

The LSM6DS3 device provides the capability to detect the orientation of the device in space, enabling easy implementation of energy-saving procedures and automatic image rotation for mobile devices.

### 4.4.1 6D orientation detection

Six orientations of the device in space can be detected; the interrupt signal is asserted when the device switches from one orientation to another. The interrupt is not re-asserted as long as the position is maintained.

6D interrupt is generated when, for two consecutive samples, only one axis exceeds a selected threshold and the acceleration values measured from the other two axes are lower than the threshold: the ZH, ZL, YH, YL, XH, XL bits of the D6D\_SRC (1Dh) register indicate which axis has triggered the 6D event.

In more detail:

**Table 29. D6D\_SRC register**

| b7 | b6     | b5 | b4 | b3 | b2 | b1 | b0 |
|----|--------|----|----|----|----|----|----|
| 0  | D6D_IA | ZH | ZL | YH | YL | XH | XL |

- D6D\_IA is set high when the device switches from one orientation to another.
- ZH (YH, XH) is set high when the face perpendicular to the Z(Y,X) axis is almost flat and the acceleration measured on the Z(Y,X) axis is positive and in the module bigger than the threshold.
- ZL (YL, XL) is set high when the face perpendicular to the Z(Y,X) axis is almost flat and the acceleration measured on the Z(Y,X) axis is negative and in the module bigger than the threshold.

The SIXD\_THS[1:0] bits of the TAP\_THS\_6D register are used to select the threshold value used to detect the change in device orientation. The threshold values given in [Table 30](#) are valid for each accelerometer full-scale value.

Table 30. Threshold for 4D/6D function

| SIXD_THS[1:0] | Threshold value [degrees] |
|---------------|---------------------------|
| 00            | 80                        |
| 01            | 70                        |
| 10            | 60                        |
| 11            | 50                        |

The low-pass filter LPF2 can also be used in 6D functionality by setting the LOW\_PASS\_ON\_6D bit of the CTRL8\_XL register to 1. The LPF2 filter has to be enabled as described in [Section 2.6](#).

This interrupt signal can be driven to the two interrupt pins by setting to 1 the INT1\_6D bit of the MD1\_CFG register or the INT2\_6D bit of the MD2\_CFG register; it can also be checked by reading the D6D\_IA bit of the D6D\_SRC register.

If latch mode is disabled (LIR bit of TAP\_CFG is set to 0), the interrupt signal is active only for  $1/\text{ODR\_XL[s]}$  then it is automatically deasserted (ODR\_XL is the accelerometer output data rate). If latch mode is enabled and the 6D interrupt signal is driven to the interrupt pins, once an orientation change has occurred and the interrupt pin is asserted, a reading of the D6D\_SRC register clears the request and the device is ready to recognize a different orientation. If the latch mode is enabled but the interrupt signal is not driven to the interrupt pins, the latch feature does not take effect.

Referring to the six possible cases illustrated in [Figure 13](#), the content of the D6D\_SRC register for each position is shown in [Table 31](#).

Figure 13. 6D recognized orientations

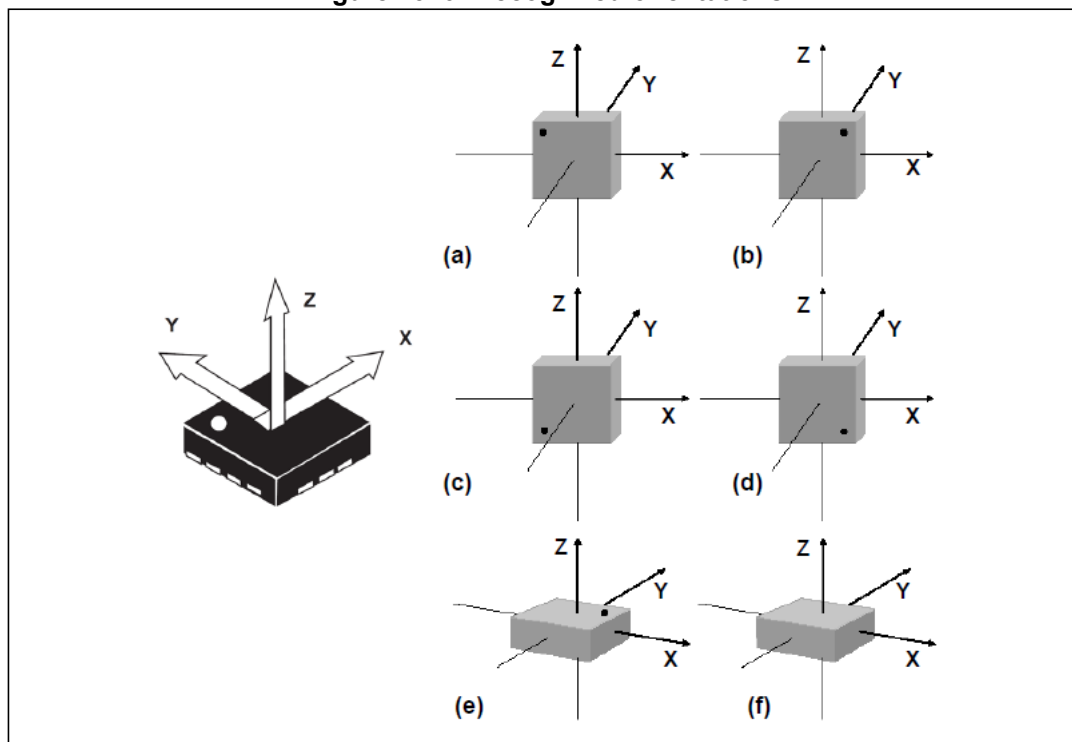


Table 31. D6D\_SRC register in 6D positions

| Case | D6D_IA | ZH | ZH | YH | YL | XH | XL |
|------|--------|----|----|----|----|----|----|
| (a)  | 1      | 0  | 0  | 1  | 0  | 0  | 0  |
| (b)  | 1      | 0  | 0  | 0  | 0  | 0  | 1  |
| (c)  | 1      | 0  | 0  | 0  | 0  | 1  | 0  |
| (d)  | 1      | 0  | 0  | 0  | 1  | 0  | 0  |
| (e)  | 1      | 1  | 0  | 0  | 0  | 0  | 0  |
| (f)  | 1      | 0  | 1  | 0  | 0  | 0  | 0  |

Hereafter an example which implements the SW routine for 6D orientation detection.

- 1 Write 60h into CTRL1\_XL // Turn on the accelerometer  
// ODR\_XL = 416 Hz, FS\_XL = 2g
- 2 Write 40h into TAP\_THS\_6D // Set 6D threshold (SIXD\_THS[1:0] = 10b = 60 degrees)
- 4 Write 10h into TAP\_CFG // Enable LPF2 filter
- 3 Write 01h into CTRL8\_XL // Apply LPF2 filter to 6D functionality
- 5 Write 04h into MD1\_CFG // 6D interrupt driven to INT1 pin

#### 4.4.2 4D orientation detection

The 4D direction function is a subset of the 6D function especially defined to be implemented in mobile devices for portrait and landscape computation. It can be enabled by setting the D4D\_EN bit of the TAP\_THS\_6D register to 1. In this configuration, the Z-axis position detection is disabled, therefore reducing position recognition to cases (a), (b), (c), and (d) of [Table 31](#).

### 4.5 Single-tap and double-tap recognition

The single-tap and double-tap recognition functions featured in the LSM6DS3 help to create a man-machine interface with little software loading. The device can be configured to output an interrupt signal on a dedicated pin when tapped in any direction.

If the sensor is exposed to a single input stimulus, it generates an interrupt request on the inertial interrupt pin INT1 and/or INT2. A more advanced feature allows the generation of an interrupt request when a double input stimulus with programmable time between the two events is recognized, enabling a mouse button-like function.

In the LSM6DS3 device the single-tap and double-tap recognition functions use the slope between two consecutive acceleration samples to detect the tap events; the slope data is computed using the following formula:

$$\text{slope}(t_n) = [ \text{acc}(t_n) - \text{acc}(t_{n-1}) ] / 2$$

This function can be fully programmed by the user in terms of expected amplitude and timing of the slope data by means of a dedicated set of registers.

Single and double-tap recognition work independently of the selected output data rate. Recommended accelerometer ODRs for these functions are 416 Hz and 833 Hz.

### 4.5.1 Single tap

If the device is configured for single-tap event detection, an interrupt is generated when the slope data on the selected channel exceed the programmed threshold, and return below it within the Shock time window.

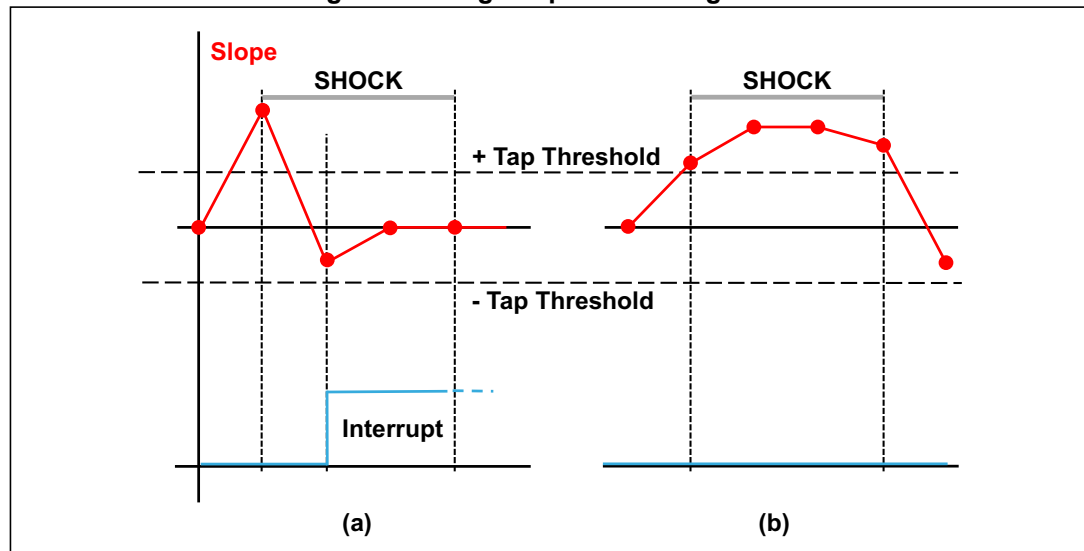
In the single-tap case, if the LIR bit of the TAP\_CFG register is set to 0, the interrupt is kept high for the duration of the Quiet window.

In order to enable the latch feature on the single-tap interrupt signal, both the LIR bit and the INT1\_DOUBLE\_TAP (or INT2\_DOUBLE\_TAP) bit of MD1\_CFG (MD2\_CFG) have to be set to 1: the interrupt is kept high until the TAP\_SRC register is read.

The SINGLE\_DOUBLE\_TAP bit of WAKE\_UP\_THS has to be set to 0 in order to enable single-tap recognition only.

In case (a) of [Figure 14](#) the single-tap event has been recognized, while in case (b) the tap has not been recognized because the slope data fall under the threshold after the Shock time window has expired.

**Figure 14. Single-tap event recognition**



### 4.5.2 Double tap

If the device is configured for double-tap event detection, an interrupt is generated when, after a first tap, a second tap is recognized. The recognition of the second tap occurs only if the event satisfies the rules defined by the Shock, the Latency and the Duration time windows.

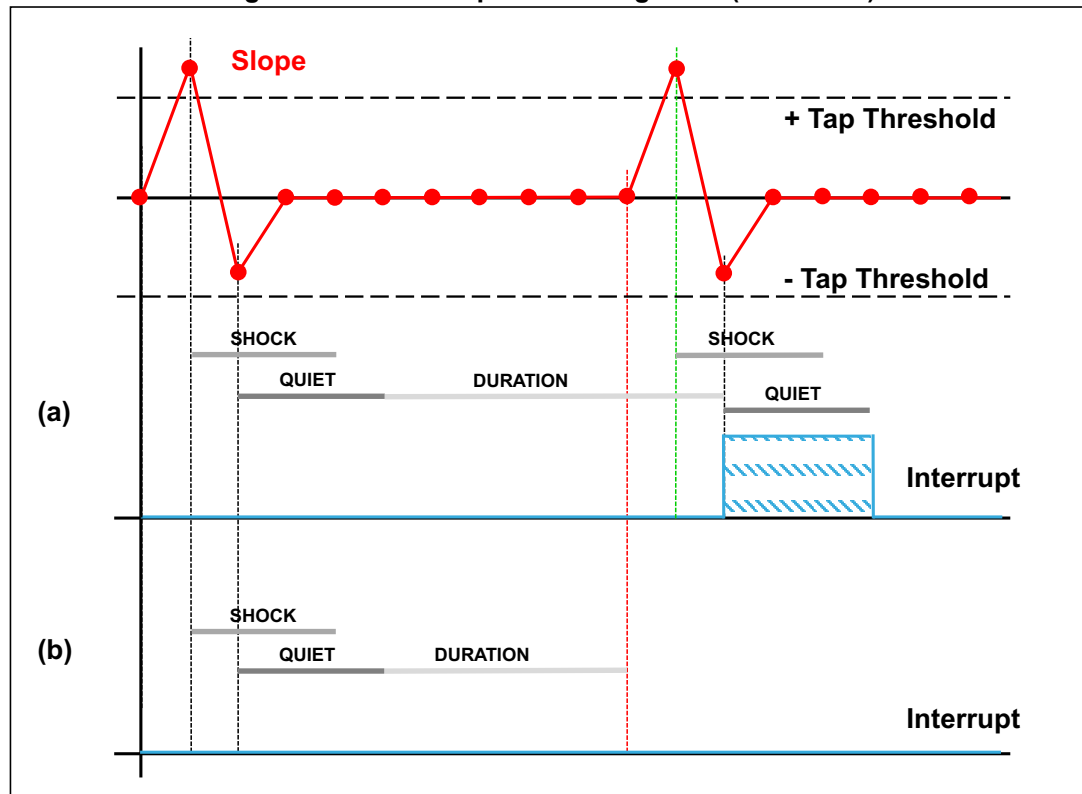
In particular, after the first tap has been recognized, the second tap detection procedure is delayed for an interval defined by the Quiet time. This means that after the first tap has been recognized, the second tap detection procedure starts only if the slope data exceed the threshold after the Quiet window but before the Duration window has expired. In case (a) of [Figure 15](#), a double-tap event has been correctly recognized, while in case (b) the interrupt has not been generated because the slope data exceed the threshold after the window interval has expired.

Once the second tap detection procedure is initiated, the second tap is recognized with the same rule as the first: the slope data must return below the threshold before the Shock window has expired.

It is important to appropriately define the Quiet window to avoid unwanted taps due to spurious bouncing of the input signal.

In the double-tap case, if the LIR bit of the TAP\_CFG register is set to 0, the interrupt is kept high for the duration of the Quiet window. If the LIR bit is set to 1, the interrupt is kept high until the TAP\_SRC register is read.

**Figure 15. Double-tap event recognition (LIR bit = 0)**



### 4.5.3 Single-tap and double-tap recognition configuration

The LSM6DS3 device can be configured to output an interrupt signal when tapped (once or twice) in any direction: the TAP\_X\_EN, TAP\_Y\_EN and TAP\_Z\_EN bits of the TAP\_CFG register must be set to 1 to enable the tap recognition on X, Y, Z directions, respectively.

Configurable parameters for tap recognition functionality are the tap threshold and the Shock, Quiet and Duration time windows.

The TAP\_THS[4:0] bits of the TAP\_THS\_6D register are used to select the unsigned threshold value used to detect the tap event. The value of 1 LSB of these 5 bits depends on the selected accelerometer full scale:  $1 \text{ LSB} = (\text{FS\_XL})/(2^5)$ . The unsigned threshold is applied to both positive and negative slope data.

The Shock time window defines the maximum duration of the overthreshold event: the acceleration must return below the threshold before the Shock window has expired, otherwise the tap event is not detected. The SHOCK[1:0] bits of the INT\_DUR2 register are

used to set the Shock time window value: the default value of these bits is 00b and corresponds to  $4 \cdot \text{ODR\_XL}$  time, where  $\text{ODR\_XL}$  is the accelerometer output data rate. If the  $\text{SHOCK}[1:0]$  bits are set to a different value, 1 LSB corresponds to  $8 \cdot \text{ODR\_XL}$  time.

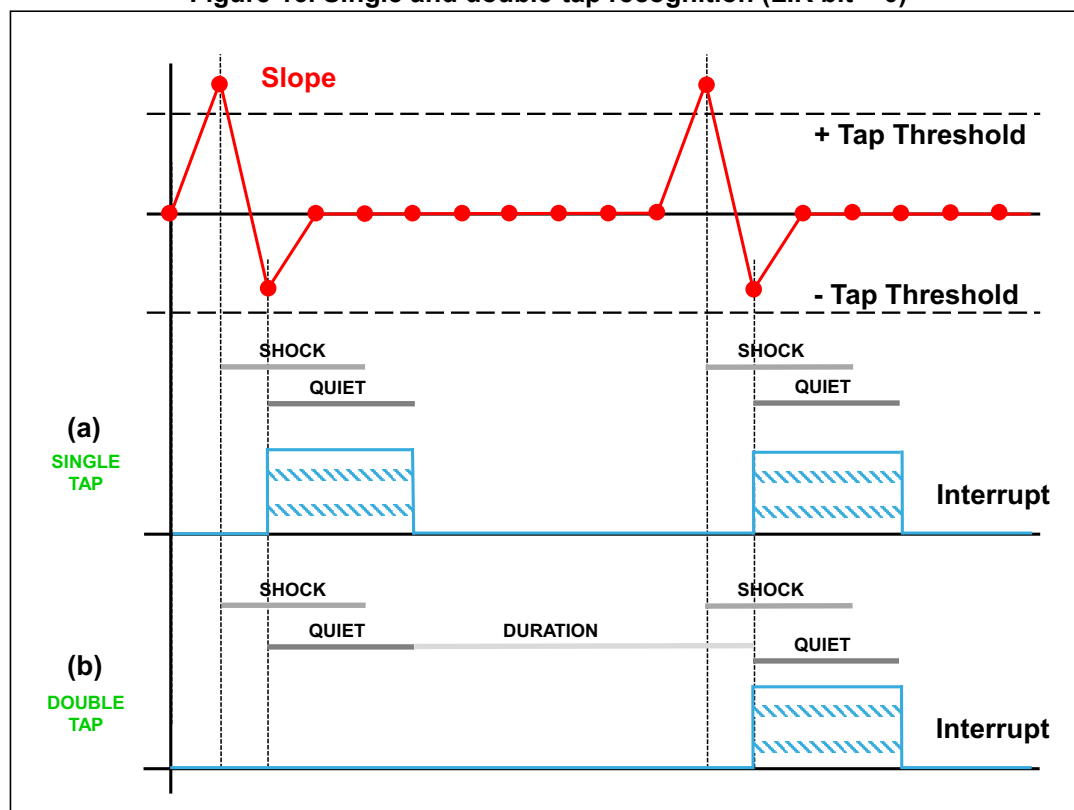
In the double-tap case, the Quiet time window defines the time after the first tap recognition in which there must not be any overthreshold. When the latch mode is disabled (LIR bit of  $\text{TAP\_CFG}$  is set to 0), the Quiet time also defines the length of the interrupt pulse (in both single and double-tap case). The  $\text{QUIET}[1:0]$  bits of the  $\text{INT\_DUR2}$  register are used to set the Quiet time window value: the default value of these bits is 00b and corresponds to  $2 \cdot \text{ODR\_XL}$  time, where  $\text{ODR\_XL}$  is the accelerometer output data rate. If the  $\text{QUIET}[1:0]$  bits are set to a different value, 1 LSB corresponds to  $4 \cdot \text{ODR\_XL}$  time.

In the double-tap case, the Duration time window defines the maximum time between two consecutive detected taps. The Duration time period starts just after the completion of the Quiet time of the first tap. The  $\text{DUR}[3:0]$  bits of the  $\text{INT\_DUR2}$  register are used to set the Duration time window value: the default value of these bits is 0000b and corresponds to  $16 \cdot \text{ODR\_XL}$  time, where  $\text{ODR\_XL}$  is the accelerometer output data rate. If the  $\text{DUR}[3:0]$  bits are set to a different value, 1 LSB corresponds to  $32 \cdot \text{ODR\_XL}$  time.

Figure 16 illustrates a single-tap event (a) and a double-tap event (b). These interrupt signals can be driven to the two interrupt pins by setting to 1 the  $\text{INT1\_SINGLE\_TAP}$  bit of the  $\text{MD1\_CFG}$  register or the  $\text{INT2\_SINGLE\_TAP}$  bit of the  $\text{MD2\_CFG}$  register for the single-tap case, and setting to 1 the  $\text{INT1\_DOUBLE\_TAP}$  bit of the  $\text{MD1\_CFG}$  register or the  $\text{INT2\_DOUBLE\_TAP}$  bit of the  $\text{MD2\_CFG}$  register for the double-tap case.

No single/double tap interrupt is generated if the accelerometer is in Inactivity status (see Section 4.6 for more details).

Figure 16. Single and double-tap recognition (LIR bit = 0)



Tap interrupt signals can also be checked by reading the TAP\_SRC (1Ch) register, described in [Table 32](#).

**Table 32. TAP\_SRC register**

| b7 | b6     | b5         | b4         | b3       | b2    | b1    | b0    |
|----|--------|------------|------------|----------|-------|-------|-------|
| 0  | TAP_IA | SINGLE_TAP | DOUBLE_TAP | TAP_SIGN | X_TAP | Y_TAP | Z_TAP |

- TAP\_IA is set high when a single-tap or double-tap event has been detected.
- SINGLE\_TAP is set high when a single tap has been detected.
- DOUBLE\_TAP is set high when a double tap has been detected.
- TAP\_SIGN indicates the acceleration sign when the tap event is detected. It is set low in case of positive sign and it is set high in case of negative sign.
- X\_TAP (Y\_TAP, Z\_TAP) is set high when the tap event has been detected on the X (Y, Z) axis

Single and double-tap recognition works independently. Setting the SINGLE\_DOUBLE\_TAP bit of WAKE\_UP\_THS to 0, only the single-tap recognition is enabled: double-tap recognition is disabled and cannot be detected. When the SINGLE\_DOUBLE\_TAP is set to 1, both single and double-tap recognition are enabled.

If the latch mode is enabled and the interrupt signal is driven to the interrupt pins, the value assigned to SINGLE\_DOUBLE\_TAP also affects the behavior of the interrupt signal: when it is set to 0, the latch mode is applied to the single-tap interrupt signal; when it is set to 1, the latch mode is applied to the double-tap interrupt signal only. The latched interrupt signal is kept high until the TAP\_SRC register is read. If the latch mode is enabled but the interrupt signal is not driven to the interrupt pins, the latch feature does not take effect.

#### 4.5.4 Single-tap example

Hereafter an example code which implements the SW routine for single-tap detection.

```

1   Write 60h into CTRL1_XL           // Turn on the accelerometer
                                     // ODR_XL = 416 Hz, FS_XL = 2g
2   Write 0Eh into TAP_CFG           // Enable tap detection on X, Y, Z axis
3   Write 09h into TAP_THS_6D        // Set tap threshold
4   Write 06h into INT_DUR2          // Set Quiet and Shock time windows
5   Write 00h into WAKE_UP_THS       // Only single tap enabled (SINGLE_DOUBLE_TAP = 0)
6   Write 40h into MD1_CFG           // Single tap interrupt driven to INT1 pin

```

In this example the TAP\_THS field of the TAP\_THS\_6D register is set to 01001b, therefore the tap threshold is 562.5 mg ( $= 9 * FS_{XL} / 2^5$ ).

The SHOCK field of the INT\_DUR2 register is set to 10b: an interrupt is generated when the slope data exceeds the programmed threshold, and returns below it within 38.5 ms ( $= 2 * 8 / ODR_{XL}$ ) corresponding to the Shock time window.

The QUIET field of the INT\_DUR2 register is set to 01b: since the latch mode is disabled, the interrupt is kept high for the duration of the Quiet window, therefore 9.6 ms ( $= 1 * 8 / ODR_{XL}$ ).

### 4.5.5 Double-tap example

The example code which implements the SW routine for single-tap detection is given below.

```

1  Write 60h into CTRL1_XL      // Turn on the accelerometer
                                // ODR_XL = 416 Hz, FS_XL = 2g
2  Write 0Eh into TAP_CFG      // Enable tap detection on X, Y, Z axis
3  Write 0Ch into TAP_THS_6D   // Set tap threshold
4  Write 7Fh into INT_DUR2     // Set Duration, Quiet and Shock time windows
5  Write 80h into WAKE_UP_THS  // Single & Double tap enabled (SINGLE_DOUBBLE_TAP = 1)
6  Write 08h into MD1_CFG      // Double tap interrupt driven to INT1 pin

```

In this example the TAP\_THS field of the TAP\_THS\_6D register is set to 01100b, therefore the tap threshold is 750 mg ( $= 12 * FS_{XL} / 2^5$ ).

For interrupt generation, during the first and the second tap the slope data must return below the threshold before the Shock window has expired. The SHOCK field of the INT\_DUR2 register is set to 11b, therefore the Shock time is 57.7 ms ( $= 3 * 8 / ODR_{XL}$ ).

For interrupt generation, after the first tap recognition there must not be any slope data overthreshold during the Quiet time window. Furthermore, since the latch mode is disabled, the interrupt is kept high for the duration of the Quiet window. The QUIET field of the INT\_DUR2 register is set to 11b, therefore the Quiet time is 28.8 ms ( $= 3 * 4 / ODR_{XL}$ ).

For the maximum time between two consecutive detected taps, the DUR field of the INT\_DUR2 register is set to 0111b, therefore the Duration time is 538.5 ms ( $= 7 * 32 / ODR_{XL}$ ).

## 4.6 Activity/Inactivity recognition

The Activity/Inactivity recognition function allows reducing system power consumption and developing new smart applications.

When the Activity/Inactivity recognition function is activated, the LSM6DS3 device is able to automatically decrease the accelerometer sampling rate to 13 Hz, increasing the accelerometer ODR and bandwidth as soon as the wake-up interrupt event has been detected. This feature is applied to the accelerometer sensor only, regardless of the selected gyroscope power mode and ODR.

With this feature the system may be efficiently switched from low-power consumption to full performance and vice-versa depending on user-selectable acceleration events, thus ensuring power saving and flexibility.

The Activity/Inactivity recognition function is enabled by setting to 1 the INACTIVITY bit of the WAKE\_UP\_THS register.

The Activity/Inactivity recognition function uses the slope between two consecutive acceleration samples to detect the Activity/Inactivity event; the slope data is computed using the following formula:

$$\text{slope}(t_n) = [ \text{acc}(t_n) - \text{acc}(t_{n-1}) ] / 2$$



This function can be fully programmed by the user in terms of expected amplitude and timing of the slope data by means of a dedicated set of registers (*Figure 17*).

The unsigned threshold value is defined using the WK\_THS[5:0] bits of dedicated set of registers WAKE\_UP\_THS register; the value of 1 LSB of these 6 bits depends on the selected accelerometer full scale:  $1 \text{ LSB} = (\text{FS\_XL})/(2^6)$ . The threshold is applied to both positive and negative slope data.

When a certain number of consecutive X,Y,Z slope data is smaller than the configured threshold, the ODR\_XL [3:0] bits of the CTRL1\_XL register are bypassed (Inactivity) and the accelerometer is internally set to 13 Hz although the content of CTRL1\_XL is left untouched. The duration of the Inactivity status to be recognized is defined by the SLEEP\_DUR[3:0] bits of the WAKE\_UP\_DUR register: 1 LSB corresponds to  $512 \cdot \text{ODR\_XL}$  time, where ODR\_XL is the accelerometer output data rate.

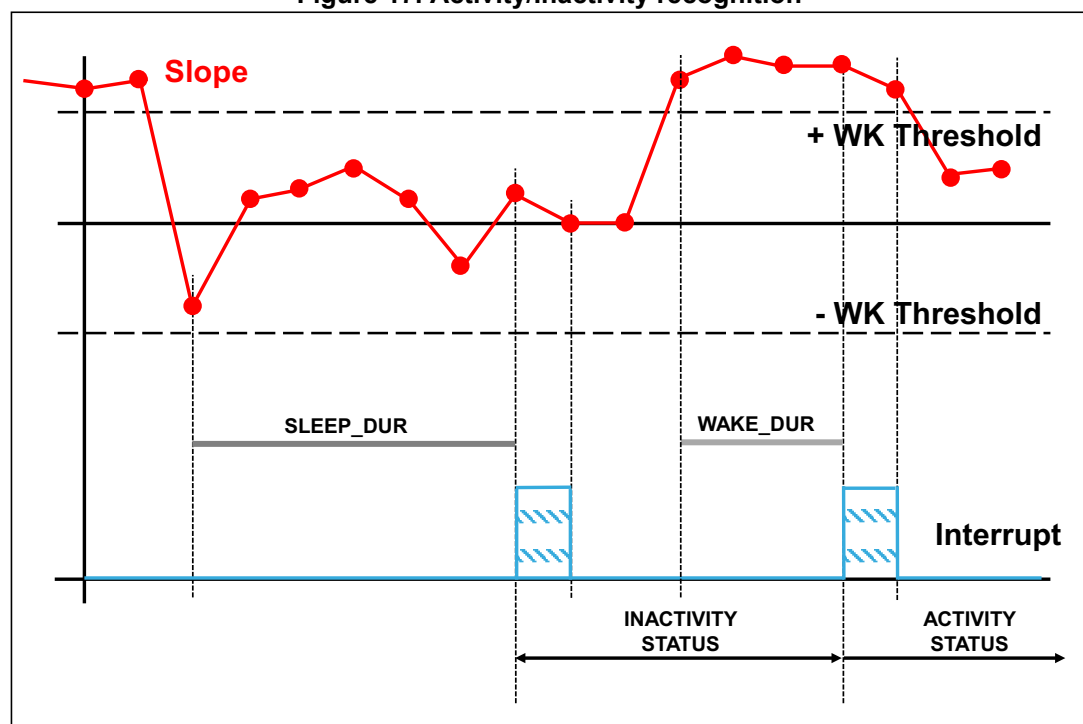
When the Inactivity status is detected, the interrupt is set high for  $1/\text{ODR\_XL}[\text{s}]$  period then it is automatically deasserted.

When a certain number of consecutive slope data on one axis become bigger than the threshold, the CTRL1\_XL register settings are immediately restored (Activity). The duration of the Activity status to be recognized is defined by the WAKE\_DUR[1:0] bits of the WAKE\_UP\_DUR register: 1 LSB corresponds to  $1 \cdot \text{ODR\_XL}$  time, where ODR\_XL is the accelerometer output data rate.

When the Activity status is detected, the interrupt is set high for  $1/\text{ODR\_XL}[\text{s}]$  period then it is automatically deasserted.

Once the Activity/Inactivity detection function is enabled, the status can be driven to the two interrupt pins by setting to 1 the INT1\_INACT\_STATE bit of the MD1\_CFG register or the INT1\_INACT\_STATE bit of the MD2\_CFG register; it can also be checked by reading the SLEEP\_STATE\_IA bit of the WAKE\_UP\_SRC register.

**Figure 17. Activity/Inactivity recognition**



The code provided below is a basic routine for Activity/Inactivity detection implementation.

```

1   Write 50h into CTRL1_XL      // Turn on the accelerometer
                                   // ODR_XL = 208 Hz, FS_XL = 2g
2   Write 42h into WAKE_UP_DUR  // Set duration for Inactivity detection
                                   // Set duration for Activity detection
3   Write 42h into WAKE_UP_THS  // Set Activity/Inactivity threshold
                                   // Enable Activity/Inactivity detection
4   Write 80h into MD1_CFG      // Activity/Inactivity interrupt driven to INT1 pin

```

In this example the WK\_THS field of the WAKE\_UP\_THS register is set to 000010b, therefore the Activity/Inactivity threshold is 62.5 mg ( $= 2 * FS\_XL / 2^6$ ).

Before Inactivity detection, the X,Y,Z slope data must be smaller than the configured threshold for a period of time defined by the SLEEP\_DUR field of the WAKE\_UP\_DUR register: this field is set to 0010b, corresponding to 4.92 s ( $= 2 * 512 / ODR\_XL$ ). After this period of time has elapsed, the accelerometer ODR is internally set to 13 Hz.

The Activity status is detected and the CTRL1\_XL register settings immediately restored if the slope data of (at least) one axis are bigger than the threshold for a period of time defined by the WAKE\_DUR field of the WAKE\_UP\_DUR register: this field is set to 10b, corresponding to 9.62 ms ( $= 2 * 1 / ODR\_XL$ ).

## 4.7 Boot status

After the device is powered up, the LSM6DS3 performs a 20 ms boot procedure to load the trimming parameters. After the boot is completed, both the accelerometer and the gyroscope are automatically configured in Power-Down mode.

After power up, the trimming parameters can be re-loaded by setting to 1 the BOOT bit of the CTRL3\_C register.

No toggle of the device power lines is required and the content of the device control registers is not modified, so the device operating mode doesn't change after boot. If the reset to the default value of the control registers is required, it can be performed by setting to 1 the SW\_RESET bit of the CTRL3\_C register.

While the boot is running, the EV\_BOOT bit of the STATUS\_REG register is set high; at the end of the boot procedure, this bit is set low again. The boot status signal can also be driven to the INT1 interrupt pin by setting to 1 the INT1\_BOOT bit of the INT1\_CTRL register.

## 5 Android embedded functions

The LSM6DS3 device implements in hardware the sensor-related functions specified in Android L; specific IP blocks with negligible power consumption and high-level performance implement the following functions using only the accelerometer:

- Pedometer functions (step detector and step counter);
- Significant motion;
- Tilt;
- Time stamp.

All these functions work at 26 Hz, so the accelerometer ODR must be set at 26 Hz or higher values.

### 5.1 Pedometer functions: step detector and step counter

A specific IP block of the LSM6DS3 device is dedicated to pedometer functions: the step detector and the step counter.

Pedometer functions work at 26 Hz, so the accelerometer ODR must be set at 26 Hz or higher values.

In order to enable the pedometer functions it is necessary to set to 1 both the FUNC\_EN bit of the CTRL10\_C register and the PEDO\_EN bit of the TAP\_CFG register.

The step detector functionality generates an interrupt every time a step is recognized. In case of interspersed step sessions, 7 consecutive steps have to be detected before the first interrupt generation (debounce functionality) in order to avoid false step detections.

This interrupt signal can be driven to the INT1 interrupt pin by setting to 1 the INT1\_STEP\_DETECTOR bit of the INT1\_CTRL register; it can also be checked by reading the STEP\_DETECTED bit of the FUNC\_SRC register.

Instead of generating an interrupt every time a step is recognized, it is possible to generate it if at least one step is detected within a certain time period. This time period is defined by setting a value higher than 00h in the STEP\_COUNT\_DELTA register. It is necessary to set the TIMER\_EN bit of the TAP\_CFG register to 1 (to enable the timer) and the TIMER\_HR bit of the WAKE\_UP\_DUR register to 0 when using this feature: in this case, 1 LSB of the value of the STEP\_COUNT\_DELTA register corresponds to 1.6384 seconds. This interrupt signal can be driven to the INT2 interrupt pin by setting to 1 the INT2\_STEP\_DELTA bit of the INT2\_CTRL register; it can also be checked by reading the STEP\_COUNT\_DELTA\_IA bit of the FUNC\_SRC register.

The step counter indicates the number of steps detected by the step detector algorithm after the pedometer function has been enabled. The step count is given by the concatenation of the STEP\_COUNTER\_H and STEP\_COUNTER\_L registers and it is represented as a 16-bit unsigned number.

The step count is not reset to zero when the accelerometer is configured in Power-Down or the pedometer is disabled; it can be reset to zero by setting the PEDO\_RST\_STEP bit of the CTRL10\_C register to 1. After the counter resets, the PEDO\_RST\_STEP bit is not automatically set back to 0.

The Step Counter overflow signal can be driven to the INT2 interrupt pin by setting to 1 the INT2\_STEP\_COUNT\_OV bit of the INT2\_CTRL register: in this case, when the step count reaches the  $2^{16}$  value, an interrupt signal is generated on the INT2 pin and the step count has to be reset to zero by setting to 1 the PEDO\_RST\_STEP bit.

If latch mode is disabled (LIR bit of TAP\_CFG is set to 0), the interrupt signal generated by the pedometer functions is pulsed: the duration of the pulse observed on the interrupt pins is about 60  $\mu$ s; the duration of the pulse observed on the bits STEP\_COUNT\_DELTA\_IA, STEP\_DETECTED\_IA and STEP\_OVERFLOW of the FUNC\_SRC register is 1/26 Hz.

If latch mode is enabled (LIR bit of TAP\_CFG is set to 1) and the interrupt signal is driven to the interrupt pins, once a step has occurred, a reading of the FUNC\_SRC register clears the request on both the pins and the STEP\_COUNT\_DELTA\_IA, STEP\_DETECTED\_IA and STEP\_OVERFLOW bits of the FUNC\_SRC register, and the device is ready to recognize the next step. If latch mode is enabled but the interrupt signal is not driven to the interrupt pins, the interrupt signal observed on the bits of the FUNC\_SRC register is pulsed, with a fixed duration of 1/26 Hz.

Step counter time stamp information is available in the STEP\_TIMESTAMP\_H and STEP\_TIMESTAMP\_L registers: when a step is detected, the value of the TIMESTAMP\_REG2 register is copied in STEP\_TIMESTAMP\_H, and the value of the TIMESTAMP\_REG1 register is copied in STEP\_TIMESTAMP\_L, providing the time stamp information of this step. For more details about LSM6DS3 time stamp counter and TIMESTAMP\_REG2/TIMESTAMP\_REG1, see [Section 5.4](#).

The step counter time stamp resolution depends on the value of the TIMER\_HR bit of the WAKE\_UP\_DUR register: when this bit is set to 0, 1 LSB of the time step count corresponds to 1638.4 ms; when this bit is set to 1, 1 LSB of the time step count corresponds to 6.4 ms.

Step counter data can be stored in FIFO as a fourth data set along with time stamp data (see [Section 8.9](#) for more details).

Hereafter a basic SW routine which shows how to enable the pedometer functions:

|   |                          |                                               |
|---|--------------------------|-----------------------------------------------|
| 1 | Write 20h into CTRL1_XL  | // Turn on the accelerometer                  |
|   |                          | // ODR_XL = 26 Hz, FS_XL = 2g                 |
| 2 | Write 3Ch into CTRL10_C  | // Enable embedded functions                  |
| 3 | Write 40h into TAP_CFG   | // Enable pedometer algorithm                 |
| 4 | Write 80h into INT1_CTRL | // Step Detector interrupt driven to INT1 pin |

The interrupt signal is generated when a step is recognized and the step count is available by reading the STEP\_COUNTER\_H / STEP\_COUNTER\_L registers.

## 5.2 Significant motion

The Significant Motion function generates an interrupt when a 'significant motion', that could be due to a change in user location, is detected: in the LSM6DS3 device this function has been implemented in hardware using only the accelerometer.

Significant Motion functionality can be used in location-based applications in order to receive a notification indicating when the user is changing location.

In order to enable Significant Motion detection it is necessary to set to 1 both the FUNC\_EN bit and the SIGN\_MOTION\_EN bit of the CTRL10\_C register.

The Significant Motion interrupt signal can be driven to the INT1 interrupt pin by setting to 1 the INT1\_SIGN\_MOTION bit of the INT1\_CTRL register; it can also be checked by reading the SIGN\_MOTION\_IA bit of the FUNC\_SRC register.

If latch mode is disabled (LIR bit of TAP\_CFG is set to 0), the interrupt signal generated by the Significant Motion function is pulsed: the duration of the pulse observed on the interrupt pins is about 60  $\mu$ s; the duration of the pulse observed on the SIGN\_MOTION\_IA bit of the FUNC\_SRC register is 1/26 Hz.

If latch mode is enabled (LIR bit of TAP\_CFG is set to 1) and the interrupt signal is driven to the interrupt pins, once a 'significant motion' is detected, a reading of the FUNC\_SRC register clears the request on both the pins and the SIGN\_MOTION\_IA bit of the FUNC\_SRC register, and the device is ready to recognize the next event. If latch mode is enabled but the interrupt signal is not driven to the interrupt pins, the interrupt signal observed on the SIGN\_MOTION\_IA bit of the FUNC\_SRC register is pulsed, with a fixed duration of 1/26 Hz.

The embedded function register (accessible by setting to 1 the FUNC\_CFG\_EN bit of FUNC\_CFG\_ACCESS) used to configure the Significant Motion threshold parameter is the SM\_THS register. The SM\_THS\_[7:0] bits of this register define the threshold value: it corresponds to the number of steps to be performed by the user upon a change of location before the Significant Motion interrupt is generated. It is expressed as an 8-bit unsigned value: the default value of this field is equal to 6 (= 00000110b).

The Significant Motion function works at 26 Hz, so the accelerometer ODR must be set at 26 Hz or higher values.

Hereafter a basic SW routine which shows how to enable the tilt detection function:

```

1  Write 80h into FUNC_CFG_ADDRESS    // Enable access to embedded functions registers
2  Write 08h into SM_THS              // Set Significant Motion threshold
3  Write 00h into FUNC_CFG_ADDRESS    // Disable access to embedded functions registers
4  Write 20h into CTRL1_XL            // Turn-on the accelerometer
                                     // ODR_XL = 26 Hz, FS_XL = 2g
5  Write 3Dh into CTRL10_C            // Enable embedded functions
                                     // Enable Significant Motion detection
6  Write 40h into INT1_CTRL           // Significant motion interrupt driven to INT1 pin
```

In this example the SM\_THS\_[7:0] bits of the SM\_THS register are set to 00001000b, therefore the Significant Motion threshold is equal to 8.

## 5.3 Tilt

The Tilt function allows detecting when an activity change occurs (e.g. when phone is in a front pocket and the user goes from sitting to standing or standing to sitting): in the LSM6DS3 device it has been implemented in hardware using only the accelerometer.

In order to enable the tilt detector it is necessary to set to 1 both the FUNC\_EN bit of the CTRL10\_C register and the TILT\_EN bit of the TAP\_CFG register.

If the device is configured for tilt event detection, an interrupt is generated when the device is tilted by an angle greater than 35 degrees from the start position. The start position is defined as the position of the device when the tilt detection is enabled or the position of the device when the last Tilt interrupt was generated.

After this function is enabled, for the generation of the first Tilt interrupt the device should be continuously tilted by an angle greater than 35 degrees since start position for a period of time of 2 seconds. After the first Tilt interrupt is generated, the Tilt interrupt signal is set high as soon as the device is tilted by an angle greater than 35 degrees from the position of the device corresponding to the last interrupt detection (no need to wait 2 seconds).

This interrupt signal can be driven to the two interrupt pins by setting to 1 the INT1\_TILT bit of the MD1\_CFG register or the INT2\_TILT bit of the MD2\_CFG register; it can also be checked by reading the TILT\_IA bit of the FUNC\_SRC register.

If latch mode is disabled (LIR bit of TAP\_CFG is set to 0), the interrupt signal generated by the Tilt function is pulsed: the duration of the pulse observed on the interrupt pins is about 60  $\mu$ s; the duration of the pulse observed on the TILT\_IA bit of FUNC\_SRC register is 1/26 Hz.

If latch mode is enabled (LIR bit of TAP\_CFG is set to 1) and the interrupt signal is driven to the interrupt pins, once a tilt is detected, a reading of the FUNC\_SRC register clears the request on both the pins and the TILT\_IA bit of FUNC\_SRC register, and the device is ready to recognize the next tilt event. If latch mode is enabled but the interrupt signal is not driven to the interrupt pins, the interrupt signal observed on the TILT\_IA bit of the FUNC\_SRC register is pulsed, with a fixed duration of 1/26 Hz.

The tilt function works at 26 Hz, so the accelerometer ODR must be set at 26 Hz or higher values.

Hereafter a basic SW routine which shows how to enable the tilt detection function:

|   |                         |                                               |
|---|-------------------------|-----------------------------------------------|
| 1 | Write 20h into CTRL1_XL | // Turn on the accelerometer                  |
|   |                         | // ODR_XL = 26 Hz, FS_XL = 2g                 |
| 2 | Write 3Ch into CTRL10_C | // Enable embedded functions                  |
| 3 | Write 20h into TAP_CFG  | // Enable tilt detection                      |
| 4 | Write 02h into MD1_CFG  | // Tilt detector interrupt driven to INT1 pin |

## 5.4 Time stamp

Together with sensor data the LSM6DS3 device can provide time stamp information.

If both the accelerometer and the gyroscope are in Power-Down mode, the time stamp counter doesn't work.

To enable this functionality the TIMER\_EN bit of the TAP\_CFG register has to be set to 1: the time step count is given by the concatenation of the TIMESTAMP\_REG2 & TIMESTAMP\_REG1 & TIMESTAMP\_REG0 registers and is represented as a 24-bit unsigned number.

The time stamp resolution can be configured using the TIMER\_HR bit of the WAKE\_UP\_DUR register: when this bit is set to 0, 1 LSB of time step count corresponds to 6.4 ms; when this bit is set to 1, 1 LSB of time step count corresponds to 25  $\mu$ s.

When the maximum value 16777215 LSB (corresponding to FFFFFFFh) is reached and low resolution (TIMER\_HR = 0) is used, the counter is automatically reset to 000000h and continues to count. When the maximum value is reached and high resolution (TIMER\_HR = 1) is used, the counter is not automatically reset to 0 and freezes at FFFFFFFh. In any case, the timer count can be reset to zero at any time by writing the reset value AAh in the TIMESTAMP\_REG2 register.

An interrupt signal is generated when the timer step count reaches the value of 16776960 LSB (corresponding to FFFF00h). This interrupt signal can be driven to the INT1 pin by setting to 1 the INT1\_TIMER bit of the MD1\_CFG register. Once the interrupt pin is asserted, it must be reset to zero by writing AAh in the TIMESTAMP\_REG2 register.

The time stamp count can be stored in FIFO as a fourth data set along with the step counter data (see [Section 8.9](#) for details).

A basic SW routine to enable the time stamp counter is shown below.

```
1   Write 50h into CTRL1_XL           // Turn on the accelerometer
                                     // ODR_XL = 208 Hz, FS_XL = 2g
2   Write 80h into TAP_CFG            // Enable time stamp count
3   Write 10h into WAKE_UP_DUR        // Time stamp resolution = 25us
4   Write 01h into MD1_CFG            // End counter interrupt driven to INT1 pin
```

## 6 Mode 2 - sensor hub mode

The hardware flexibility of the LSM6DS3 allow connecting the pins with different mode connections to external sensors to expand functionalities such as adding a sensor hub and an auxiliary SPI.

Three different mode connections are available:

- **Mode 1 - Slave-only mode:** I<sup>2</sup>C/SPI(3- and 4-wire) slave interface is available;
- **Mode 2 - Sensor Hub mode:** I<sup>2</sup>C/SPI(3- and 4-wire) slave interface and I<sup>2</sup>C master interface for the connection of external sensors are available;
- **Mode 3 - Auxiliary SPI mode:** I<sup>2</sup>C slave interface and auxiliary SPI (3-wire) interface for external device connection (i.e. EIS application) are available (see [Chapter 7](#)).

The pin connections and function for each mode are described in the LSM6DS3 datasheet; Mode 2 connection mode is described in detail in the following paragraphs.

### 6.1 Sensor hub mode description

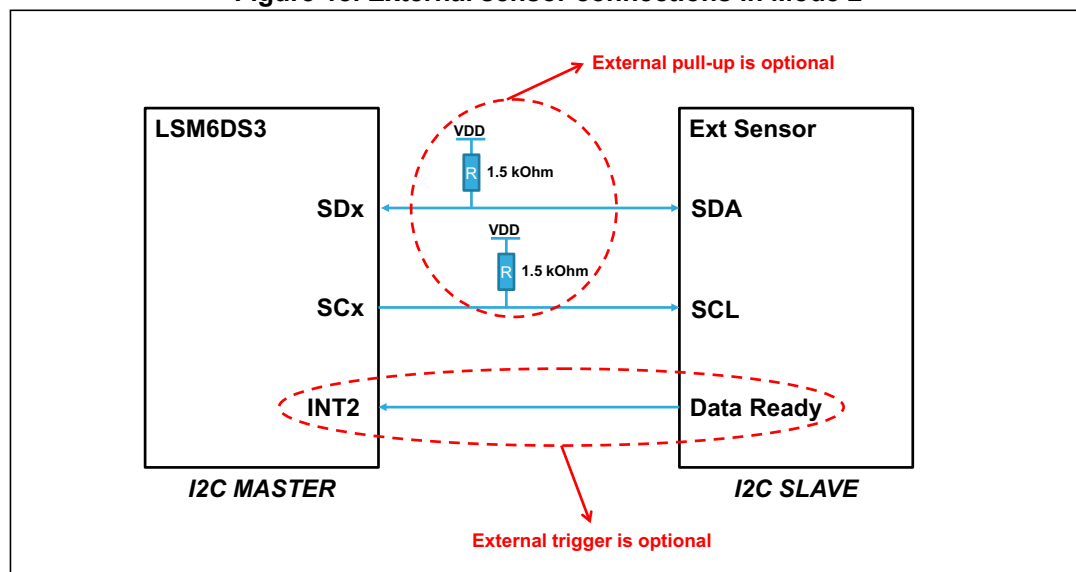
In sensor hub mode (Mode 2) up to 4 external sensors can be connected to the I<sup>2</sup>C master interface of LSM6DS3 device.

External sensor data can also be stored in FIFO with a configurable decimation factor (see [Chapter 8](#) for details).

If both the accelerometer and the gyroscope are in Power-Down mode, the sensor hub doesn't work.

All external sensors have to be connected in parallel to the SDx/SCx pins of the device, as illustrated in [Figure 18](#) for a single external sensor. External pull-up resistors and the external trigger signal connection are optional and depend on the configuration of the registers.

**Figure 18. External sensor connections in Mode 2**





## 6.2 Sensor hub mode registers

In order to enable the embedded functionalities of registers LSM6DS3, the FUNC\_EN bit of the CTRL10\_C register has to be set to 1; after enabling the sensor hub functionality, the MASTER\_CONFIG register has to be used for the configuration of the I<sup>2</sup>C master interface.

A set of registers SLVx\_ADD, SLVx\_SUBADD, SLAVEx\_CONFIG is dedicated to the configuration of the 4 slave interfaces associated to the 4 connectable external sensors. An additional register, DATAWRITE\_SRC\_MODE\_SUB\_SLV0, is associated to slave #0 only: it can be used to implement the writing and the source mode conditioned reading of the registers of the external sensor associated to slave #0.

Finally, 18 registers (from SENSORHUB1\_REG to SENSORHUB18\_REG) are available to store the data read from the external sensors.

### 6.2.1 CTRL10\_C (19h)

Table 33. CTRL10\_C register

| b7 | b6 | b5 | b4 | b3 | b2      | b1 | b0 |
|----|----|----|----|----|---------|----|----|
| 0  | 0  | X  | X  | X  | FUNC_EN | X  | X  |

- FUNC\_EN must be set to 1 in order to enable the embedded functionalities of the LSM6DS3 (pedometer, tilt, significant motion, ironing).

### 6.2.2 MASTER\_CONFIG (1Ah)

This register is used to configure the I<sup>2</sup>C master interface.

Table 34. MASTER\_CONFIG register

| b7           | b6 | b5 | b4           | b3         | b2                 | b1 | b0        |
|--------------|----|----|--------------|------------|--------------------|----|-----------|
| DRDY_ON_INT1 | X  | 0  | START_CONFIG | PULL_UP_EN | PASS_TH ROUGH_MODE | X  | MASTER_ON |

- DRDY\_ON\_INT1 bit has to be set to 1 to drive on the INT1 pin the I<sup>2</sup>C master Data-Ready signal (corresponding to the behavior of the SENSORHUB\_END\_OP bit of FUNC\_SRC register). The I<sup>2</sup>C master Data-Ready signal indicates when the sensor hub routine is complete and the external sensor data are available to be read on the SENSORHUBx\_REG registers (depending on the configuration of the SLVx\_ADD, SLVx\_SUBADD, SLAVEx\_CONFIG registers).
- START\_CONFIG bit selects the sensor hub trigger signal.
  - When this bit is set to 0, the accelerometer sensor has to be active (not in Power Down mode) and the sensor hub trigger signal is the accelerometer data-ready signal, with a frequency corresponding to the accelerometer ODR up to 100 Hz.
  - When this bit is set to 1, at least one sensor between the accelerometer and the gyroscope has to be active and the sensor hub trigger signal is the INT2 pin; in fact, when both the MASTER\_ON bit and START\_CONFIG bit are set to 1, the INT2 pin is configured as an input signal. In this case, the INT2 pin has to be

connected to the Data-Ready pin of the external sensor (*Figure 18*) in order to trigger the reading/writing operations on the external sensor registers.

- PULL\_UP\_EN bit enables/disables the internal pull-up on the auxiliary I<sup>2</sup>C line. When this bit is set to 0, the internal pull-up is disabled and the external pull-up resistors on the SDx/SCx pins are required, as shown in *Figure 18*. When this bit is set to 1, the internal pull-up is enabled and the external pull-up resistors on the SDx/SCx pins are not required.
- PASS\_THROUGH\_MODE bit is used to enable/disable the I<sup>2</sup>C interface pass-through. When this bit is set to 1, the main I<sup>2</sup>C line (e.g. connected to an external microcontroller) is short-circuited with the auxiliary one, in order to implement a direct access to the external sensor registers. It is recommended to use this feature when configuring the external sensors.
- MASTER\_ON bit has to be set to 1 to enable the auxiliary I<sup>2</sup>C master of the LSM6DS3 device (sensor hub mode).

### 6.2.3 FUNC\_SRC (53h)

Table 35. FUNC\_SRC register

| b7 | b6 | b5 | b4 | b3 | b2 | b1 | b0                    |
|----|----|----|----|----|----|----|-----------------------|
| 0  | X  | X  | X  | 0  | 0  | X  | SENSOR HUB<br>_END_OP |

- SENSORHUB\_END\_OP bit is set high when the sensor hub routine is completed and the external sensor data are available to be read from the SENSORHUBx\_REG registers (depending on the configuration of the SLVx\_ADD, SLVx\_SUBADD, SLAVEx\_CONFIG registers). This signal can be driven to the INT1 interrupt pin by setting to 1 the DRDY\_ON\_INT1 bit of the MASTER\_CONFIG register. The SENSORHUB\_END\_OP bit is cleared by reading the FUNC\_SRC register.

### 6.2.4 SLV0\_ADD (02h), SLV0\_SUBADD (03h), SLAVE0\_CONFIG (04h)

The embedded function registers (accessible when the FUNC\_CFG\_EN bit of the FUNC\_CFG\_ACCESS register is set to 1) used to configure the I<sup>2</sup>C slave interface associated to the first external sensor are described hereafter.

Table 36. SLV0\_ADD register

| b7              | b6              | b5              | b4              | b3              | b2              | b1              | b0   |
|-----------------|-----------------|-----------------|-----------------|-----------------|-----------------|-----------------|------|
| Slave0<br>_add6 | Slave0<br>_add5 | Slave0<br>_add4 | Slave0<br>_add3 | Slave0<br>_add2 | Slave0<br>_add1 | Slave0<br>_add0 | rw_0 |

- Slave0\_add[6:0] bits are used to indicate the I<sup>2</sup>C slave address of the first external sensor.
- rw\_0 bit configures the read/write operation to be performed on the first external sensor (0: write operation; 1: read operation). The read/write operation is executed when the next sensor hub trigger event occurs.

Table 37. SLV0\_SUBADD register

| b7          | b6          | b5          | b4          | b3          | b2          | b1          | b0          |
|-------------|-------------|-------------|-------------|-------------|-------------|-------------|-------------|
| Slave0_reg7 | Slave0_reg6 | Slave0_reg5 | Slave0_reg4 | Slave0_reg3 | Slave0_reg2 | Slave0_reg1 | Slave0_reg0 |

- Slave0\_reg[7:0] bits are used to indicate the address of the register of the first external sensor to be written (if the rw\_0 bit of the SLV0\_ADD register is set to 0) or the address of the first register to be read (if the rw\_0 bit of the SLV0\_ADD register is set to 1).

Table 38. SLAVE0\_CONFIG register

| b7           | b6           | b5           | b4           | b3       | b2            | b1            | b0            |
|--------------|--------------|--------------|--------------|----------|---------------|---------------|---------------|
| Slave0_rate1 | Slave0_rate0 | Aux_sens_on1 | Aux_sens_on0 | Src_mode | Slave0_numop2 | Slave0_numop1 | Slave0_numop0 |

- Slave0\_rate[1:0] bits are used to define the decimation factor applied to read operations on the first external sensor starting from the sensor hub trigger:
  - 00: no decimation
  - 01: update every 2 sensor hub trigger events
  - 10: update every 4 sensor hub trigger events
  - 11: update every 8 sensor hub trigger events
- Aux\_sens\_on[1:0] bits have to be used to indicate the number of external sensors to be managed by the sensor hub:
  - 00: one external sensor
  - 01: two external sensors
  - 10: three external sensors
  - 11: four external sensors
- Src\_mode bit enables/disables Source Mode conditioned reading. When this bit is set to 1, Source Mode conditioned reading is enabled; before proceeding with the reading of the register address indicated in the SLV0\_SUBADD register, the content of the register at the address specified in DATAWRITE\_SRC\_MODE\_SUB\_SLV0 is checked: if the content is non-zero the operation continues, else the reading operation is interrupted. Source Mode conditioned reading is available on slave 0 only.
- Slave0\_numop[2:0] bits are dedicated to define the number of consecutive read operations to be performed on the first external sensor starting from the register address indicated in the SLV0\_SUBADD register.

### 6.2.5 SLV1\_ADD (05h), SLV1\_SUBADD (06h), SLAVE1\_CONFIG (07h)

The embedded function registers (accessible when the FUNC\_CFG\_EN bit of the FUNC\_CFG\_ACCESS register is set to 1) used to configure the I<sup>2</sup>C slave interface associated to the second external sensor are described hereafter.

**Table 39. SLV1\_ADD register**

| b7          | b6          | b5          | b4          | b3          | b2          | b1          | b0  |
|-------------|-------------|-------------|-------------|-------------|-------------|-------------|-----|
| Slave1_add6 | Slave1_add5 | Slave1_add4 | Slave1_add3 | Slave1_add2 | Slave1_add1 | Slave1_add0 | r_1 |

- Slave1\_add[6:0] bits are used to indicate the I<sup>2</sup>C slave address of the second external sensor.
- r\_1 bit enables/disables the read operation to be performed on the second external sensor (0: read operation disabled; 1: read operation enabled). The read operation is executed when the next sensor hub trigger event occurs.

**Table 40. SLV1\_SUBADD register**

| b7          | b6          | b5          | b4          | b3          | b2          | b1          | b0          |
|-------------|-------------|-------------|-------------|-------------|-------------|-------------|-------------|
| Slave1_reg7 | Slave1_reg6 | Slave1_reg5 | Slave1_reg4 | Slave1_reg3 | Slave1_reg2 | Slave1_reg1 | Slave1_reg0 |

- Slave1\_reg[7:0] bits are used to indicate the address of the register of the second external sensor to be read when the r\_1 bit of SLV1\_ADD register is set to 1.

**Table 41. SLAVE1\_CONFIG register**

| b7           | b6           | b5 | b4 | b3 | b2            | b1            | b0            |
|--------------|--------------|----|----|----|---------------|---------------|---------------|
| Slave1_rate1 | Slave1_rate0 | 0  | 0  | 0  | Slave1_numop2 | Slave1_numop1 | Slave1_numop0 |

- Slave1\_rate[1:0] bits are used to define the decimation factor applied to read operations on the second external sensor starting from the sensor hub trigger:
  - 00: no decimation
  - 01: update every 2 sensor hub trigger events
  - 10: update every 4 sensor hub trigger events
  - 11: update every 8 sensor hub trigger events
- Slave1\_numop[2:0] bits are dedicated to define the number of consecutive read operations to be performed on the second external sensor starting from the register address indicated in SLV1\_SUBADD register.

### 6.2.6 SLV2\_ADD (08h), SLV2\_SUBADD (09h), SLAVE2\_CONFIG (0Ah)

The embedded function registers (accessible when the FUNC\_CFG\_EN bit of the FUNC\_CFG\_ACCESS register is set to 1) used to configure the I<sup>2</sup>C slave interface associated to the third external sensor are described hereafter.

**Table 42. SLV2\_ADD register**

| b7          | b6          | b5          | b4          | b3          | b2          | b1          | b0  |
|-------------|-------------|-------------|-------------|-------------|-------------|-------------|-----|
| Slave2_add6 | Slave2_add5 | Slave2_add4 | Slave2_add3 | Slave2_add2 | Slave2_add1 | Slave2_add0 | r_2 |

- Slave2\_add[6:0] bits are used to indicate the I<sup>2</sup>C slave address of the third external sensor.
- r\_2 bit enables/disables the read operation to be performed on the third external sensor (0: read operation disabled; 1: read operation enabled). The read operation is executed when the next sensor hub trigger event occurs.

**Table 43. SLV2\_SUBADD register**

| b7          | b6          | b5          | b4          | b3          | b2          | b1          | b0          |
|-------------|-------------|-------------|-------------|-------------|-------------|-------------|-------------|
| Slave2_reg7 | Slave2_reg6 | Slave2_reg5 | Slave2_reg4 | Slave2_reg3 | Slave2_reg2 | Slave2_reg1 | Slave2_reg0 |

- Slave2\_reg[7:0] bits are used to indicate the address of the register of the third external sensor to be read when the r\_2 bit of SLV2\_ADD register is set to 1.

**Table 44. SLAVE2\_CONFIG register**

| b7           | b6           | b5 | b4 | b3 | b2            | b1            | b0            |
|--------------|--------------|----|----|----|---------------|---------------|---------------|
| Slave2_rate1 | Slave2_rate0 | 0  | 0  | 0  | Slave2_numop2 | Slave2_numop1 | Slave2_numop0 |

- Slave2\_rate[1:0] bits are used to define the decimation factor applied to read operations on the third external sensor starting from the sensor hub trigger:
  - 00: no decimation
  - 01: update every 2 sensor hub trigger events
  - 10: update every 4 sensor hub trigger events
  - 11: update every 8 sensor hub trigger events
- Slave2\_numop[2:0] bits are dedicated to define the number of consecutive read operations to be performed on the third external sensor starting from the register address indicated in the SLV2\_SUBADD register.

### 6.2.7 SLV3\_ADD (0Bh), SLV3\_SUBADD (0Ch), SLAVE3\_CONFIG (0Dh)

The embedded function registers (accessible when the FUNC\_CFG\_EN bit of FUNC\_CFG\_ACCESS register is set to 1) used to configure the I<sup>2</sup>C slave interface associated to the fourth external sensor are described hereafter.

**Table 45. SLV3\_ADD register**

| b7          | b6          | b5          | b4          | b3          | b2          | b1          | b0  |
|-------------|-------------|-------------|-------------|-------------|-------------|-------------|-----|
| Slave3_add6 | Slave3_add5 | Slave3_add4 | Slave3_add3 | Slave3_add2 | Slave3_add1 | Slave3_add0 | r_3 |

- Slave3\_add[6:0] bits are used to indicate the I<sup>2</sup>C slave address of the fourth external sensor.
- r\_3 bit enables/disables the read operation to be performed on the fourth external sensor (0: read operation disabled; 1: read operation enabled). The read operation is executed when the next sensor hub trigger event occurs.

**Table 46. SLV3\_SUBADD register**

| b7          | b6          | b5          | b4          | b3          | b2          | b1          | b0          |
|-------------|-------------|-------------|-------------|-------------|-------------|-------------|-------------|
| Slave3_reg7 | Slave3_reg6 | Slave3_reg5 | Slave3_reg4 | Slave3_reg3 | Slave3_reg2 | Slave3_reg1 | Slave3_reg0 |

- Slave3\_reg[7:0] bits are used to indicate the address of the register of the fourth external sensor to be read when the r\_3 bit of the SLV3\_ADD register is set to 1.

**Table 47. SLAVE3\_CONFIG register**

| b7           | b6           | b5 | b4 | b3 | b2            | b1            | b0            |
|--------------|--------------|----|----|----|---------------|---------------|---------------|
| Slave3_rate1 | Slave3_rate0 | 0  | 0  | 0  | Slave3_numop2 | Slave3_numop1 | Slave3_numop0 |

- Slave3\_rate[1:0] bits are used to define the decimation factor applied to read operations on the fourth external sensor starting from the sensor hub trigger:
  - 00: no decimation
  - 01: update every 2 sensor hub trigger events
  - 10: update every 4 sensor hub trigger events
  - 11: update every 8 sensor hub trigger events
- Slave3\_numop[2:0] bits are dedicated to define the number of consecutive read operations to be performed on the fourth external sensor starting from the register address indicated in the SLV3\_SUBADD register.

## 6.2.8 DATAWRITE\_SRC\_MODE\_SUB\_SLV0 (0Eh)

Table 48. DATAWRITE\_SRC\_MODE\_SUB\_SLV0 register

| b7           | b6           | b5           | b4           | b3           | b2           | b1           | b0           |
|--------------|--------------|--------------|--------------|--------------|--------------|--------------|--------------|
| Slave_dataw7 | Slave_dataw6 | Slave_dataw5 | Slave_dataw4 | Slave_dataw3 | Slave_dataw2 | Slave_dataw1 | Slave_dataw0 |

- Slave\_dataw[7:0] bits are dedicated, when the rw\_0 bit of SLV0\_ADD register is set to 0 (write operation), to indicate the data to be written to the first external sensor at the address specified in the SLV0\_SUBADD register. During read operations (rw\_0 = 1), this register is used if the Source Mode conditioned reading is enabled (Src\_mode bit = 1 in the SLAVE0\_CONFIG register) and it indicates the address of the external sensor register to be checked before proceeding with the reading operation.

## 6.2.9 SENSORHUBx\_REG registers

Once the auxiliary I<sup>2</sup>C master is enabled, for each of the external sensors it reads a number of registers equal to the value of the Slavex\_numop (x = 0, 1, 2, 3) field, starting from the register address specified in SLVx\_SUBADD (x = 0, 1, 2, 3) register. The number of external sensors to be managed is specified in the Aux\_sens\_on bits of the SLAVE0\_CONFIG register.

Read data are consecutively stored (in the same order they are read) in the LSM6DS3 registers starting from the SENSORHUB1\_REG register, as in the example in [Figure 19](#); 18 registers, from SENSORHUB1\_REG to SENSORHUB18\_REG, are available to store the data read from the external sensors.

The values of the registers from SENSORHUB1\_REG to SENSORHUB6\_REG can be saved in the FIFO buffer as a third data set; the values of the registers from SENSORHUB7\_REG to SENSORHUB12\_REG can be saved in the FIFO buffer as a fourth data set (see [Chapter 8](#) for details).

Figure 19. SENSORHUBx\_REG allocation example

|           |   |                                                                      |                 |                  |           |
|-----------|---|----------------------------------------------------------------------|-----------------|------------------|-----------|
| Sensor #1 | { | SLV0_SUBADD(03h) = 28h<br>SLAVE0_CONFIG(04h) – Slave0_numop[2:0] = 3 | SENSORHUB1_REG  | Value of reg 28h | Sensor #1 |
|           |   |                                                                      | SENSORHUB2_REG  | Value of reg 29h |           |
|           |   |                                                                      | SENSORHUB3_REG  | Value of reg 2Ah |           |
| Sensor #2 | { | SLV1_SUBADD(06h) = 00h<br>SLAVE1_CONFIG(07h) – Slave1_numop[2:0] = 6 | SENSORHUB4_REG  | Value of reg 00h | Sensor #2 |
|           |   |                                                                      | SENSORHUB5_REG  | Value of reg 01h |           |
|           |   |                                                                      | SENSORHUB6_REG  | Value of reg 02h |           |
| Sensor #3 | { | SLV2_SUBADD(09h) = 20h<br>SLAVE2_CONFIG(0Ah) – Slave2_numop[2:0] = 4 | SENSORHUB7_REG  | Value of reg 03h | Sensor #3 |
|           |   |                                                                      | SENSORHUB8_REG  | Value of reg 04h |           |
|           |   |                                                                      | SENSORHUB9_REG  | Value of reg 05h |           |
| Sensor #4 | { | SLV3_SUBADD(0Ch) = 40h<br>SLAVE3_CONFIG(0Dh) – Slave3_numop[2:0] = 5 | SENSORHUB10_REG | Value of reg 20h | Sensor #3 |
|           |   |                                                                      | SENSORHUB11_REG | Value of reg 21h |           |
|           |   |                                                                      | SENSORHUB12_REG | Value of reg 22h |           |
|           |   |                                                                      | SENSORHUB13_REG | Value of reg 23h | Sensor #4 |
|           |   |                                                                      | SENSORHUB14_REG | Value of reg 40h |           |
|           |   |                                                                      | SENSORHUB15_REG | Value of reg 41h |           |
|           |   |                                                                      | SENSORHUB16_REG | Value of reg 42h | Sensor #4 |
|           |   |                                                                      | SENSORHUB17_REG | Value of reg 43h |           |
|           |   |                                                                      | SENSORHUB18_REG | Value of reg 44h |           |

### 6.3 Sensor hub mode example

The configuration of the external sensors should be performed using the pass-through feature: this feature can be enabled by setting the PASS\_THROUGH\_MODE bit of the MASTER\_CONFIG register to 1 and implements a direct access to the external sensor registers, allowing quick configuration.

The code provided below gives a basic routine to configure the LSM6DS3 in sensor hub mode. Furthermore, this sequence configures the LIS3MDL external magnetometer sensor (refer to the datasheet for additional details) in continuous-conversion mode and reads the magnetometer output registers, saving their values in the SENSORHUB1\_REG to SENSORHUB6\_REG registers. The pass-through feature is not used in this example.

```

1  Write 80h into FUNC_CFG_ADDRESS    // Enable access to embedded functions registers

2  Write 38h into SLV0_ADD             // LIS3MDL slave address = 0011100b (if SDO=0)
                                     // Enable write operation (rw_0=0)

3  Write 22h into SLV0_SUBADD         // 22h is the LIS3MDL register to be written

4  Write 00h into                     // 00h is the value to be written in register 22h of
   DATAWRITE_SRC_MODE_SUB_SLV0      // LIS3MDL to configure it in continuous
                                     // conversion mode

5  Write 00h into FUNC_CFG_ADDRESS    // Disable access to embedded functions registers

6  Write 3Ch into CTRL10_C            // Enable embedded functions

                                     // Enable internal pull-up on SDx/SCx lines
7  Write 09h into MASTER_CONFIG      // Sensor hub trigger signal is XL Data Ready
                                     // Enable auxiliary I2C master

8  Write 80h into CTRL1_XL            // Turn-on the accelerometer (for trigger signal)

9  Write 38h into CTRL10_C            // Disable embedded functions

10 Write 00h into MASTER_CONFIG       // Disable auxiliary I2C master

11 Write 00h into CTRL1_XL            // Turn-off the accelerometer


12 Write 80h into FUNC_CFG_ADDRESS    // Enable access to embedded functions registers

13 Write 39h into SLV0_ADD             // LIS3MDL slave address = 0011100b (if SDO=0)
                                     // Enable read operation (rw_0=1)

14 Write 28h into SLV0_SUBADD         // 28h is the first LIS3MDL output register to be
                                     // read

                                     // No decimation
15 Write 06h into SLAVE0_CONFIG       // 1 external sensor connected
                                     // Number of registers to read = 6

16 Write 00h into FUNC_CFG_ADDRESS    // Disable access to embedded functions registers

17 Write 3Ch into CTRL10_C            // Enable embedded functions

                                     // Enable internal pull-up on SDx/SCx lines
18 Write 09h into MASTER_CONFIG      // Sensor hub trigger signal is XL Data Ready
                                     // Enable auxiliary I2C master

19 Write 80h into CTRL1_XL            // Turn on the accelerometer (for trigger signal)

```



## 6.4 Magnetometer hard-iron / soft-iron correction

The LSM6DS3 device supports the data acquisition of an external magnetometer with soft-iron and hard-iron correction features. For this purpose, it is required to set the MASTER\_ON bit of the MASTER\_CONFIG register to 1 to enable the sensor hub mode, to associate the external magnetometer to slave 0 registers (SLV0\_ADD, SLV0\_SUBADD and SLAVE0\_CONFIG) and to set the Slave0\_numop field of SLAVE0\_CONFIG to 6.

The FUNC\_EN bit of CTRL10\_C register has to be set to 1 in order to enable the embedded ironing functionalities. Then, distortion correction algorithms can be enabled as described in [Table 49](#): the IRON\_EN bit of MASTER\_CONFIG and the SOFT\_EN bit of CTRL9\_XL are used to enable hard-iron correction only or both hard-iron and soft-iron corrections. In the latter case, both calibrated (hard-iron & soft-iron) and uncalibrated (soft-iron only) magnetometer data are available.

**Table 49. Ironing configuration**

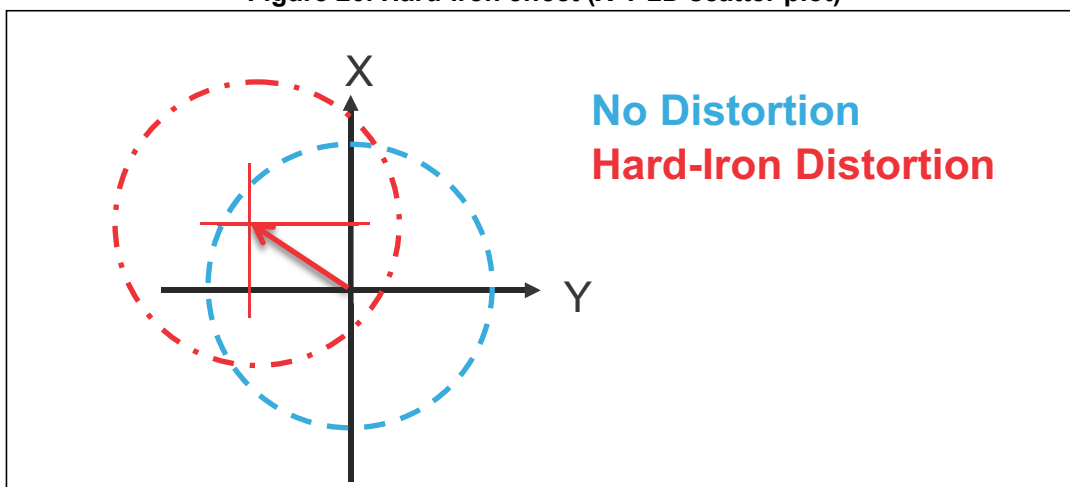
| CTRL9_XL<br>SOFT bit | MASTER_CONFIG<br>IRON_EN bit | Ironing<br>configuration          |
|----------------------|------------------------------|-----------------------------------|
| 0                    | 0                            | No correction applied             |
| 0                    | 1                            | Hard-iron only                    |
| 1                    | 1                            | Hard-iron + soft-iron corrections |

### 6.4.1 Hard-iron correction

Hard-iron distortion is normally generated by ferromagnetic material with permanent magnetic fields that are part of the object (e.g. a tablet) in use; these materials could be permanent magnets or magnetized iron or steel. They are time invariant and deform the local geomagnetic field with different offset on different directions.

Generally, if the user performs many 3D rotations of the object in an ideal environment (no hard-iron/soft-iron distortion) and plots the collected magnetic sensor raw data, the result will be a perfect sphere with no offset. The hard-iron distortion effect is to offset the sphere along the x, y and z axes; in the x-y plane, the hard-iron distortion is identified by an offset of the origin of the ideal circle from (0, 0), as shown in [Figure 20](#).

**Figure 20. Hard-iron effect (X-Y 2D scatter plot)**



In the LSM6DS3 device, the 3x1 hard-iron vector containing the X, Y, Z magnetic offset values calculated by the user have to be indicated in dedicated registers: the MAG\_OFFX\_L and MAG\_OFFX\_H registers are dedicated to the X-axis offset, the MAG\_OFFY\_L and MAG\_OFFY\_H registers are dedicated to the Y-axis offset, the MAG\_OFFZ\_L and MAG\_OFFZ\_H registers are dedicated to the Z-axis offset. These registers values are expressed as a 16-bit word in two's complement; the sensitivity [LSB/Gauss] to be applied to calculate the hard-iron register values corresponds to that of the external magnetometer.

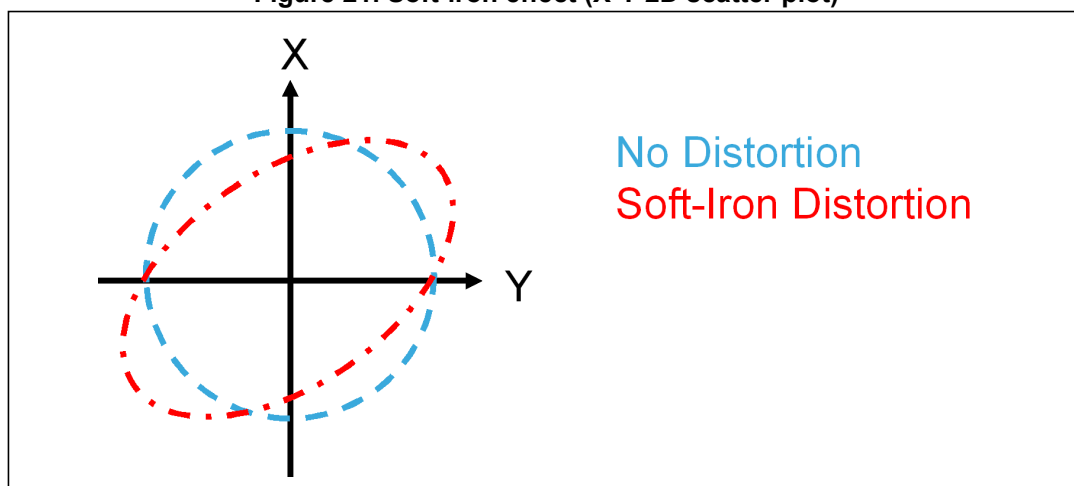
The hard-iron registers are accessible when the FUNC\_CFG\_EN bit of the FUNC\_CFG\_ACCESS register is set to 1. In order to enable the hard-iron correction algorithm, it is necessary to set to 1 both the FUNC\_EN bit of the CTRL10\_C register and the IRON\_EN bit of the MASTER\_CONFIG register ([Table 49](#)).

#### 6.4.2 Soft-iron correction

Soft-iron distortion is generated by magnetically soft materials or current carrying PCB traces. While the hard-iron distortion is constant regardless of the orientation, the soft-iron distortion changes with the orientation of the object in the Earth's field. Basically, the local geomagnetic field is deformed with different gain on different directions.

The effect of the soft-iron distortion is to make the ideal full round sphere become a tilted ellipsoid; in the x-y plane, the soft-iron distortion is identified by a tilted ellipse with the origin in (0, 0), as shown in [Figure 21](#).

Figure 21. Soft-iron effect (X-Y 2D scatter plot)



In the LSM6DS3 device, the 3x3 soft-iron rotation matrix calculated by the user has to be indicated in 9 dedicated registers: MAG\_SI\_XX, MAG\_SI\_XY, MAG\_SI\_XZ, MAG\_SI\_YX, MAG\_SI\_YY, MAG\_SI\_YZ, MAG\_SI\_ZX, MAG\_SI\_ZY, MAG\_SI\_ZZ. These register values are expressed as an 8-bit word in sign-module format; for these registers 1 LSB corresponds to 1/8, so the matrix parameters calculated by the user must be multiplied by 8 before writing them in the soft-iron registers.

The soft-iron registers are accessible when the FUNC\_CFG\_EN bit of the FUNC\_CFG\_ACCESS register is set to 1. In order to enable the soft-iron correction algorithm it is necessary to set to 1 the FUNC\_EN bit of the CTRL10\_C register, the IRON\_EN bit of the MASTER\_CONFIG register and the SOFT\_EN bit of the CTRL9\_XL register ([Table 49](#)).

### 6.4.3 Getting compensated magnetometer data

The status of magnetometer data acquisition and hard-iron/soft correction can be checked using the FUNC\_SRC register:

- SENSORHUB\_END\_OP bit is set high when the sensor hub routine is completed. The acquired magnetometer raw data are available in registers from address 66h (OUT\_MAG\_RAW\_X\_L) to 6Bh (OUT\_MAG\_RAW\_Z\_H).
- SI\_END\_OP bit is set high when the execution of the enabled hard-iron and soft-iron algorithms has completed. If the soft-iron correction is enabled, the magnetometer uncalibrated data (with soft-iron only applied) are available in registers from address 4Dh (SENSORHUB13\_REG) to 52h (SENSORHUB18\_REG). The magnetometer calibrated data, with both hard-iron (if enabled) and soft-iron (if enabled) correction are available in registers from address 2Eh (SENSORHUB1\_REG) to 33h (SENSORHUB6\_REG).

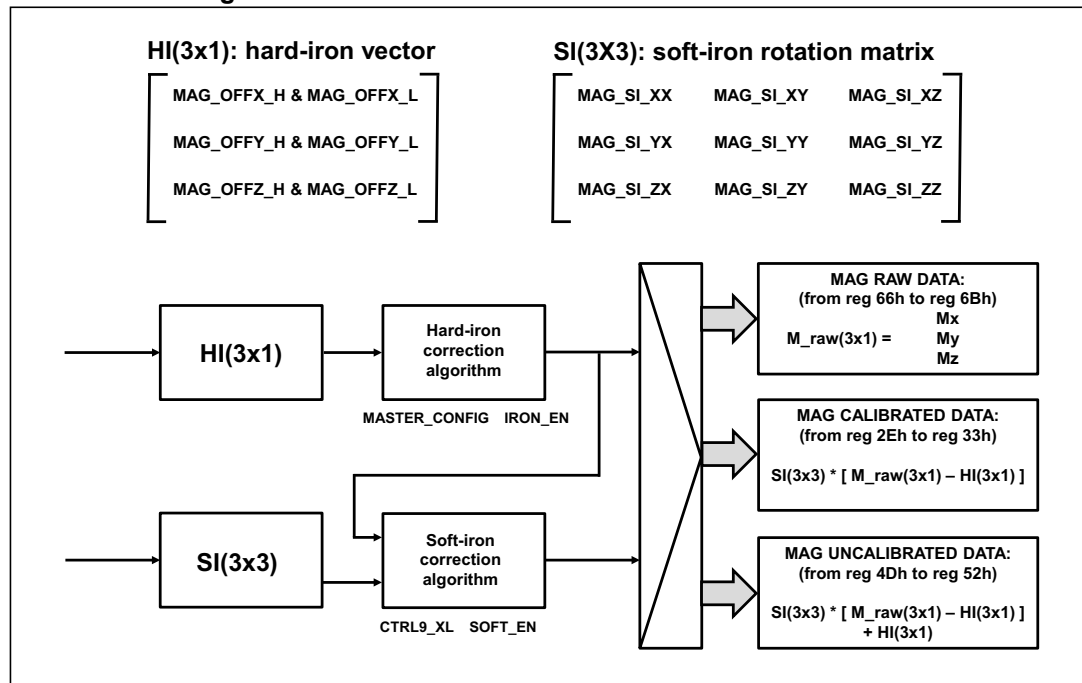
If latch mode is disabled (LIR bit of TAP\_CFG is set to 0), the SENSORHUB\_END\_OP and SI\_END\_OP bits are active only for 1/100 Hz, then they are automatically deasserted. If latch mode is enabled, these two bits are cleared by reading the FUNC\_SRC register.

The SENSORHUB\_END\_OP signal can be driven to the INT1 interrupt pin by setting to 1 the DRDY\_ON\_INT1 bit of the MASTER\_CONFIG register. The SI\_END\_OP signal can be driven to the INT2 interrupt pin by setting to 1 the INT2\_IRON bit of the MD2\_CFG register.

A schematic representation of hard-iron and soft-iron correction feature is illustrated in [Figure 22](#) below.

If the soft-iron correction is enabled and the soft-iron registers still have the default zero value, then the magnetometer calibrated data and the magnetometer uncalibrated data will be equal to zero too. As a consequence, when the soft-iron correction is enabled, the soft-iron rotation matrix must be at least initialized to the identity matrix multiplied by 8, setting the value of the MAG\_SI\_XX, MAG\_SI\_YY and MAG\_SI\_ZZ registers to 08h.

**Figure 22. Hard-iron / soft-iron correction block scheme**



### 6.4.4 Ironing example

The following example demonstrates how to define the values to be assigned to hard-iron and soft-iron correction registers starting from the calculated hard-iron vector and soft-iron rotation matrix. This example refers to the usage of the LIS3MDL magnetometer sensor with 4 gauss of configured full-scale (sensitivity = 6842 LSB/gauss).

- Hard-iron (X,Y,Z) offset values vector (gauss):

$$HI(3 \times 1) = \begin{bmatrix} -0.335605 \\ 0.126487 \\ -0.114722 \end{bmatrix}$$

These three offset values must be multiplied by the LIS3MDL sensitivity value (6842 LSB/gauss for full-scale = 4 gauss) in order to get the LSB values to be written in the hard-iron correction registers ([Table 50](#)):

**Table 50. Hard-iron register values**

|   | Offset values [LSB] | Register values                      |
|---|---------------------|--------------------------------------|
| X | -2296 (F708h)       | MAG_OFFX_H = F7h<br>MAG_OFFX_L = 08h |
| Y | +865 (0361h)        | MAG_OFFY_H = 03h<br>MAG_OFFY_L = 61h |
| Z | -785 (FCEh)         | MAG_OFFZ_H = FCh<br>MAG_OFFZ_L = EFh |

- Soft-iron rotation matrix:

$$SI(3 \times 3) = \begin{bmatrix} 1.229006 & 0.173917 & 0.052327 \\ 0.173917 & 1.033307 & -0.130089 \\ 0.052327 & -0.130089 & 1.243645 \end{bmatrix}$$

These soft-iron matrix elements must be multiplied by 8 in order to get the LSB values to be written in the soft-iron correction registers ([Table 51](#)):

**Table 51. Soft-iron register values**

|    | Soft-iron matrix elements | Register values |
|----|---------------------------|-----------------|
| XX | +1.229006                 | MAG_SI_XX = 0Ah |
| XY | +0.173917                 | MAG_SI_XY = 01h |
| XZ | +0.052327                 | MAG_SI_XZ = 00h |
| YX | +0.173917                 | MAG_SI_YX = 01h |
| YY | +1.033307                 | MAG_SI_YY = 08h |

Table 51. Soft-iron register values (continued)

|    | Soft-iron matrix elements | Register values |
|----|---------------------------|-----------------|
| YZ | -0.130089                 | MAG_SI_YZ = 81h |
| ZX | +0.052327                 | MAG_SI_ZX = 00h |
| ZY | -0.130089                 | MAG_SI_ZY = 81h |
| ZZ | +1.243645                 | MAG_SI_ZZ = 0Ah |

The code provided below gives a basic routine to configure the LIS3MDL external magnetometer sensor (refer to the datasheet for additional details) in continuous-conversion mode, initialize the hard-iron and soft-iron correction registers and read the magnetometer output registers. In this case, the pass-through feature is not used for the magnetometer configuration.

```

1  Write 80h into FUNC_CFG_ADDRESS // Enable access to embedded functions registers
2  Write 38h into SLV0_ADD          // LIS3MDL slave address = 0011100b (if SDO=0)
                                   // Enable write operation (rw_0=0)
3  Write 22h into SLV0_SUBADD       // 22h is the LIS3MDL register to be written
                                   // 00h is the value to be written in register 22h of
4  Write 00h into                   // LIS3MDL to configure it in continuous
  DATAWRITE_SRC_MODE_SUB_SLV0     // conversion mode
5  Write 00h into FUNC_CFG_ADDRESS // Disable access to embedded functions registers
6  Write 3Ch into CTRL10_C          // Enable embedded functions
                                   // Enable internal pull-up on SDx/SCx lines
7  Write 09h into MASTER_CONFIG     // Sensor hub trigger signal is XL Data Ready
                                   // Enable auxiliary I2C master
8  Write 80h into CTRL1_XL          // Turn-on the accelerometer (for trigger signal)
9  Write 38h into CTRL10_C          // Disable embedded functions
10 Write 00h into MASTER_CONFIG     // Disable auxiliary I2C master
11 Write 00h into CTRL1_XL          // Turn-off the accelerometer

12 Write 80h into FUNC_CFG_ADDRESS // Enable access to embedded functions registers
13 Write 39h into SLV0_ADD          // LIS3MDL slave address = 0011100b (if SDO=0)
                                   // Enable read operation (rw_0=1)
14 Write 28h into SLV0_SUBADD       // 28h is the first LIS3MDL output register to be
                                   // read
                                   // No decimation
15 Write 06h into SLAVE0_CONFIG     // 1 external sensor connected
                                   // Number of registers to read = 6
16 Write F7h into MAG_OFFX_H        // X offset value initialization
17 Write 08h into MAG_OFFX_L        // X offset value initialization

```

- 18 Write 03h into MAG\_OFFY\_H // Y offset value initialization
- 19 Write 61h into MAG\_OFFY\_L // Y offset value initialization
- 20 Write FCh into MAG\_OFFZ\_H // Z offset value initialization
- 21 Write EFh into MAG\_OFFZ\_L // Z offset value initialization
- 22 Write 0Ah into MAG\_SI\_XX // XX soft-iron element
- 23 Write 01h into MAG\_SI\_XY // XY soft-iron element
- 24 Write 00h into MAG\_SI\_XZ // XZ soft-iron element
- 25 Write 01h into MAG\_SI\_YX // YX soft-iron element
- 26 Write 08h into MAG\_SI\_YY // YY soft-iron element
- 27 Write 81h into MAG\_SI\_YZ // YZ soft-iron element
- 28 Write 00h into MAG\_SI\_ZX // ZX soft-iron element
- 29 Write 81h into MAG\_SI\_ZY // ZY soft-iron element
- 30 Write 0Ah into MAG\_SI\_ZZ // ZZ soft-iron element
- 31 Write 00h into FUNC\_CFG\_ADDRESS // Disable access to embedded functions registers
- 32 Write 3Ch into CTRL10\_C // Enable embedded functions
  - // Enable internal pull-up on SDx/SCx lines
  - // Sensor hub trigger signal is XL data-ready
- 33 Write 0Bh into MASTER\_CONFIG // Enable hard-iron correction
  - // Enable auxiliary I<sup>2</sup>C master
- 34 Write 3Ch into CTRL9\_XL // Enable soft-iron correction
- 35 Write 80h into CTRL1\_XL // Turn-on the accelerometer (for trigger signal)

The acquired magnetometer raw data are available in registers from address 66h (OUT\_MAG\_RAW\_X\_L) to 6Bh (OUT\_MAG\_RAW\_Z\_L).

The magnetometer uncalibrated data (with soft-iron only applied) are available in registers from address 4Dh (SENSORHUB13\_REG) to 52h (SENSORHUB18\_REG).

The magnetometer calibrated data, with both hard-iron and soft-iron correction are available in registers from address 2Eh (SENSORHUB1\_REG) to 33h (SENSORHUB6\_REG).

## 7 Mode 3 - Auxiliary SPI mode

The Auxiliary SPI mode (Mode 3) allows accessing the LSM6DS3 from multiple external devices: when this mode is enabled, both an I<sup>2</sup>C slave interface and an auxiliary SPI (3-wire) interface are available for connecting external devices.

It can be used, for example, in Electrical Image Stabilization (EIS) applications to access the device from both the application processor and the EIS controller at the same time. The EIS controller continuously gets the sensor data at a high rate for its image stabilization algorithms.

### 7.1 Auxiliary SPI mode description

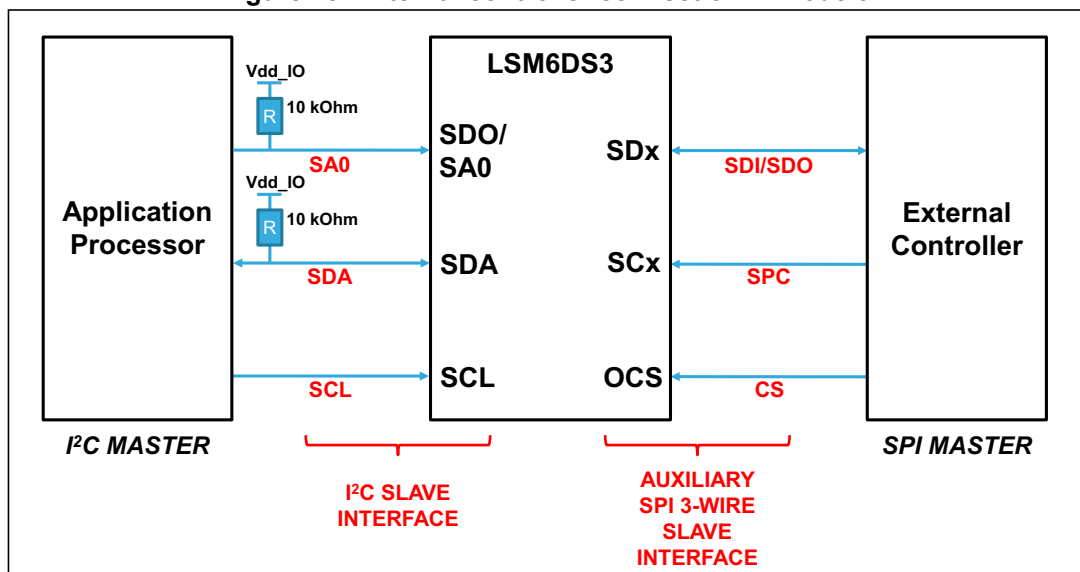
In order to enable the Auxiliary SPI mode (Mode 3), the MODE3\_EN bit of the CTRL4\_C register must be set to 1. The function of the LSM6DS3 pins after Mode 3 enable is indicated in [Table 52](#).

**Table 52. Mode 3 pin description**

|                                             | Pin     | Mode 3 function                                                                     |
|---------------------------------------------|---------|-------------------------------------------------------------------------------------|
| <b>I<sup>2</sup>C slave interface</b>       | SDO/SA0 | I <sup>2</sup> C least significant bit of the device address (SA0)                  |
|                                             | SCL     | I <sup>2</sup> C serial clock (SCL)                                                 |
|                                             | SDA     | I <sup>2</sup> C serial data (SDA)                                                  |
| <b>Auxiliary SPI 3-wire slave interface</b> | SDx     | Auxiliary SPI 3-wire interface serial data input (SDI) and serial data output (SDO) |
|                                             | SCx     | Auxiliary SPI 3-wire interface serial port clock (SPC)                              |
|                                             | OCS     | Auxiliary SPI 3-wire enable                                                         |

The external master devices have to be connected to the LSM6DS3 as illustrated in [Figure 23](#).

**Figure 23. External controller connection in Mode 3**



Both the external master devices can enable the Auxiliary SPI mode (Mode 3); once enabled, the device connected to the auxiliary SPI interface can only access the LSM6DS3 to read the registers, and only the device connected on the I<sup>2</sup>C interface can write them.

The registers of the LSM6DS3 can be read at the same time from both master devices (e.g. to get the sensor data from output data registers). Concurrent writing of LSM6DS3 registers is not allowed in order to avoid write conflicts.



## 8 First-in first-out (FIFO) buffer

In order to limit intervention by the host processor and facilitate post-processing data for event recognition, the LSM6DS3 embeds an 8 kbyte first-in first-out buffer (FIFO).

The FIFO can be configured to store the following data:

- gyroscope sensor data;
- accelerometer sensor data;
- external sensors (connected to sensor hub interface) data;
- step counter and time stamp data;
- temperature sensor data.

Saving data in the FIFO buffer is based on four 'FIFO data set' consisting of 6 bytes each:

- The 1<sup>st</sup> FIFO data set is reserved for gyroscope data;
- The 2<sup>nd</sup> FIFO data set is reserved for accelerometer data;
- The 3<sup>rd</sup> FIFO data set is reserved for the external sensor data stored in the registers from SENSORHUB1\_REG to SENSORHUB6\_REG (see [Section 6.2.9](#) for details on the SENSORHUBx\_REG);
- The 4<sup>th</sup> FIFO data set can be alternately associated to the external sensor data stored in the registers from SENSORHUB7\_REG to SENSORHUB12\_REG, to the step counter and time stamp info, or to the temperature sensor data.

All these data sets can be stored in FIFO at different ODRs, by setting the decimation factors in the FIFO\_CTRL3 and FIFO\_CTRL4 registers. Decimation factors are also used to select which FIFO data sets have to be stored in FIFO.

Five different FIFO operating modes can be chosen through the FIFO\_MODE\_[2:0] bits of the FIFO\_CTRL5 register:

- Bypass mode;
- FIFO mode;
- Continuous mode;
- Continuous-to-FIFO mode;
- Bypass-to-Continuous mode.

Data are retrieved from the FIFO through two dedicated registers: FIFO\_DATA\_OUT\_L and FIFO\_DATA\_OUT\_H. In this way, data can be read either from the FIFO (at a slower ODR) or from the device output registers (at the normal ODR).

To monitor the FIFO status (full, empty, number of sample stored, etc), four dedicated registers are available: FIFO\_STATUS1, FIFO\_STATUS2, FIFO\_STATUS3, FIFO\_STATUS4.

Programmable FIFO thresholds can be set in FIFO\_CTRL1 and FIFO\_CTRL2 using the FTH [11:0] bits.

FIFO full, FIFO threshold and FIFO overrun events can be enabled to generate dedicated interrupts on the two interrupt pins (INT1 and INT2) through the INT1\_FULL\_FLAG, INT1\_FTH and INT1\_OVR bits of the INT1\_CTRL register, and through the INT2\_FULL\_FLAG, INT2\_FTH and INT2\_OVR bits of the INT2\_CTRL register.

In order to increase the number of samples which can be stored in the FIFO, it is also possible to store (as 1<sup>st</sup> FIFO data set) only the 8 most significant bits of the accelerometer and gyroscope data by setting the bit ONLY\_HIGH\_DATA in the FIFO\_CTRL4 register.

Writing data in the FIFO can be triggered by the accelerometer/gyroscope data-ready; it can also be triggered by the sensor hub data-ready (corresponding to the behavior of the SENSORHUB\_END\_OP bit of FUNC\_SRC register): in this case the DATA\_VALID\_SEL\_FIFO bit of the MASTER\_CONFIG register must be set to 1. Moreover, if DATA\_VALID\_SEL\_FIFO is set to 0 and the TIMER\_PEDO\_FIFO\_DRDY bit of the FIFO\_CTRL2 register is set to 1, the data are stored in FIFO every time a step is detected.

## 8.1 FIFO registers

The FIFO buffer is managed by:

- five control registers (from FIFO\_CTRL1 to FIFO\_CTRL5);
- four status registers (from FIFO\_STATUS1 to FIFO\_STATUS4);
- two data output registers (FIFO\_DATA\_OUT\_L and FIFO\_DATA\_OUT\_H);
- some additional bits to enable threshold usage (STOP\_ON\_FTH) and route FIFO full, threshold or overrun events to the two interrupt lines (bits: INT1\_FULL\_FLAG, INT2\_FULL\_FLAG, INT1\_FTH, INT2\_FTH, INT1\_FIFO\_OVR, INT2\_FIFO\_OVR).

### 8.1.1 FIFO\_CTRL1 (06h)

The FIFO\_CTRL1 register contains the lower part of the 12-bit FIFO threshold level. For the complete threshold level configuration, consider also the FTH\_[11:8] bits of the FIFO\_CTRL2 register. The value of the FIFO threshold level is referred to data having 16-bit format.

The FIFO watermark flag (FTH bit in FIFO\_STATUS2 register) rises when the number of bytes stored in the FIFO is equal to or higher than the threshold level.

In order to limit the FIFO depth to the watermark level, the STOP\_ON\_FTH bit must be set to 1 in the CTRL4\_C register.

**Table 53. FIFO\_CTRL1 register**

| b7    | b6    | b5    | b4    | b3    | b2    | b1    | b0    |
|-------|-------|-------|-------|-------|-------|-------|-------|
| FTH_7 | FTH_6 | FTH_5 | FTH_4 | FTH_3 | FTH_2 | FTH_1 | FTH_0 |

### 8.1.2 FIFO\_CTRL2 (07h)

**Table 54. FIFO\_CTRL2 register**

| b7                 | b6                   | b5 | b4 | b3     | b2     | b1    | b0    |
|--------------------|----------------------|----|----|--------|--------|-------|-------|
| TIMER_PEDO_FIFO_EN | TIMER_PEDO_FIFO_DRDY | 0  | 0  | FTH_11 | FTH_10 | FTH_9 | FTH_8 |

- **TIMER\_PEDO\_FIFO\_EN** enables step counter and time stamp data to be stored as the 4<sup>th</sup> FIFO data set. The content of the 6 bytes stored in the FIFO when this bit is set to 1 is described in [Section 8.8](#).
- **TIMER\_PEDO\_FIFO\_DRDY**. When this bit is set to 1 and the **DATA\_VALID\_SEL\_FIFO** bit in the **MASTER\_CONFIG** register is set to 0, all the data are stored in the FIFO every time a new step has been detected by the step counter. See [Section 8.3](#) for details.
- **FTH\_[11:8]** contains the upper part of the FIFO threshold level. For the complete threshold level configuration, consider also the **FTH\_[7:0]** bits in the **FIFO\_CTRL1** register.

### 8.1.3 FIFO\_CTRL3 (08h)

The **FIFO\_CTRL3** register contains the accelerometer and gyroscope FIFO decimation factors, used to choose if the data of these sensors have to be stored in the FIFO and at which rate they are stored.

When the **DEC\_FIFO\_GYRO[2:0]** bits are set to 000b, the 1<sup>st</sup> FIFO data set (reserved for gyroscope data) is not stored in the FIFO. When the **DEC\_FIFO\_XL[2:0]** bits are set to 000b, the 2<sup>nd</sup> FIFO data set (reserved for accelerometer data) is not stored in the FIFO.

*Note: It's required to set at least one of the four decimation factors to 1 (no decimation).*

**Table 55. FIFO\_CTRL3 register**

| b7 | b6 | b5             | b4             | b3             | b2           | b1           | b0           |
|----|----|----------------|----------------|----------------|--------------|--------------|--------------|
| 0  | 0  | DEC_FIFO_GYRO2 | DEC_FIFO_GYRO1 | DEC_FIFO_GYRO0 | DEC_FIFO_XL2 | DEC_FIFO_XL1 | DEC_FIFO_XL0 |

**Table 56. Gyroscope FIFO decimation setting**

| DEC_FIFO_GYRO [2:0] | Configuration                |
|---------------------|------------------------------|
| 000                 | Gyroscope sensor not in FIFO |
| 001                 | No decimation                |
| 010                 | Decimation with factor 2     |
| 011                 | Decimation with factor 3     |
| 100                 | Decimation with factor 4     |
| 101                 | Decimation with factor 8     |
| 110                 | Decimation with factor 16    |
| 111                 | Decimation with factor 32    |

Table 57. Accelerometer FIFO decimation setting

| DEC_FIFO_XL [2:0] | Configuration                    |
|-------------------|----------------------------------|
| 000               | Accelerometer sensor not in FIFO |
| 001               | No decimation                    |
| 010               | Decimation with factor 2         |
| 011               | Decimation with factor 3         |
| 100               | Decimation with factor 4         |
| 101               | Decimation with factor 8         |
| 110               | Decimation with factor 16        |
| 111               | Decimation with factor 32        |

#### 8.1.4 FIFO\_CTRL4 (09h)

The FIFO\_CTRL4 register contains the decimation factors used to define at which data rate the data associated to the 3<sup>rd</sup> FIFO data set and the 4<sup>th</sup> FIFO are stored in the FIFO.

When the DEC\_DS3\_FIFO[2:0] bits are set to 000b, the 3<sup>rd</sup> FIFO data set is not stored in the FIFO. When the DEC\_DS4\_FIFO[2:0] bits are set to 000b, the 4<sup>th</sup> FIFO data set is not stored in the FIFO.

*Note: It's required to set at least one of the four decimation factors to 1 (no decimation).*

The FIFO\_CTRL4 register also contains the bit ONLY\_HIGH\_DATA, which allows storing in the FIFO only the upper part (Most Significant Byte) of accelerometer and gyroscope data, in order to increase the maximum number of accelerometer and gyroscope samples in the FIFO. See [Section 8.7](#) for more details about this functionality.

Table 58. FIFO\_CTRL4 register

| b7 | b6             | b5            | b4            | b3            | b2            | b1            | b0            |
|----|----------------|---------------|---------------|---------------|---------------|---------------|---------------|
| 0  | ONLY_HIGH_DATA | DEC_DS4_FIFO2 | DEC_DS4_FIFO1 | DEC_DS4_FIFO0 | DEC_DS3_FIFO2 | DEC_DS3_FIFO1 | DEC_DS3_FIFO0 |

Table 59. 3<sup>rd</sup> FIFO data set decimation setting

| DEC_DS3_FIFO [2:0] | Configuration                             |
|--------------------|-------------------------------------------|
| 000                | 3 <sup>rd</sup> FIFO data set not in FIFO |
| 001                | No decimation                             |
| 010                | Decimation with factor 2                  |
| 011                | Decimation with factor 3                  |
| 100                | Decimation with factor 4                  |
| 101                | Decimation with factor 8                  |
| 110                | Decimation with factor 16                 |
| 111                | Decimation with factor 32                 |

Table 60. 4<sup>th</sup> FIFO data set decimation setting

| DEC_DS4_FIFO [2:0] | Configuration                             |
|--------------------|-------------------------------------------|
| 000                | 4 <sup>th</sup> FIFO data set not in FIFO |
| 001                | No decimation                             |
| 010                | Decimation with factor 2                  |
| 011                | Decimation with factor 3                  |
| 100                | Decimation with factor 4                  |
| 101                | Decimation with factor 8                  |
| 110                | Decimation with factor 16                 |
| 111                | Decimation with factor 32                 |

### 8.1.5 FIFO\_CTRL5 (0Ah)

The FIFO\_CTRL5 register contains the FIFO operating mode bits (FIFO\_MODE\_[2:0]) and the FIFO output data rate bits (ODR\_FIFO\_[3:0]).

FIFO operating modes ([Table 63](#)) are described in [Section 8.2](#).

When the internal trigger (accelerometer/gyroscope data-ready) is used, the ODR\_FIFO\_[3:0] bits define the maximum data rate at which data are stored in FIFO. Data can be stored in FIFO at a lower data rate using the FIFO decimation factors.

For more information on how to configure the FIFO trigger and the FIFO ODR see [Section 8.3](#).

Table 61. FIFO\_CTRL5 register

| b7 | b6         | b5         | b4         | b3         | b2          | b1          | b0          |
|----|------------|------------|------------|------------|-------------|-------------|-------------|
| 0  | ODR_FIFO_3 | ODR_FIFO_2 | ODR_FIFO_1 | ODR_FIFO_0 | FIFO_MODE_2 | FIFO_MODE_1 | FIFO_MODE_0 |

Table 62. FIFO ODR selection setting

| ODR_FIFO [3:0] | Configuration              |
|----------------|----------------------------|
| 0000           | FIFO disabled              |
| 0001           | FIFO ODR is set to 10 Hz   |
| 0010           | FIFO ODR is set to 25 Hz   |
| 0011           | FIFO ODR is set to 50 Hz   |
| 0100           | FIFO ODR is set to 100 Hz  |
| 0101           | FIFO ODR is set to 200 Hz  |
| 0110           | FIFO ODR is set to 400 Hz  |
| 0111           | FIFO ODR is set to 800 Hz  |
| 1000           | FIFO ODR is set to 1600 Hz |
| 1001           | FIFO ODR is set to 3300 Hz |
| 1010           | FIFO ODR is set to 6600 Hz |

Table 63. FIFO mode selection

| FIFO_MODE [2:0] | Configuration                                                                  |
|-----------------|--------------------------------------------------------------------------------|
| 000             | Bypass mode. FIFO disabled.                                                    |
| 001             | FIFO mode. Stops collecting data when FIFO is full.                            |
| 010             | Reserved                                                                       |
| 011             | Continuous mode until trigger is deasserted, then FIFO mode.                   |
| 100             | Bypass mode until trigger is deasserted, then Continuous mode.                 |
| 101             | Reserved                                                                       |
| 110             | Continuous mode. If the FIFO is full, the new sample overwrites the older one. |
| 111             | Reserved                                                                       |

### 8.1.6 FIFO\_STATUS1 (3Ah)

The FIFO\_STATUS1 register, together with the FIFO\_STATUS2 register, provides information about the number of samples stored in the FIFO. Each sample is represented as 16-bit data.

Table 64. FIFO\_STATUS1 register

| b7          | b6          | b5          | b4          | b3          | b2          | b1          | b0          |
|-------------|-------------|-------------|-------------|-------------|-------------|-------------|-------------|
| DIFF_FIF0_7 | DIFF_FIF0_6 | DIFF_FIF0_5 | DIFF_FIF0_4 | DIFF_FIF0_3 | DIFF_FIF0_2 | DIFF_FIF0_1 | DIFF_FIF0_0 |

### 8.1.7 FIFO\_STATUS2 (3Bh)

The FIFO\_STATUS2 register, together with the FIFO\_STATUS1 register, provides information about the number of samples stored in the FIFO and about the current status (threshold, overrun, full, empty) of the FIFO buffer.

Table 65. FIFO\_STATUS2 register

| b7  | b6            | b5        | b4         | b3           | b2           | b1          | b0          |
|-----|---------------|-----------|------------|--------------|--------------|-------------|-------------|
| FTH | FIFO_OVER_RUN | FIFO_FULL | FIFO_EMPTY | DIFF_FIF0_11 | DIFF_FIF0_10 | DIFF_FIF0_9 | DIFF_FIF0_8 |

- FTH represents the watermark status. This bit is set high when the number of bytes stored in the FIFO is equal to or higher than the watermark level (each sample is represented as 16-bit data). The watermark status can be driven to the two interrupt pins by setting to 1 the INT1\_FTH bit of the INT1\_CTRL register or the INT2\_FTH bit of the INT2\_CTRL register.
- FIFO\_OVER\_RUN is set high when the FIFO is completely filled and at least one sample has already been overwritten to store the new data. This signal can be driven to

the two interrupt pins by setting to 1 the INT1\_FIFO\_OVR bit of the INT1\_CTRL register or the INT2\_FIFO\_OVR bit of the INT2\_CTRL register.

- FIFO\_FULL is set high when the next set of data that will be stored in FIFO will make the FIFO full. The next stored sample will replace the oldest FIFO sample. This signal can be driven to the two interrupt pins by setting to 1 the INT1\_FULL\_FLAG bit of the INT1\_CTRL register or the INT2\_FULL\_FLAG bit of the INT2\_CTRL register.
- FIFO\_EMPTY is set high when the FIFO is empty.
- DIFF\_FIFO\_[11:8] contains the upper part of the number of unread words (16-bit data) stored in the FIFO. The lower part is represented by the DIFF\_FIFO\_[7:0] bits in FIFO\_STATUS1. The value of DIFF\_FIFO\_[11:0] field corresponds to the number of samples in the FIFO (each sample is represented as 16-bit data). When a FIFO full event occurs (FIFO\_FULL bit is set high), the value of the DIFF\_FIFO\_[11:0] field is set to 0.

Register content is updated synchronously to the FIFO write and read operation, as illustrated in [Table 66](#).

**Table 66. FIFO\_STATUS2 behavior**

| FIFO_OVER_RUN | FIFO_FULL | FIFO_EMPTY | DIFF_FIFO_[11:0] | Number of FIFO samples           | FIFO trigger timing |
|---------------|-----------|------------|------------------|----------------------------------|---------------------|
| 0             | 0         | 1          | 0                | 0                                | t0                  |
| 0             | 0         | 0          | 1                | 1                                | t1                  |
| 0             | 0         | 0          | 2                | 2                                | t2                  |
| ...           | ...       | ...        | ...              | ...                              | ...                 |
| 0             | 0         | 0          | 4095             | 4095                             | t_full - 1          |
| 0             | 1         | 0          | 0                | 4096                             | t_full              |
| 1             | 1         | 0          | 0                | 4096<br>(old sample overwritten) | t_full + 1          |

### 8.1.8 FIFO\_STATUS3 (3Ch)

The FIFO\_STATUS3 register, together with FIFO\_STATUS4 register, specifies which axis of which sensor data will be read at the next reading. For more information on how to retrieve data from the FIFO see [Section 8.5](#).

**Table 67. FIFO\_STATUS3 register**

| b7             | b6             | b5             | b4             | b3             | b2             | b1             | b0             |
|----------------|----------------|----------------|----------------|----------------|----------------|----------------|----------------|
| FIFO_PATTERN_7 | FIFO_PATTERN_6 | FIFO_PATTERN_5 | FIFO_PATTERN_4 | FIFO_PATTERN_3 | FIFO_PATTERN_2 | FIFO_PATTERN_1 | FIFO_PATTERN_0 |

### 8.1.9 FIFO\_STATUS4 (3Dh)

The FIFO\_STATUS4 register, together with the FIFO\_STATUS3 register, specifies which axis of which sensor data will be read at the next reading. For more information on how to retrieve data from the FIFO see [Section 8.5](#).

**Table 68. FIFO\_STATUS4 register**

| b7 | b6 | b5 | b4 | b3 | b2 | b1             | b0             |
|----|----|----|----|----|----|----------------|----------------|
| 0  | 0  | 0  | 0  | 0  | 0  | FIFO_PATTERN_9 | FIFO_PATTERN_8 |

### 8.1.10 FIFO\_DATA\_OUT\_L (3Eh)

The FIFO\_DATA\_OUT\_L register is the least significant byte of the FIFO output data. The most significant byte is stored in the FIFO\_DATA\_OUT\_H register. For more information on how to retrieve data from the FIFO, see [Section 8.4](#).

**Table 69. FIFO\_DATA\_OUT\_L register**

| b7                | b6                | b5                | b4                | b3                | b2                | b1                | b0                |
|-------------------|-------------------|-------------------|-------------------|-------------------|-------------------|-------------------|-------------------|
| DATA_OUT_FIFO_L_7 | DATA_OUT_FIFO_L_6 | DATA_OUT_FIFO_L_5 | DATA_OUT_FIFO_L_4 | DATA_OUT_FIFO_L_3 | DATA_OUT_FIFO_L_2 | DATA_OUT_FIFO_L_1 | DATA_OUT_FIFO_L_0 |

### 8.1.11 FIFO\_DATA\_OUT\_H (3Fh)

The FIFO\_DATA\_OUT\_H register is the most significant byte of the FIFO output data. The least significant byte is stored in the FIFO\_DATA\_OUT\_L register. For more information on how to retrieve data from the FIFO, see [Section 8.4](#).

**Table 70. FIFO\_DATA\_OUT\_H register**

| b7                | b6                | b5                | b4                | b3                | b2                | b1                | b0                |
|-------------------|-------------------|-------------------|-------------------|-------------------|-------------------|-------------------|-------------------|
| DATA_OUT_FIFO_H_7 | DATA_OUT_FIFO_H_6 | DATA_OUT_FIFO_H_5 | DATA_OUT_FIFO_H_4 | DATA_OUT_FIFO_H_3 | DATA_OUT_FIFO_H_2 | DATA_OUT_FIFO_H_1 | DATA_OUT_FIFO_H_0 |

## 8.2 FIFO modes

The LSM6DS3 FIFO buffer can be configured to operate in five different modes selectable through the FIFO\_MODE\_[2:0] field of the FIFO\_CTRL5 register. The available configurations ensure a high level of flexibility and extend the number of functions usable in application development.

Bypass, FIFO, Continuous, Continuous-to-FIFO and Bypass-to-Continuous modes are described in the following paragraphs.



### 8.2.1 Bypass mode

When Bypass mode is enabled, the FIFO is not used, the buffer content is cleared, and it remains empty until another mode is selected.

Bypass mode is selected when the FIFO\_MODE\_[2:0] bits are set to 000b. When this mode is enabled, the FIFO\_STATUS2 register contains the value 10h (FIFO empty).

Bypass mode must be used in order to stop and reset the FIFO buffer when a different mode is operating. Note that placing the FIFO buffer into Bypass mode, the whole buffer content is cleared.

### 8.2.2 FIFO mode

In FIFO mode, the buffer continues filling until it becomes full. Then it stops collecting data and the FIFO content remains unchanged until a different mode is selected.

To guarantee the correct acquisition of data during the switching into and out of FIFO mode, the first set of data acquired must be discarded.

Follow these steps for FIFO mode configuration (if accelerometer/gyroscope data-ready is used as FIFO trigger):

1. Choose the decimation factor for each sensor through the decimation bits in the FIFO\_CTRL3 and FIFO\_CTRL4 registers (see [Section 8.3](#) for details);
2. Choose the FIFO ODR through the ODR\_FIFO\_[3:0] bits in the FIFO\_CTRL5 register;
3. Set the FIFO\_MODE\_[2:0] bits in the FIFO\_CTRL5 register to 001b to enable the FIFO mode.

When this mode is selected, the FIFO starts collecting data. The FIFO\_STATUS1 and FIFO\_STATUS2 registers are updated according to the number of samples stored.

When the FIFO is full, the FIFO\_FULL bit of the FIFO\_STATUS2 register will be set to 1. Data can be retrieved after the FIFO\_FULL event, by reading the FIFO\_DATA\_OUT\_L and FIFO\_DATA\_OUT\_H registers for the number of times specified by the DIFF\_FIFO\_[11:0] bits of the FIFO\_STATUS1 and FIFO\_STATUS2 registers.

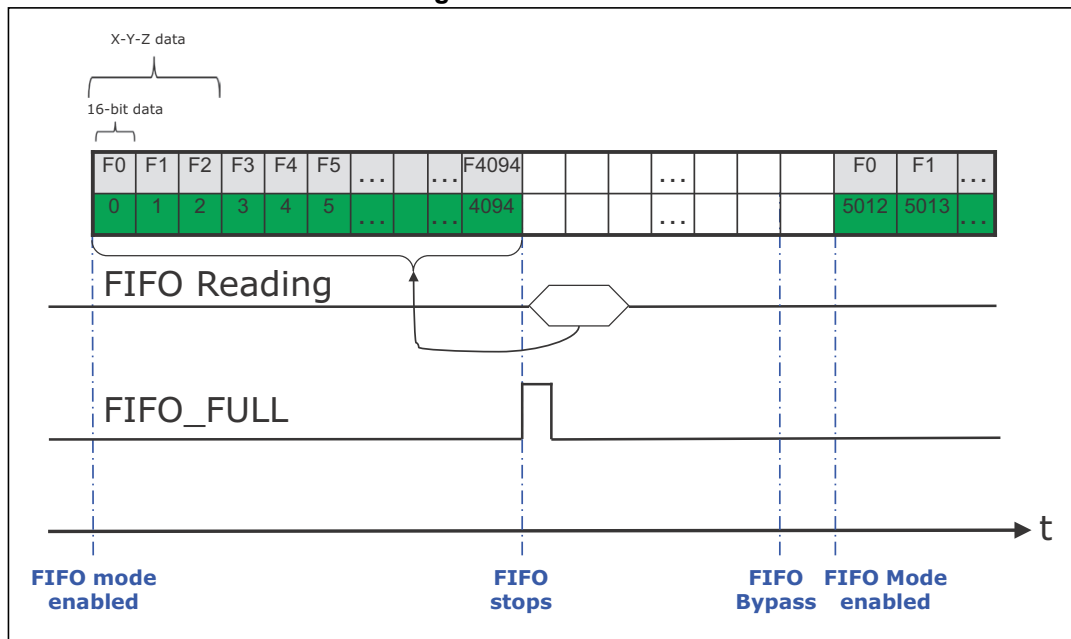
Using the FTH bit of the FIFO\_STATUS2 register, data can also be retrieved when a threshold level (FTH\_[11:0] in FIFO\_CTRL1 and FIFO\_CTRL2 registers) is reached, if the application requires a lower number of samples in the FIFO.

If the STOP\_ON\_FTH bit of the CTRL4\_C register is set to 1, the FIFO size is limited to the value of the FTH\_[11:0] bits in the FIFO\_CTRL1 and FIFO\_CTRL2 registers: in this case, the FIFO\_FULL bit of the FIFO\_STATUS2 register is set high when the number of samples in FIFO will reach or exceed the FTH\_[11:0] value on the next FIFO write operation.

Communication speed is not very important in FIFO mode because the data collection is stopped and there is no risk of overwriting data already acquired. Before restarting the FIFO mode, it is necessary to set to Bypass mode first, in order to completely clear the FIFO content.

[Figure 24](#) shows an example of FIFO mode usage. In the example X-Y-Z data (green cells indicate the sample number) from just one sensor are stored in the FIFO. In these conditions, the number of samples stored is 4095: when the FIFO buffer is completely filled the FIFO\_FULL bit of the FIFO\_STATUS2 register is set high.

### Figure 24. FIFO mode



### 8.2.3 Continuous mode

In Continuous mode, the FIFO continues filling. When the buffer is full, the FIFO index restarts from the beginning, and older data are replaced by the new data. The oldest values continue to be overwritten until a read operation frees FIFO slots. The host processor's reading speed is important in order to free slots faster than new data is made available. To stop this configuration, the Bypass mode must be selected.

Follow these steps for Continuous mode configuration (if accelerometer/gyroscope data-ready is used as FIFO trigger):

1. Choose the decimation factor for each sensor through the decimation bits in the FIFO\_CTRL3 and FIFO\_CTRL4 registers (see [Section 8.3](#) for details);
2. Choose the FIFO ODR through the ODR\_FIFO\_[3:0] bits in the FIFO\_CTRL5 register;
3. Set the FIFO\_MODE\_[2:0] bits in the FIFO\_CTRL5 register to 110b to enable FIFO Continuous mode.

When this mode is selected, the FIFO collects data continuously. The FIFO\_STATUS1 and FIFO\_STATUS2 registers are updated according to the number of samples stored.

The FIFO\_FULL bit in the FIFO\_STATUS2 register indicates when there is no more space available for other data. The FIFO\_OVER\_RUN bit in the FIFO\_STATUS2 register indicates when at least one sample has been overwritten to store the new data.

Data can be retrieved after the FIFO\_FULL event, by reading the FIFO\_DATA\_OUT\_L and FIFO\_DATA\_OUT\_H registers for a number of times specified by the DIFF\_FIFO\_[11:0] bits in the FIFO\_STATUS1 and FIFO\_STATUS2 registers.

Using the FTH bit of the FIFO\_STATUS2 register, data can also be retrieved when a threshold level (FTH [11:0] in FIFO\_CTRL1 and FIFO\_CTRL2 registers) is reached.

If the STOP\_ON\_FTH bit of CTRL4\_C register is set to 1, the FIFO size is limited to the value of the FTH [11:0] bits in the FIFO\_CTRL1 and FIFO\_CTRL2 registers: in this case,

the FTH bit of the FIFO\_STATUS2 register is set high when the number of samples in FIFO will reach or exceed the FTH\_[11:0] value on the next FIFO write operation.

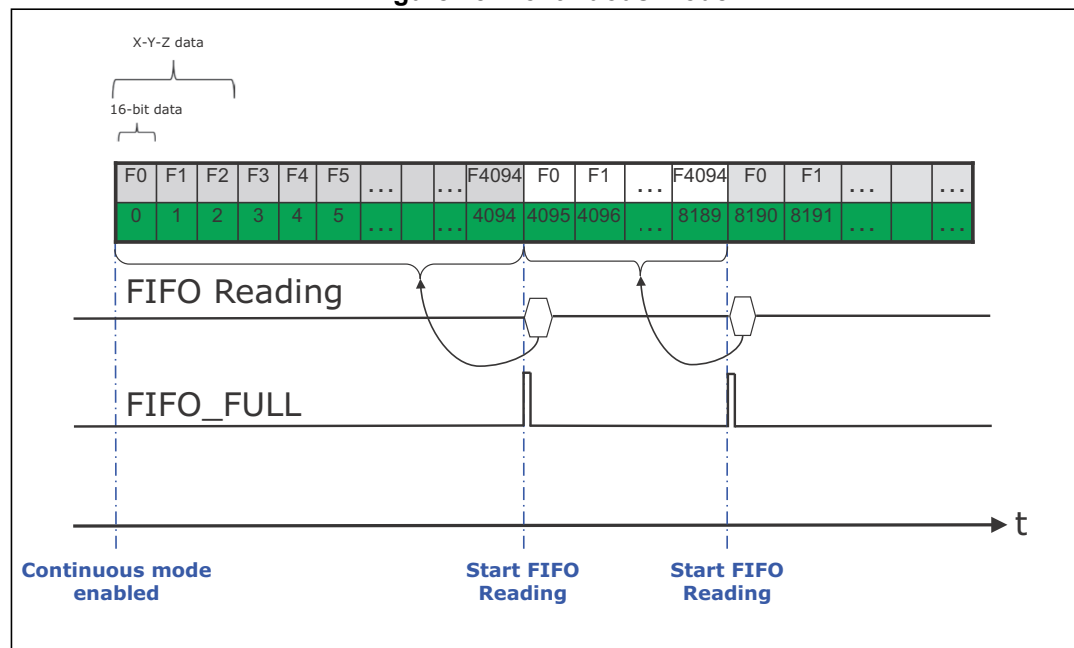
It is recommended to read faster than  $1 \times \text{ODR}$  the number of samples stored in the FIFO equal to at least three times the number of the enabled data set, in order to free FIFO slots for the new data: this allows avoiding loss of data.

If the read procedure is not fast enough, three different cases can be observed:

1. FIFO sample set is read faster than  $1 \times \text{ODR}$ : data are correctly retrieved because a free slot is made available before new data is generated.
2. FIFO sample set is read synchronous to  $1 \times \text{ODR}$ : data are correctly retrieved because a free slot is made available before new data is generated but FIFO benefits are not exploited. This case is equivalent to reading data on the data-ready interrupt and does not reduce the intervention of the host processor compared to the standard accelerometer reading.
3. FIFO sample set is read slower than  $1 \times \text{ODR}$ : in this case some data are lost because data reading is not fast enough to free slots for new data. The number of correctly recovered samples is related to the difference between the current ODR and the FIFO sample set reading rate.

*Figure 25* shows an example of the Continuous mode usage. In the example, X-Y-Z data (green cells indicate the sample number) from just one sensor are stored in the FIFO. In these conditions, the number of samples stored is 4095. In this example the FIFO samples are read faster than  $1 \times \text{ODR}$  and no data is lost.

**Figure 25. Continuous mode**



### 8.2.4 Continuous-to-FIFO mode

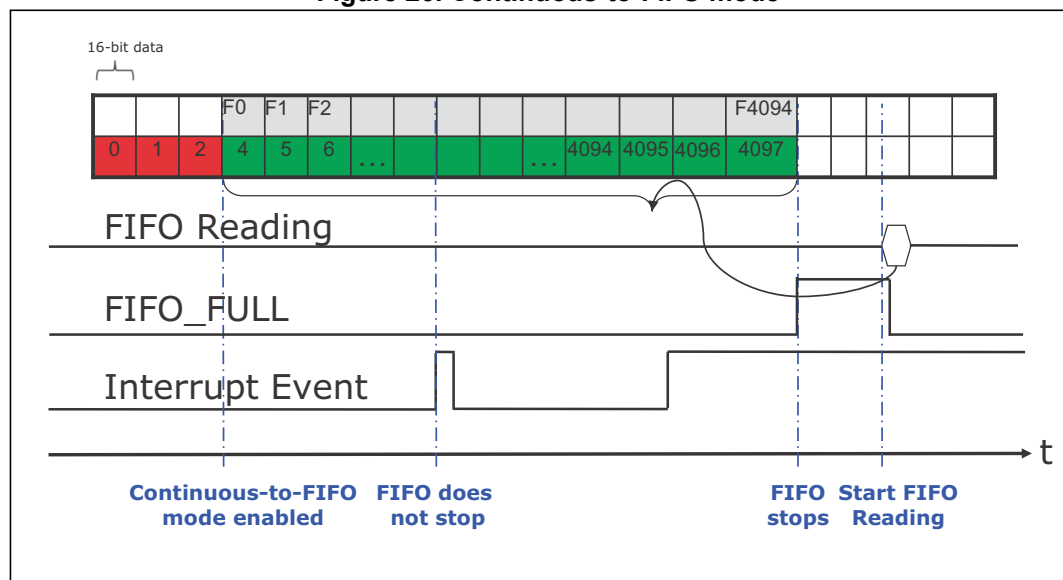
This mode is a combination of the Continuous and FIFO modes previously described. In Continuous-to-FIFO mode, the FIFO buffer starts operating in Continuous mode and switches to FIFO mode when an event condition occurs.

The event condition can be one of the following:

- Significant Motion: event detection has to be configured and the INT1\_SIG\_MOT bit of the INT1\_CTRL register has to be set to 1;
- Tilt: event detection has to be configured and the INT2\_TILT bit of the MD2\_CFG register has to be set to 1;
- Step detection: event detection has to be configured and the INT1\_STEP\_DETECTOR bit of the INT1\_CTRL register has to be set to 1;
- Single tap: event detection has to be configured and the INT2\_SINGLE\_TAP bit of the MD2\_CFG register has to be set to 1;
- Double tap: event detection has to be configured and the INT2\_DOUBLE\_TAP bit of the MD2\_CFG register has to be set to 1;
- Free-fall: event detection has to be configured and the INT2\_FF bit of the MD2\_CFG register has to be set to 1;
- Wake-up: event detection has to be configured and the INT2\_WU bit of the MD2\_CFG register has to be set to 1;
- 6D: event detection has to be configured and the INT2\_6D bit of the MD2\_CFG register has to be set to 1.

Continuous-to-FIFO mode is sensitive to the level of the interrupt signal and not to the edge, which means that if Continuous-to-FIFO is in FIFO mode and the interrupt condition disappears, the FIFO buffer returns to Continuous mode. It is recommended to latch the interrupt signal used as the FIFO event in order to avoid losing interrupt events (the interrupt signal has to be driven to the interrupt pin so that the latch function takes effect).

**Figure 26. Continuous-to-FIFO mode**



Follow these steps for Continuous-to-FIFO mode configuration (if the accelerometer/gyroscope data-ready is used as the FIFO trigger):

1. Configure one of the events as previously described;
2. Choose the decimation factor for each sensor through the decimation bits in the FIFO\_CTRL3 and FIFO\_CTRL4 registers (see [Section 8.3](#) for details);
3. Choose the FIFO ODR through the ODR\_FIFO\_[3:0] bits in the FIFO\_CTRL5 register;
4. Set the FIFO\_MODE\_[2:0] bits in the FIFO\_CTRL5 register to 011b to enable FIFO Continuous-to-FIFO mode.

In Continuous-to-FIFO mode the FIFO buffer continues filling; when the buffer is full, the FIFO\_FULL bit is set high.

If the STOP\_ON\_FTH bit of the CTRL4\_C register is set to 1, the FIFO size is limited to the value of the FTH\_[11:0] bits in the FIFO\_CTRL1 and FIFO\_CTRL2 registers: in this case, the FIFO\_FULL bit of the FIFO\_STATUS2 register is set high when the number of samples in FIFO will reach or exceed the FTH\_[11:0] value on the next FIFO write operation.

When the trigger event occurs, two different cases can be observed:

1. If the FIFO buffer is already full (FIFO\_FULL = 1), it stops collecting data at the first sample after the event trigger. The FIFO content is composed of the samples collected before the event.
2. If FIFO buffer is not full yet (initial transient), it continues filling until it becomes full (FIFO\_FULL = 1) and then, if the trigger is still present, it stops collecting data.

Continuous-to-FIFO can be used in order to analyze the history of the samples which have generated an interrupt; the standard operation is to read the FIFO content when the FIFO mode is triggered and the FIFO buffer is full and stopped.

### 8.2.5 Bypass-to-Continuous mode

This mode is a combination of the Bypass and Continuous modes previously described. In Bypass-to-Continuous mode, the FIFO buffer starts operating in Bypass mode and switches to Continuous mode when a trigger condition occurs.

The event condition can be one of the following:

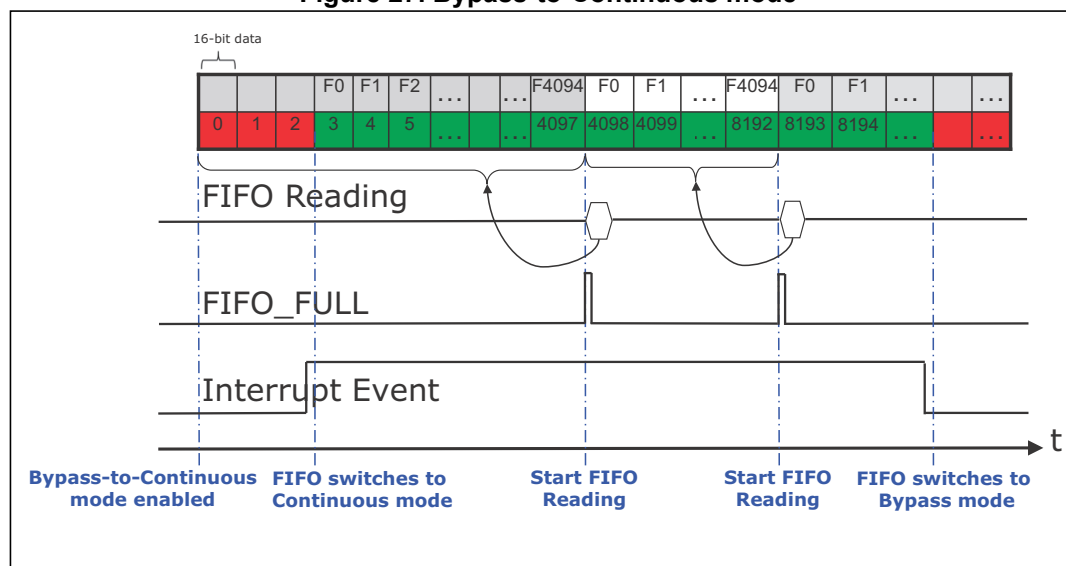
- Significant Motion: event detection has to be configured and the INT1\_SIG\_MOT bit of the INT1\_CTRL register has to be set to 1;
- Tilt: event detection has to be configured and the INT2\_TILT bit of the MD2\_CFG register has to be set to 1;
- Step detection: event detection has to be configured and the INT1\_STEP\_DETECTOR bit of the INT1\_CTRL register has to be set to 1;
- Single tap: event detection has to be configured and the INT2\_SINGLE\_TAP bit of MD2\_CFG register has to be set to 1;
- Double tap: event detection has to be configured and the INT2\_DOUBLE\_TAP bit of the MD2\_CFG register has to be set to 1;
- Free-fall: event detection has to be configured and the INT2\_FF bit of the MD2\_CFG register has to be set to 1;
- Wake-up: event detection has to be configured and the INT2\_WU bit of the MD2\_CFG register has to be set to 1;
- 6D: event detection has to be configured and the INT2\_6D bit of the MD2\_CFG register has to be set to 1.

Bypass-to-Continuous mode is sensitive to the level of the interrupt signal and not to the edge, which means that if Bypass-to-Continuous is in Continuous mode and the interrupt condition disappears, the FIFO buffer returns to Bypass mode. It is recommended to latch the interrupt signal used as the FIFO event in order to avoid losing data (the interrupt signal has to be driven to the interrupt pin so that the latch function takes effect).

Follow these steps for Bypass-to-Continuous mode configuration (if the accelerometer / gyroscope data-ready is used as the FIFO trigger):

1. Configure one of the events as previously described;
2. Choose the decimation factor for each sensor through the decimation bits in the FIFO\_CTRL3 and FIFO\_CTRL4 registers (see [Section 8.3](#) for details);
3. Choose the FIFO ODR through the ODR\_FIFO\_[3:0] bits in the FIFO\_CTRL5 register.
4. Set the FIFO\_MODE\_[2:0] bits in the FIFO\_CTRL5 register to 100b to enable FIFO Bypass-to-Continuous mode.

**Figure 27. Bypass-to-Continuous mode**



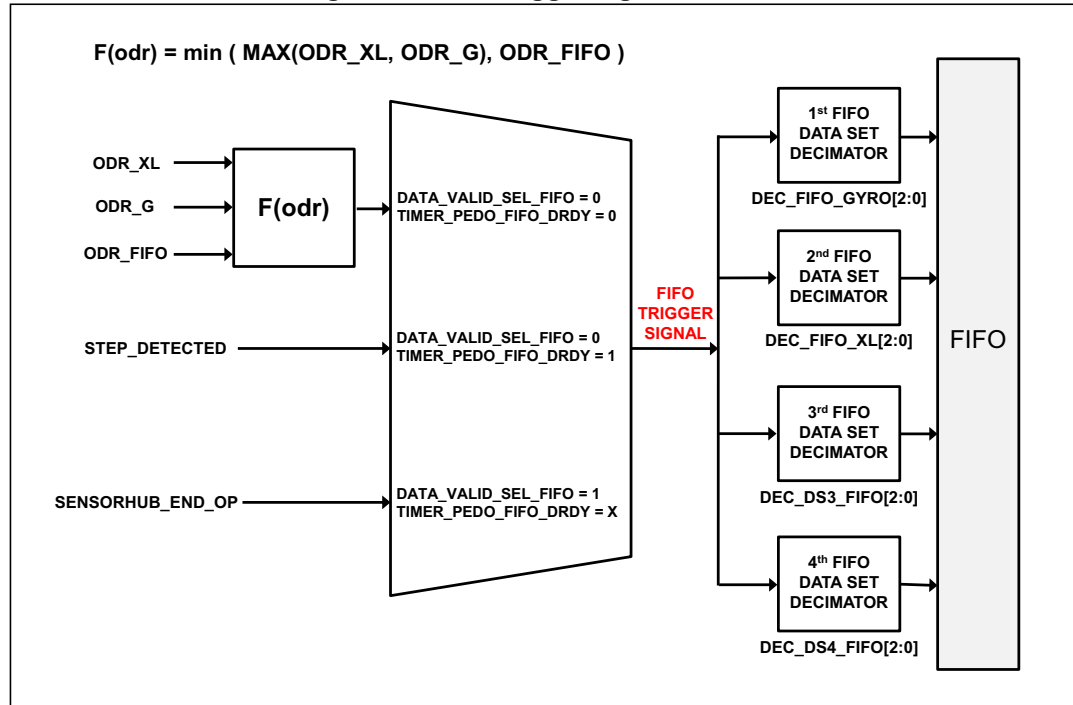
Once the trigger condition appears and the buffer switches to Continuous mode, the FIFO buffer continues filling. When the buffer is full, the FIFO\_FULL bit is set high.

Bypass-to-Continuous can be used in order to start the acquisition when the configured interrupt is generated.

### 8.3 Setting the FIFO trigger, FIFO ODR and decimation factors

Writing data in the FIFO can be configured to be triggered by three different sources.

Figure 28. FIFO trigger signal selection



As described in [Figure 28](#), the DATA\_VALID\_SEL\_FIFO bit of the MASTER\_CONFIG register and the TIMER\_PEDO\_FIFO\_DRDY bit of the FIFO\_CTRL2 register are used for this purpose:

- if both the DATA\_VALID\_SEL\_FIFO bit and the TIMER\_PEDO\_FIFO\_DRDY bit are set to 0, writing data in the FIFO is triggered by the accelerometer/gyroscope data-ready. The ODR\_FIFO[3:0] bits of FIFO\_CTRL5 define the maximum data rate at which data are stored in FIFO; the latter is limited to the maximum value between the accelerometer ODR (defined by the ODR\_XL[3:0] bits of the CTRL1\_XL register) and the gyroscope ODR (defined by the ODR\_G[3:0] bits of the CTRL2\_G register);
- if the DATA\_VALID\_SEL\_FIFO bit is set to 0 and the TIMER\_PEDO\_FIFO\_DRDY bit is set to 1, writing data in the FIFO is triggered by step detection (corresponding to the behavior of the STEP\_DETECTED bit of the FUNC\_SRC register): the data are stored in FIFO every time a step is detected;
- if the DATA\_VALID\_SEL\_FIFO bit is set to 1, writing data in the FIFO is triggered by the sensor hub (corresponding to the behavior of the SENSORHUB\_END\_OP bit of the FUNC\_SRC register), regardless of the configuration of the TIMER\_PEDO\_FIFO\_DRDY bit: the data are stored in FIFO when the sensor hub routine is complete.

Using the FIFO decimation factors, data can be stored in FIFO at a rate lower than the rate of the FIFO trigger signal. Four decimation factors can be configured, one for each FIFO data set:

- the DEC\_FIFO\_G[2:0] bits of the FIFO\_CTRL3 register define if the gyroscope data (associated to the 1<sup>st</sup> FIFO data set) are stored in FIFO and the relative rate;
- the DEC\_FIFO\_XL[2:0] bits of the FIFO\_CTRL3 register define if the accelerometer data (associated to the 2<sup>nd</sup> FIFO data set) are stored in FIFO and the relative rate;
- the DEC\_DS3\_G[2:0] bits of the FIFO\_CTRL4 register define if the data associated to the 3<sup>rd</sup> FIFO data set are stored in FIFO and the relative rate;
- the DEC\_DS4\_G[2:0] bits of the FIFO\_CTRL4 register define if the data associated to the 4<sup>th</sup> FIFO data set are stored in FIFO and the relative rate.

**Note:** *It's required to set at least one of the four decimation factors to 1 (no decimation).*

When using the internal trigger (accelerometer/gyroscope data-ready), the recommended procedure to configure the FIFO trigger is the following:

1. Set the ODR\_FIFO bits of the FIFO\_CTRL5 register to the value corresponding to the maximum ODR between the gyroscope and accelerometer;
2. Set to 1 the decimation factor of the FIFO data set associated to the sensor having the maximum ODR.

## 8.4 Retrieving data from the FIFO

**Note:** *When data are stored in the FIFO, the configuration must not be changed in order to be able to retrieve data correctly.*

When FIFO is enabled and the mode is different from Bypass, reading the FIFO output registers (FIFO\_DATA\_OUT\_L and FIFO\_DATA\_OUT\_H) returns the oldest FIFO sample set. Whenever these registers are read, their content is moved to the SPI/I<sup>2</sup>C output buffer. FIFO slots are ideally shifted up one level in order to release room for a new sample, and the FIFO output registers load the current oldest value stored in the FIFO buffer.

The recommended way to retrieve data from the FIFO is the following:

1. Read the FIFO\_STATUS1 and FIFO\_STATUS2 registers to check how many words (16-bit data) are stored in the FIFO. This information is contained in the DIFF\_FIFO\_[11:0] bits.
2. Read the FIFO\_STATUS3 and FIFO\_STATUS4 registers. The FIFO\_PATTERN\_[9:0] bits allows understanding which sensor and which couple of bytes is being read (see [Section 8.5](#) for more details).
3. Read the FIFO\_DATA\_OUT\_L and FIFO\_DATA\_OUT\_H registers to retrieve the oldest sample (16-bits format) in the FIFO. They are respectively the lower and the upper part of the oldest sample.

**Note:** *In case of SPI interface usage, reading of the FIFO\_STATUS registers (points 1 and 2 above) must be synchronized either on the data-ready interrupt or on watermark interrupt signal; the interrupt signal has to be driven on an interrupt pin.*

The entire FIFO content is retrieved by performing a certain number of read operations from the FIFO output registers until the buffer becomes empty (FIFO\_EMPTY bit of FIFO\_STATUS2 register is set high). Every other read operation returns the same value (the latest sample).



It is recommended to read all FIFO slots faster than  $1 \times \text{ODR}$ , in order to avoid losing data. In this case, the number of read operations to be performed to empty the FIFO buffer corresponds to the value of the `DIFF_FIFO_[11:0]` bits.

The rounding function (see [Section 3.6](#) for details) is automatically enabled when applying a multiple read operation to the FIFO output registers `FIFO_DATA_OUT_L` and `FIFO_DATA_OUT_H`.

## 8.5 FIFO pattern

Data are stored in the FIFO without any tag in order to maximize the number of samples stored. To understand which couple of data and which FIFO data set is going to be read, it is necessary to check the content of the `FIFO_PATTERN_[9:0]` bits in the `FIFO_STATUS3` and `FIFO_STATUS4` registers.

Data are written to the FIFO with a specific pattern (for example GyroX, GyroY, GyroZ, AccX, AccY, AccZ). This pattern changes depending on the ODRs and decimation factors assigned to the four FIFO data sets. The `FIFO_PATTERN_[9:0]` bits contain a number from 0 to the index of the last sample of the pattern, then the pattern is repeated in all FIFO content.

The first sequence of data stored in FIFO buffer contains the data of all the enabled FIFO data sets, from the first one to the fourth one. Then, data are repeated depending on the value of the decimation factor set for each FIFO data set.

The examples in the next sections explain how to use the information contained in the `FIFO_PATTERN_[9:0]` bits.

### 8.5.1 Example 1

Supposing the FIFO is storing data from the gyroscope and accelerometer at the same ODR:

- Gyroscope ODR = 104 Hz, Accelerometer ODR = 104 Hz.

If the internal trigger (accelerometer/gyroscope data-ready) is used, it's recommended to set the `ODR_FIFO_[3:0]` bits of the `FIFO_CTRL5` register to 0100b in order to set the FIFO trigger ODR to 104 Hz.

Both the `DEC_FIFO_GYRO[2:0]` and the `DEC_FIFO_XL[2:0]` fields of the `FIFO_CTRL3` register have to be set to 001b (no decimation).

The following data pattern is repeated every 6 samples (each sample is represented as 16-bit data):

- Gx Gy Gz XLx XLy XLz (Gyroscope and Accelerometer data)

The `FIFO_PATTERN_[9:0]` bits will contain a number from 0 to 5, as shown in [Table 71](#).

**Table 71. Example 1: FIFO\_PATTERN\_[9:0] bits and next reading**

| Time | FIFO_PATTERN_[9:0] | Next reading from FIFO output registers |
|------|--------------------|-----------------------------------------|
| t0   | 0                  | Gx                                      |
| t0   | 1                  | Gy                                      |
| t0   | 2                  | Gz                                      |
| t0   | 3                  | XLx                                     |
| t0   | 4                  | XLy                                     |
| t0   | 5                  | XLz                                     |

### 8.5.2 Example 2

Supposing the FIFO is storing data from the gyroscope and accelerometer at different ODRs:

- Gyroscope ODR = 208 Hz, Accelerometer ODR = 104 Hz.

If the internal trigger (accelerometer/gyroscope data-ready) is used, it's recommended to set the ODR\_FIFO\_[3:0] bits of the FIFO\_CTRL5 register to 0101b in order to set the FIFO trigger ODR to 208 Hz.

The DEC\_FIFO\_GYRO[2:0] field of the FIFO\_CTRL3 register has to be set to 001b (no decimation applied to gyroscope data) and the DEC\_FIFO\_XL[2:0] field has to be set to 010b (decimation with factor 2 applied to accelerometer data).

Since the gyroscope ODR is twice the accelerometer ODR, the following data pattern is repeated every 9 samples (each sample is represented as 16-bit data):

- Gx Gy Gz XLx XLy XLz Gx Gy Gz

The FIFO\_PATTERN\_[9:0] bits will contain a number from 0 to 8, as shown in [Table 72](#).

**Table 72. Example 2: FIFO\_PATTERN\_[9:0] bits and next reading**

| Time | FIFO_PATTERN_[9:0] | Next reading from FIFO output registers |
|------|--------------------|-----------------------------------------|
| t0   | 0                  | Gx                                      |
| t0   | 1                  | Gy                                      |
| t0   | 2                  | Gz                                      |
| t0   | 3                  | XLx                                     |
| t0   | 4                  | XLy                                     |
| t0   | 5                  | XLz                                     |
| t1   | 6                  | Gx                                      |
| t1   | 7                  | Gy                                      |
| t1   | 8                  | Gz                                      |

### 8.5.3 Example 3

Supposing the FIFO is storing data from the gyroscope, accelerometer and magnetometer at different ODRs:

Gyroscope ODR = 104 Hz, Accelerometer ODR = 208 Hz, Magnetometer ODR = 52 Hz.

If the internal trigger (accelerometer/gyroscope data-ready) is used, it's recommended to set the ODR\_FIFO\_[3:0] bits of the FIFO\_CTRL5 register to 0101b in order to set the FIFO trigger ODR to 208 Hz.

The DEC\_FIFO\_GYRO[2:0] field of FIFO\_CTRL3 register has to be set to 010b (decimation with factor 2 applied to gyroscope data) and the DEC\_FIFO\_XL[2:0] field has to be set to 001b (no decimation applied to accelerometer data). Assuming that the magnetometer is associated to the 3<sup>rd</sup> FIFO data set, the DEC\_DS3\_FIFO[2:0] field of the FIFO\_CTRL4 register has to be set to 100b (decimation with factor 4 applied to magnetometer data).

The following data pattern is repeated every 21 samples:

- Gx Gy Gz XLx XLy XLz Mx My Mz (gyroscope, accelerometer, magnetometer data - 9 samples)
- XLx XLy XLz (accelerometer data - 3 samples)
- Gx Gy Gz XLx XLy XLz (gyroscope and accelerometer data - 6 samples)
- XLx XLy XLz (accelerometer data - 3 samples)

The FIFO\_PATTERN\_[9:0] bits will contain a number from 0 to 20, as shown in [Table 73](#).

**Table 73. Example 3: FIFO\_PATTERN\_[9:0] bits and next reading**

| Time | FIFO_PATTERN_[9:0] | Next reading from FIFO output registers |
|------|--------------------|-----------------------------------------|
| t0   | 0                  | Gx                                      |
| t0   | 1                  | Gy                                      |
| t0   | 2                  | Gz                                      |
| t0   | 3                  | XLx                                     |
| t0   | 4                  | XLy                                     |
| t0   | 5                  | XLz                                     |
| t0   | 6                  | Mx                                      |
| t0   | 7                  | My                                      |
| t0   | 8                  | Mz                                      |
| t1   | 9                  | XLx                                     |
| t1   | 10                 | XLy                                     |
| t1   | 11                 | XLz                                     |
| t2   | 12                 | Gx                                      |
| t2   | 13                 | Gy                                      |
| t2   | 14                 | Gz                                      |
| t2   | 15                 | XLx                                     |
| t2   | 16                 | XLy                                     |
| t2   | 17                 | XLz                                     |
| t3   | 18                 | XLx                                     |
| t3   | 19                 | XLy                                     |
| t3   | 20                 | XLz                                     |

## 8.6 FIFO threshold

The FIFO threshold is a functionality of the LSM6DS3 FIFO which can be used to check when the number of samples in the FIFO reaches a defined threshold level.

The bits FTH[11:0] in the FIFO\_CTRL1 and FIFO\_CTRL2 registers contain the threshold level. The resolution of the FTH[11:0] field is two bytes (1 LSB = 2 Bytes, each sample is represented as 16-bit data). So, the user can select the desired level in a range between 0 and 4095.

The bit FTH in the FIFO\_STATUS2 register represents the watermark status. This bit is set high if after the next FIFO write operation the number of samples in the FIFO reaches or exceeds the watermark level (each sample is represented as 16-bit data). When the FIFO is configured in Continuous mode or in Bypass-to-Continuous mode, the FTH bit can be used to catch when the number of samples in the FIFO reaches the threshold level. In the other FIFO modes, the FIFO\_FULL bit in the FIFO\_STATUS2 register has to be used for this purpose.

FIFO size can be limited to the threshold level by setting the STOP\_ON\_FTH bit in the CTRL4\_C register to 1.

**Figure 29. FIFO threshold (STOP\_ON\_FTH = 0)**

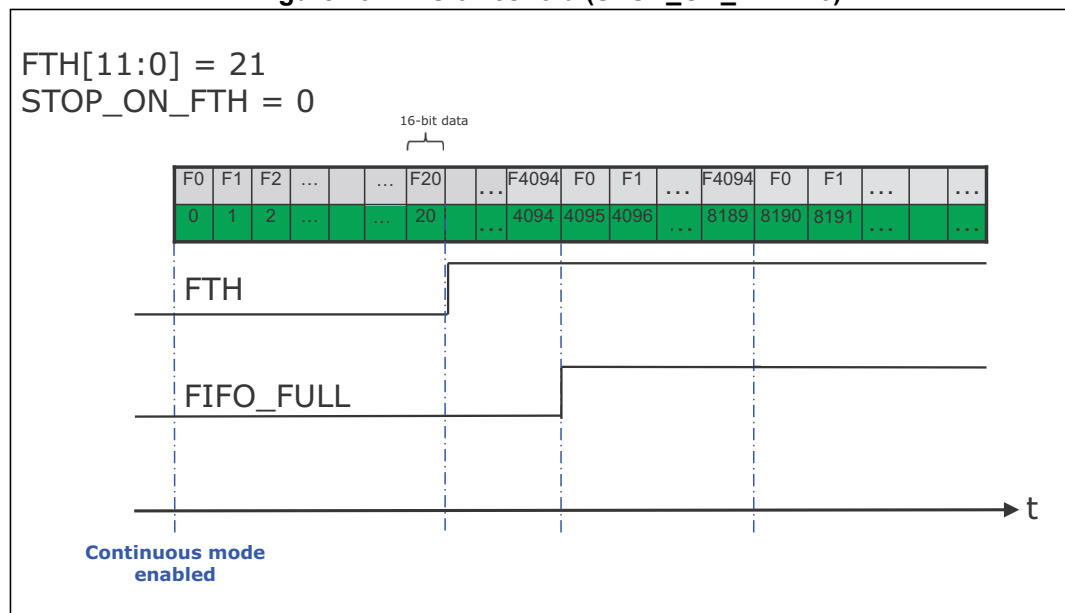


Figure 30. FIFO threshold (STOP\_ON\_FTH = 1) in Continuous mode

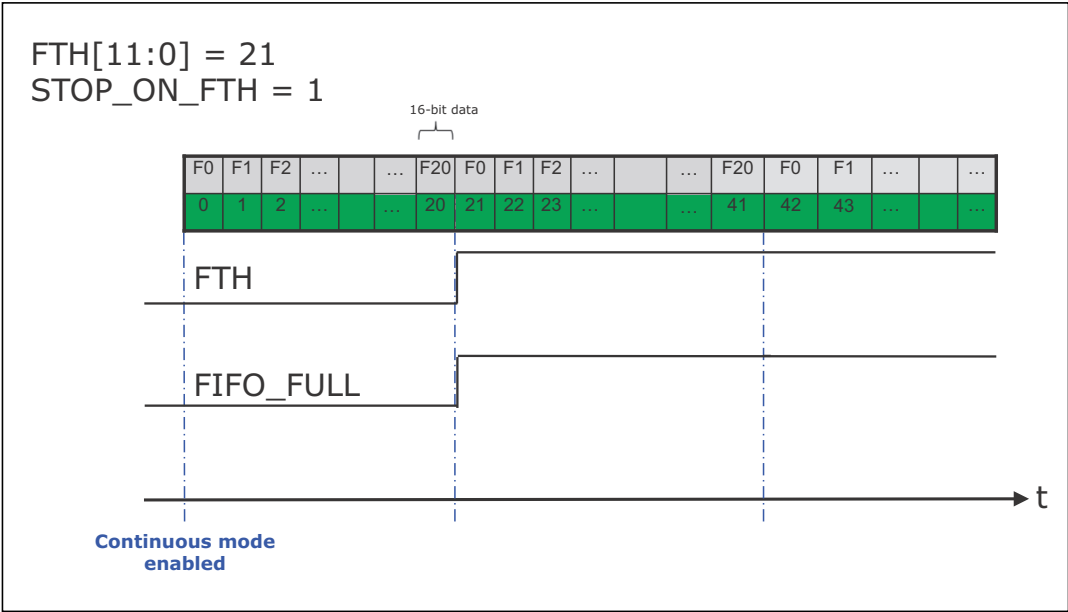


Figure 30 shows an example of FIFO threshold level usage in Continuous mode with the STOP\_ON\_FTH bit set to 1 in the CTRL4\_C register. The threshold level is set to 21 through the FTH[11:0] bits. The FTH bit of the FIFO\_STATUS2 register rises after the level 21 has been reached (21 samples in the FIFO). Since, the STOP\_ON\_FTH bit is set to 1, this time the FIFO will stop storing data at the 21<sup>st</sup> sample. So, it contains 21 samples, from 0 to 20.

## 8.7 High part of gyroscope and accelerometer data

It is possible to increase the number of samples stored in the FIFO by storing just the high part (8 bits) of gyroscope and accelerometer data. This feature is not valid for the other (external) sensors.

To enable this feature, the bit ONLY\_HIGH\_DATA must be set to 1 in the FIFO\_CTRL4 register. Gyroscope and accelerometer data will be written in the FIFO at the same ODR, in the order shown in Table 74:

Table 74. High part of gyroscope and accelerometer data in FIFO

| Byte 1   | Byte 2    | Byte 3   | Byte 4    | Byte 5   | Byte 6    |
|----------|-----------|----------|-----------|----------|-----------|
| Gyro_X_H | Accel_X_H | Gyro_Y_H | Accel_Y_H | Gyro_Z_H | Accel_Z_H |

When this feature is enabled, the 6 bytes containing the high part (8 bits) of gyroscope and accelerometer data are associated to the 1<sup>st</sup> FIFO data set and the 2<sup>nd</sup> FIFO data set is not used.

The DEC\_FIFO\_G[2:0] field of the FIFO\_CTRL3 register has to be set to a value different from 000b (1<sup>st</sup> FIFO data set stored in FIFO).

The DEC\_FIFO\_XL[2:0] field of FIFO\_CTRL3 register has to be set to 000b (2<sup>nd</sup> FIFO data set not in FIFO).

## 8.8 Step counter and time stamp data in FIFO

It is possible to store timestamp and step counter data in the FIFO. These data are stored as a 4<sup>th</sup> FIFO data set in the 6-byte data format shown in [Table 75](#):

- 3 bytes for the time stamp;
- 1 byte is not used;
- 2 bytes for the number of steps.

**Table 75. Timestamp and pedometer data in FIFO**

| Byte 1              | Byte 2               | Byte 3 | Byte 4             | Byte 5         | Byte 6          |
|---------------------|----------------------|--------|--------------------|----------------|-----------------|
| TIMESTAMP<br>[15:8] | TIMESTAMP<br>[23:16] | -      | TIMESTAMP<br>[7:0] | STEPS<br>[7:0] | STEPS<br>[15:8] |

To enable this feature, the bit `TIMER_PEDO_FIFO_EN` must be set to 1 in the `FIFO_CTRL2` register.

When this feature is enabled, the 6 bytes containing the timestamp and Step Counter data are associated to the 4<sup>th</sup> FIFO data set: the `DEC_DS4_FIFO[2:0]` field of `FIFO_CTRL4` register has to be used to define the decimator factor.

When this feature is enabled and the `DATA_VALID_SEL_FIFO` bit of the `MASTER_CONFIG` register is set to 0, data can be stored in the FIFO in two ways, depending on the configuration of the `TIMER_PEDO_FIFO_DRDY` bit in `FIFO_CTRL2`:

- When the `TIMER_PEDO_FIFO_DRDY` bit is set to 0, data are written to the FIFO at the `ODR_FIFO` rate set in the `FIFO_CTRL5` register.
- When the `TIMER_PEDO_FIFO_DRDY` bit is set to 1, data are stored in the FIFO every time a new step is detected.

Follow these steps to store time stamp and pedometer data in the FIFO using either the internal trigger (accelerometer/gyroscope data ready) or the 'step detected' method:

1. Turn on the accelerometer;
2. Enable the time stamp and pedometer (see [Section 5.1](#) and [Section 5.4](#));
3. Choose the decimation factor for the 4<sup>th</sup> FIFO data set through the `DEC_DS4_FIFO[2:0]` bits of the `FIFO_CTRL4` register;
4. Set to 1 the `TIMER_PEDO_FIFO_EN` bit in the `FIFO_CTRL2` register;
5. Configure the bit `TIMER_PEDO_FIFO_DRDY` in the `FIFO_CTRL2` register, in order to choose when storing data in the FIFO (internal trigger or every step detected);
6. If an internal trigger is used, choose the FIFO ODR through the `ODR_FIFO_[3:0]` bits of the `FIFO_CTRL5` register. If 'step detected' trigger is used, no need to set the `ODR_FIFO_[3:0]` bits;
7. Configure the FIFO operating mode through the `FIFO_MODE_[2:0]` field of the `FIFO_CTRL5` register.

## 8.9 Temperature data in FIFO

It is possible to store only temperature data as the 4<sup>th</sup> FIFO data set.

To enable this feature:

- the bit `TIMER_PEDO_FIFO_EN` of the `FIFO_CTRL2` register has to be set to 0;
- the bit `FIFO_TEMP_EN` of the `CTRL4_C` register has to be set to 1.

Temperature samples (16-bit) are stored in FIFO in the 6-byte data format shown in [Table 76](#):

**Table 76. Temperature data in FIFO**

| Byte 1 | Byte 2 | Byte 3     | Byte 4      | Byte 5 | Byte 6 |
|--------|--------|------------|-------------|--------|--------|
| -      | -      | TEMP [7:0] | TEMP [15:8] | -      | -      |

Follow these steps to store 16-bit temperature data in the FIFO using the internal trigger (accelerometer/gyroscope data ready):

1. Turn on the accelerometer or the gyroscope;
2. Choose the decimation factor (different from 000b) for the 4<sup>th</sup> FIFO data set through the `DEC_DS4_FIFO[2:0]` bits in the `FIFO_CTRL4` register;
3. Set to 1 the `FIFO_TEMP_EN` bit in the `CTRL4_C` register and to 0 the bit `TIMER_PEDO_FIFO_EN` of the `FIFO_CTRL2` register;
4. Choose the FIFO ODR through the `ODR_FIFO_[3:0]` bits of the `FIFO_CTRL5` register;
5. Configure the FIFO operating mode through the `FIFO_MODE_[2:0]` field of the `FIFO_CTRL5` register.

## 9 Temperature sensor

The LSM6DS3 is provided with an internal temperature sensor that is suitable for ambient temperature measurement.

If both the accelerometer and the gyroscope sensors are in Power-Down mode, the temperature sensor is off.

The maximum output data rate of temperature sensor is 52 Hz and its value depends on how the accelerometer and gyroscope sensors are configured:

- If the gyroscope is in Power-Down mode:
  - the temperature data rate is equal to 13 Hz if the accelerometer ODR is equal to 13 Hz (in both Low-Power and High-Performance mode);
  - the temperature data rate is equal to 26 Hz if the accelerometer configuration is 26 Hz Low-Power mode;
  - the temperature data rate is equal to 52 Hz for all other accelerometer configurations.
- If the accelerometer is in Power-Down mode:
  - the temperature data rate is equal to 13 Hz if the gyroscope configuration is 13 Hz Low-Power mode;
  - the temperature data rate is equal to 2 Hz if the gyroscope configuration is 26 Hz Low-Power mode;
  - the temperature data rate is equal to 52 Hz for all other gyroscope configurations.
- In combo mode:
  - if the gyroscope is configured in High-Performance mode, the temperature data rate is equal to 52 Hz regardless of the gyroscope ODR and the accelerometer configuration;
  - if the gyroscope is configured in Low-Power / Normal mode, the temperature data rate is equal to the maximum value between the accelerometer ODR and gyroscope ODR, while remaining below the 52 Hz value.

For the temperature sensor, the data-ready signal is represented by the TDA bit of the STATUS\_REG register. The signal can be driven to the INT2 pin by setting to 1 the INT2\_DRDY\_TEMP bit of the INT2\_CTRL register.

The temperature data is given by the concatenation of the OUT\_TEMP\_H and OUT\_TEMP\_L registers and it is represented as a number of 16 bits in two's complement format, with a sensitivity of +16 LSB/°C. The output zero level corresponds to 25 °C.

The LSM6DS3 allows swapping, by setting the BLE bit of the CTRL3\_C register set to 1, the content of the lower and the upper part of the temperature output data registers (i.e. OUT\_TEMP\_H with OUT\_TEMP\_L).

Temperature sensor data can also be stored in FIFO with a configurable decimation factor (see [Section 8.9](#) for details).



## 9.1 Example of temperature data calculation

*Table 77* provides a few basic examples of the data that is read in the temperature data registers at different ambient temperature values. The values listed in this table are given under the hypothesis of perfect device calibration (i.e. no offset, no gain error,...).

**Table 77. Output data registers content vs. temperature**

| Temperature values | BLE = 0             |                     | BLE = 1             |                     |
|--------------------|---------------------|---------------------|---------------------|---------------------|
|                    | Register address    |                     |                     |                     |
|                    | OUT_TEMP_H<br>(21h) | OUT_TEMP_L<br>(20h) | OUT_TEMP_L<br>(20h) | OUT_TEMP_H<br>(21h) |
| 0°C                | FEh                 | 70h                 | 70h                 | FEh                 |
| 25°C               | 00h                 | 00h                 | 00h                 | 00h                 |
| 50°C               | 01h                 | 90h                 | 90h                 | 01h                 |

## 10 Self-test

The embedded self-test functions allows checking the device functionality without moving it.

### 10.1 Accelerometer self-test

When the accelerometer self-test is enabled, an actuation force is applied to the sensor, simulating a definite input acceleration. In this case the sensor outputs exhibit a change in their DC levels which are related to the selected full scale through the sensitivity value.

The accelerometer self-test function is off when the ST\_XL[1:0] bits of the CTRL5\_C register are programmed to 00b; it is enabled when the ST\_XL bits are set to 01b (positive sign self-test) or 10b (negative sign self-test).

When the accelerometer self-test is activated, the sensor output level is given by the algebraic sum of the signals produced by the acceleration acting on the sensor and by the electrostatic test-force. The complete accelerometer self-test procedure is indicated in [Figure 31](#).

### 10.2 Gyroscope self-test

The gyroscope self-test allows testing of the mechanical and electrical part of the gyroscope sensor: when it is activated, an actuation force is applied to the sensor, emulating a definite Coriolis force, and the seismic mass is moved by means of this electrostatic test-force. In this case the sensor output exhibits an output change.

The gyroscope self-test function is off when the ST\_G[1:0] bits of the CTRL5\_C register are programmed to 00b; it is enabled when the ST\_G bits are set to 01b (positive sign self-test) or 11b (negative sign self-test).

When the gyroscope self-test is active, the sensor output level is given by the algebraic sum of the signals produced by the velocity acting on the sensor and by the electrostatic test-force. The complete gyroscope self-test procedure is indicated in [Figure 32](#).

Figure 31. Accelerometer self-test procedure

**Note: Keep the device still during the self-test procedure**

Write 30h to CTRL1\_XL (10h)  
 Write 00h to CTRL2\_G (11h)  
 Write 44h to CTRL3\_C (12h)  
 Write 00h to CTRL4\_C (13h)  
 Write 00h to CTRL5\_C (14h)  
 Write 00h to CTRL6\_G (15h)  
 Write 00h to CTRL7\_G (16h)  
 Write 00h to CTRL8\_XL (17h)  
 Write 38h to CTRL9\_XL (18h)  
 Write 00h to CTRL10\_C (19h)

→ Initialize Sensor, turn on sensor, enable X/Y/Z axes.  
 → Set BDU=1, FS=2G, ODR = 52Hz

**Power up, wait for 200ms for stable output**

Check XLDA in STATUS\_REG (1Eh) – Acc Data Ready Bit  
 → Reading OUTX\_XL/OUTY\_XL/OUTZ\_XL clears XLDA, Wait for the first sample  
 Read OUTX\_XL (28h/29h), OUTY\_XL (2Ah/2Bh), OUTZ\_XL (2Ch/2Dh)  
 → **Discard data**

**Read the output registers after checking XLDA bit \*5 times**

Read OUTX\_XL\_L (28h), OUTX\_XL\_H (29h): Store data in OUTX\_NOST  
 Read OUTY\_XL\_L (2Ah), OUTY\_XL\_H (2Bh): Store data in OUTY\_NOST  
 Read OUTZ\_XL\_L (2Ch), OUTZ\_XL\_H (2Dh): Store data in OUTZ\_NOST  
 The 16 bit data is expressed in two's complement.

**Average the stored data on each axis.**

Write 01h to CTRL5\_C (14h) → Enable Acc Self Test

**Wait for 200ms for stable output**

Check XLDA in STATUS\_REG (1Eh) – Acc Data Ready Bit

→ Reading OUTX\_XL/OUTY\_XL/OUTZ\_XL clears XLDA, Wait for the first sample

Read OUTX\_XL (28h/29h), OUTY\_XL (2Ah/2Bh), OUTZ\_XL (2Ch/2Dh)

→ **Discard data**

**Read the output registers after checking XLDA bit \* 5 times**

Read OUTX\_XL\_L (28h), OUTX\_XL\_H (29h): Store data in OUTX\_ST

Read OUTY\_XL\_L (2Ah), OUTY\_XL\_H (2Bh): Store data in OUTY\_ST

Read OUTZ\_XL\_L (2Ch), OUTZ\_XL\_H (2Dh): Store data in OUTZ\_ST

The 16 bit data is expressed in two's complement.

**Average the stored data on each axis**

$|\text{Min}(\text{ST\_X})| \leq |\text{OUTX\_ST} - \text{OUTX\_NOST}| \leq |\text{Max}(\text{ST\_X})|$

AND

$|\text{Min}(\text{ST\_Y})| \leq |\text{OUTY\_ST} - \text{OUTY\_NOST}| \leq |\text{Max}(\text{ST\_Y})|$

AND

$|\text{Min}(\text{ST\_Z})| \leq |\text{OUTZ\_ST} - \text{OUTZ\_NOST}| \leq |\text{Max}(\text{ST\_Z})|$

**YES (PASS)**

**NO (FAIL)**

Write 00h to CTRL1\_XL (10h): Disable sensor

Write 00h to CTRL5\_C (14h): Disable self test

Figure 32. Gyroscope self-test procedure

**Note: Keep the device still during the self-test procedure**

Write 00h to CTRL1\_XL (10h)  
 Write 5Ch to CTRL2\_G (11h)  
 Write 44h to CTRL3\_C (12h)  
 Write 00h to CTRL4\_C (13h)  
 Write 00h to CTRL5\_C (14h)  
 Write 00h to CTRL6\_G (15h)  
 Write 00h to CTRL7\_G (16h)  
 Write 00h to CTRL8\_XL (17h)  
 Write 00h to CTRL9\_XL (18h)  
 Write 38h to CTRL10\_C (19h)

→ Initialize Sensor, turn on sensor, enable P/R/Y.  
 → Set BDU=1, ODR=208Hz, FS=2000dps

**Power up, wait for 800ms for stable output**

**Check GDA in STATUS\_REG (1Eh) – Gyro Data Ready Bit**

→ Reading OUTX\_G/OUTY\_G/OUTZ\_G clears GDA, Wait for the first sample

Read OUTX\_G(22h/23h), OUTY\_G(24h/25h), OUTZ\_G(26h/27h)

→ **Discard data**

**Read the output registers after checking GDA bit \*5 times**

Read OUTX\_G\_L(22h), OUTX\_G\_H(23h): Store data in OUTX\_NOST

Read OUTY\_G\_L(24h), OUTY\_G\_H(25h): Store data in OUTY\_NOST

Read OUTZ\_G\_L(26h), OUTZ\_G\_H(27h): Store data in OUTZ\_NOST

The 16 bit data is expressed in two's complement.

**Average the stored data on each axis.**

Write 04h to CTRL5\_C (14h) → Enable Gyro Self Test

**Wait for 60 ms**

**Check GDA in STATUS\_REG (1Eh) – Gyro Data Ready Bit**

→ Reading OUTX\_G/OUTY\_G/OUTZ\_G clears GDA, Wait for the first sample

Read OUTX\_G(22h/23h), OUTY\_G(24h/25h), OUTZ\_G(26h/27h)

→ **Discard data**

**Read the output registers after checking GDA bit \* 5 times**

Read OUTX\_G\_L(22h), OUTX\_G\_H(23h): Store data in OUTX\_ST

Read OUTY\_G\_L(24h), OUTY\_G\_H(25h): Store data in OUTY\_ST

Read OUTZ\_G\_L(26h), OUTZ\_G\_H(27h): Store data in OUTZ\_ST

The 16 bit data is expressed in two's complement.

**Average the stored data on each axis**

$|\text{Min}(\text{ST\_X})| \leq |\text{OUTX\_ST} - \text{OUTX\_NOST}| \leq |\text{Max}(\text{ST\_X})|$

AND

$|\text{Min}(\text{ST\_Y})| \leq |\text{OUTY\_ST} - \text{OUTY\_NOST}| \leq |\text{Max}(\text{ST\_Y})|$

AND

$|\text{Min}(\text{ST\_Z})| \leq |\text{OUTZ\_ST} - \text{OUTZ\_NOST}| \leq |\text{Max}(\text{ST\_Z})|$

**YES (PASS)**

**NO (FAIL)**

Write 00h to CTRL2\_G (11h): Disable sensor

Write 00h to CTRL5\_C (14h): Disable self test

## 11 Revision history

**Table 78. Document revision history**

| Date        | Revision | Changes          |
|-------------|----------|------------------|
| 13-Feb-2015 | 1        | Initial release. |

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